Spark
POLYTECHNIC

قسم
هندسة الطاقة
الكهربائية
فريق سبارك

ملخص :

دوائر كهربائية ١

Circuit 1

Made by: Laith Awwad

Spark Team www.Sparkpoly.com

جروب طلاب البور والاتصالات - البوليتكنك

Topic (المواضيع)

- Chapter 1 (basic concepts)

Units , definition , circuit element

- Chapter 2 (basic laws)

ohm's low , Kirchhoff's laws , series parallel , star-dalta connection passive circuit element , voltage and current dividers

- Chapter 3 (nodel analysis)

Circuit analysis methods , nodal analysis methods , mesh analysis

- Chapter 4 (circuit theorems)

Linearity poperty , super position , source transformation , thevenin and norton theorems , maximum power transfer

- Chapter 6 (Capacitor and Inductors)

Inductance and capacitance

- Chapter 7 (First-order circuit)

Source free R C circuit , Source free R L circuit , unit step function , Driven R C circuit , Driven R L circuit

- Chapter 8 (Second order circuit)

Source free paralle R L C circuit , Source free series R L C circuit , The complete Response of the R L C circuit

مواضيع مادة دوائر كهربائية 1

Chapter 1 (Basic concept) مفاهيم أساسية

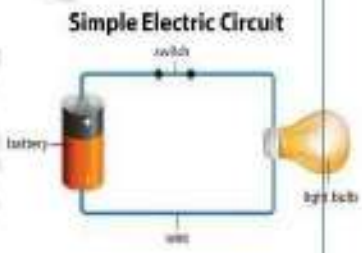
Dr. Khoulaid Adam Awad

Electric Circuit الدارة الكهربائية

اولاً : بدنا نعرف ايش هي الدارة الكهربائية ؟؟
تنتج دارة كهربائية عن طريق توصيل عدة أجهزة ثنائية الأقطاب مع بعضها بحيث الدارة الكهربائية تكون شبكة مغلقة لكي تعمل. ولتبسيط ذلك نأخذ دارة بسيطة تتكون من بطارية ومصباح ومفتاح.
.عند غلق الدائرة تلاحظ إضاءة المصباح وذلك بسبب مرور تيار كهربائي
سميت الدارة الكهربائية بدائرة لأن تيار كهربائي لا يسري في الدائرة إلا إذا كانت الدائرة مغلقة مثل الدائرة. وسميت بكهربائية لأن عملها يعتمد على الكهرباء.

- An electric circuit is an interconnection of electrical element

وتقسم الدوائر الكهربائية
دوائر بسيطة وا دوائر معقدة



- **Units** (الوحدات)

عند اجراء القياسات , يجب علينا استخدام الوحدات لتحديد القيم
نستخدم الانظمة الدولية للوحدات

- When taking measurements, we must use units to quantify values
- We use the International Systems of Units (SI for short)

مراجعة بسيطة

Systems of Units **(انظمة الوحدات)**

Quantity	Basic Unit	Symbol
Length	**meter**	**m**
Mass	**kilogram**	**kg**
Time	**second**	**s**
Electric current	**ampere**	**A**
Thermodynamic Temperature	**kelvin**	**K**
Luminous intensity	**candela**	**cd**

SI prefixes **(الوحدات الدولية)**

Spark POLYTECHNIC

Prefixes on SI units allow for easy relationships between large and small value

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

TABLE 1.2

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

❖ Charge (الشحنة)

- Charge is a basic SI unit, measured in Coulombs (C)
- Charge of single electron is 1.602*10-19 C
- One Coulomb is quite large, 6.24*1018 electrons.

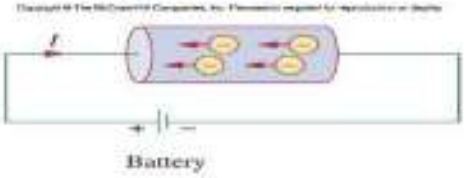
الشحنة تقاس بالكولوم وشحنة الكترون واحد تساوي 1.6×10^{-19} وا وحدة كولوم كبيرة جداً أي تساوي 6.24×10^{18} الكترون , وفي المختبر يرى المرء عادةً الشحنة ب بيكو كولوم او نانو كولوم او مايكرو كولوم , والشحنة هي مضاعف لشحنة الكترونات حسب قانون كولوم اي لايوجد نصف شحنة الكترون , ولايمكن انشاء الشحنة او اتلافها , يمكن نقلها فقط .

قانون تكمية الشحنة $q=n \times e$

❖ Electric current (التيار الكهربائي)

- The movement of charge is called a current
- Historically the moving charges were thought to be positive
- Thus we always note the direction of the equivalent positive charges, even if the moving charges are negative

تعريف التيار وهو حركة الشحنة في الموصل
تاريخاً , كان يعتقد ان الشحنات المتحركة هي الموجبة
لكن بعد فترة من الزمن أكتشفو العلماء ان هذا الكلام خطأ
وان حركة الشحنات هي شحنات السالبة .



❖ Electric current

Spark POLYTECHNIC

- Current, i, is measured as charge moved per unit time through an element.

$$i \equiv \frac{dq}{dt}$$

where

i = current in amperes

q = charge in coulombs

t = time in seconds

- Unit is Ampere (A), is one Coulomb/second

اذاً التيار نوعان اصطلاحي (ما يستخدام الان) والتيار الحقيقي

1)الاصطلاحي وهو من الموجب الى السالب

2) الحقيقي هو من السالب الى الموجب

يتم قياس التيار الكهربائي عند تحريك الشحنة لكل وحدة زمنية عبر العنصر

وحدة التيار امبير وتكافئ كولوم / ثانية

- The charge transferred between time t_o and t

الشحنة المنقولة بين الفترة الزمنية أو في الفترة الزمنية

$$q = \int_{t_o}^{t} i\,dt$$

❖ AC Vs DC (التيار المتردد وا التيار الثابت)

- A current that remains constant with time is called Direct Current (DC)
- Such current is represented by the capital I, time varying current uses the lowercase, i.
- A common source of DC is a battery.
- A current that varies sinusoidally with time is called Alternating Current (AC)
- Mains power is an example of AC

Laith Abdulsalam AlWad

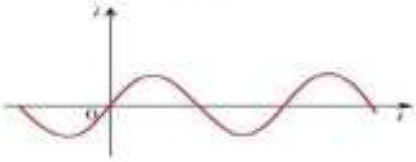
يسمى التيار الذي يبقى ثابت بمرور الوقت
التيار المباشر او الثابت



(a)

يتم تمثيل التيار المباشر بالرمز (I) والتيار المتردد ب (i)

التيار المتردد هو الذي يتغير جيبياً مع الوقت واكبر مثال على تيار المتردد هو التيار الذي يستخدم في المنازل



(b)

❖ Direction of current

- The sign of the current indicates the direction in which the charge is moving with reference to the direction of interest we define.
- We need not use the direction that the charge moves in as our reference, and often have no choice in the matter.
- A positive current through a component is the same as a negative current flowing in the opposite direction.

تشير اشارة التيار الكهربائي الى الاتجاه الذي تتحرك فيه الشحنات ولا نحتاج الى استخدام الاتجاه الذي تتحرك فيه الشحنة كمرجع لنا
التيار الموجب عبر العنصر هو نفسه التيار السالب المتدفق في الاتجاه المعاكس اي يعني ان كان التيار بسالب ف يكون الاتجاه الحقيقي هو العكس لاتجاه هذا التيار

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



(a) (b)

❖ Voltage (الجهد)

- Electrons move when there is a difference in charge between two locations.
- This difference is expressed at the potential difference, or voltage (V).
- It is always expressed with reference to two locations

where
v = voltage in volts
w = energy in Joules
q = charge in coulombs

$$v = \frac{dw}{dq}$$

Spark POLYTECHNIC

- It is equal to the energy needed to move a unit charge between the locations.
- Positive charge moving from a higher potential to a lower yields energy.
- Moving from negative to positive requires energy.

الجهد الكهربائي او فرق الجهد او الفولتية او القوة الدافعة الكهربائية وهي كمية الطاقة الدافعة لل الاكترونات من القطب السالب الى القطب الموجب وينتج عن هذه الحركة تحويل الطاقة الكهربائية الى انواع اخرى من انواع الطاقة

❖ Power (الطاقة)

- Power: time rate of expending or absorbing energy
- Denoted by p
- Circuit Elements that absorb power have a positive value of p
- Circuit Elements that produce power have a negative value of p

$$p = \frac{dw}{dt} \qquad p = \mp vi$$

$p =$ power in watts (W = J/s)
$w =$ energy in joules (J)
$t =$ time in seconds (s)
$v =$ voltage in volts (V)
$i =$ current in amperes (A)

الطاقة وهي المعدل الزمني لانفاق او امتصاص الطاقة الكهربائية
عناصر الدارة التي تمتص الطاقة لها قيمة موجبة +
عناصر الدارة التي تنتج الطاقة لها قيمة سالبة –

Absorb power → p+ produce power →p –

❖ Energy (الطاقة)

- *Law of Conservation of Energy*: the net power absorbed by a circuit is equal to 0.
- In other words, the total energy produced in a circuit is equal to the total energy absorbed
- *Energy*: capacity to do work, measured in joules (J)

$$w = \int_{t_0}^{t} p\,dt = \int_{t_0}^{t} (\pm vi)\,dt$$

If current and voltage are constant (DC),

$$w = \int_{t_0}^{t} p\,dt = p(t - t_0)$$

قانون حفظ الطاقة : صافي القدرة التي تمتصها الدارة يساوي صفر و في عبارة اخرى , اجمالي الطاقة المنتجة في الدارة يساوي اجمالي الطاقة الممتصة او المستهلكة والطاقة هنا هي القدرة على القيام بالعمل وتقاس بالجول.

Ibrahim Abdulsalam Awad

Question : prove that p=i×v ?

$$p = \frac{dw}{dt} \rightarrow p = \frac{dw}{dt} \times \frac{dq}{dq} \rightarrow p = \frac{dw}{dq} \times \frac{dq}{dt} \rightarrow p = i \times V$$

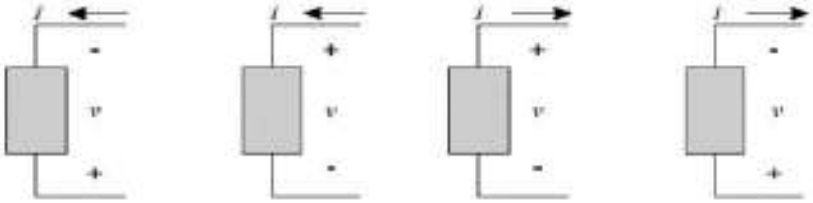
❖ Passive sign convention (اصطلاح الاشارة السلبية)

- Passive Sign Convention (PSC): Current enters the positive terminal of an element
- Most two-terminal circuit elements (e.g. batteries, light bulbs, resistors, switches) are characterized by a single equation that relates voltage to current: $v = \pm f(i)$ or $i = \pm g(v)$
- The PSC determines the sign of the relationship
  - If PSC is satisfied: $v = f(i)$ or $i = g(v)$
  - If PSC is not satisfied: $v = -f(i)$ or $i = -g(v)$
- This is also true of power
  - If PSC is satisfied: $p = vi$
  - If PSC is not satisfied: $p = -vi$

اصطلاح الاشارة السلبية وهي دخول التيار الى الطرف الموجب للعنصر ف اذا كان الجهد والتيار موجب او سالب ف هو satisfied

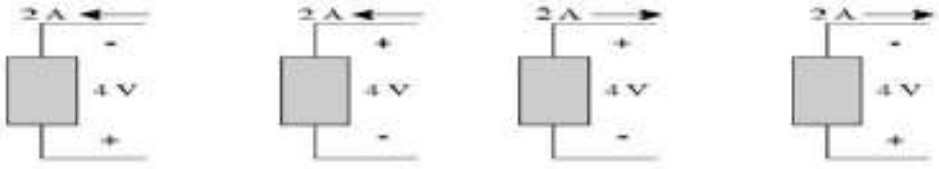
و اذا كان الجهد او التيار سالب ف هو not satisfied

Suppose the circuit element shown below is characterized by $v = \pm f(i)$ and $i = \pm g(v)$. Determine whether the PSC is satisfied and write the equations for the voltage, current, and power for each f the diagrams below.



Not saisfied satisfed not satisfied satisfied

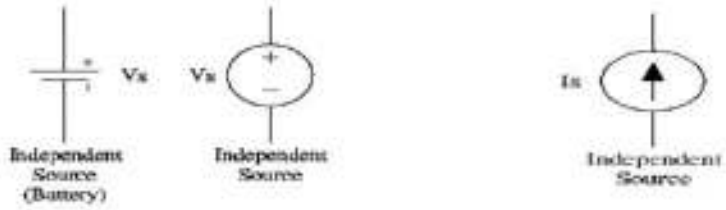
Find the power (absorbed) for each device.



P=-8 (produce) p=8(absorb) p=-8(produce) p=8(produce)

❖ Circuit Elements (عناصر الدارة)

- Ideal Independent Source: provides a specified voltage or current that is completely independent of other circuit variables



Ideal independet voltage source

ideal independent current source

مصدر جهد مستقل

مصدر تيار مستقل

- Ideal dependent source : is a voltage source or a current source whose value depends on a voltage or current else where in the network .

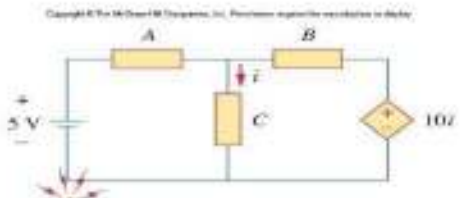


Ideal dependent voltage source
1)voltege controlled Vs (VCVS)
2)current controlled Vs (CCVS)



ideal dependent current source
1) voltage controlled Is (VCCS)
2)current controlled Is (CCCS)

- The circuit shown below is an example of using a dependent source.
- The source on the right is controlled by the current passing through element C.



Luai Abdulsalam Awad

❖Examples

Example 1.1 How much charge is represented by 4,600 electrons?

Sol: Each electron has -1.602×10^{-19} C. Hence 4,600 electrons will have -1.602×10^{-19} C/electron × 4,600 electrons = -7.369×10^{-16} C

Practice problem 1.1 Calculate the amount of charge represented by 6.667 billion protons ?

Sol: Each electron has -1.602×10^{-19} C . Hence 6.667×10^{6} electrons will have -1.602×10^{-19} C/electron $\times 6.667 \times 10^{6}$ electrons $= 1.0681 \times 10^{-9}$C

Example 1.2 The total char ge entering a terminal is gi ven by q = 5 t sin 4 πt mC. Calculate the current at t = 0.5 s.

Sol: $i = \frac{dq}{dt} = \frac{d}{dt}(5t \sin 4\pi t) \times 10^{-3}$

$i = (5\sin 4\pi t + 20\pi t \cos 4\pi t)mA \quad i(0.5) = 0 + 10\pi = 31.42mA$

Practice Problem 1.2 If in Example 1.2, q = (10 − 10e^{-2t}) mC, find the current at t = 1.0 s.

Sol: $q = \int i\,dt = 20e^{-2t} \rightarrow$ i(1s)=2.707mA

Example 1.3 Determine the total charge entering a terminal between t = 1 s and t = 2 s if the current passing the terminal is i = (3t^2 − t) A.

Sol: $q = \int_{t=1}^{t=2} (3t^2 - t)dt = 5.5\,c$

Practic Problem 1.3 The current flowing through an element is

$$i = \begin{cases} 4A & 0 < t < 1 \\ 4t^2 & t > 1 \end{cases}$$

Calculate the charge entering the element from t = 0 to t = 2 s.

Sol : $q = \int_0^1 4\,dt + \int_1^2 4t^2 dt$ =13.333c

Example 1.4 An energy source forces a constant current of 2 A for 10 s to flow through a light bulb. If 2.3 kJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb.

Lalit Ahad Salim Awad

Example 1.4

Sol : The total charge is

$\Delta q = i\,\Delta t = 2 \times 10 = 20$ C

The voltage drop

$$V = \frac{\Delta w}{\Delta q} = \frac{2.3 \times 10^3}{20} = 115\,v$$

Parctice problem 1.4 To move charge q from point b to point a requires 25 J. Find the voltage drop v_{ab} (the voltage at a positive with respect to b) if: (a)q=5C, (b) q = −10 C

Sol : $V = \frac{\Delta w}{\Delta q} = \frac{25}{5} = 5v$ $V = \frac{\Delta w}{\Delta q} = \frac{25}{-10} = -2.5v$

Example 1.5 Find the power delivered to an element at t = 3 ms if the current entering its positive terminal is i = 5 cos 60π t A and the voltage is: (a) v = 3i , (b) v = 3 di/dt.

Sol: a) The voltage is v = 3i = 15 cos 60π t; hence, the power is

$p = v \times i = 75 \cos^2 60\pi t$ W

At t = 3 ms , $p = 75 \cos 2(60\pi \times 3 \times 10^{-3}) = 75 \cos^2 0.18\pi = 53.48$ W

(b) We find the voltage and the power as

$v = 3\,di/dt = 3(-60\pi)5 \sin 60\pi t = -900\pi \sin 60\pi t$ V

$p = vi = -4500\pi \sin 60\pi t \cos 60\pi t$ W

At t = 3 ms,

$p = -4500\pi \sin 0.18\pi \cos 0.18\pi$ W

$= -14137.167 \sin 32.4° \cos 32.4° = -6.396$ kW

Practice problem 1.5 Find the power delivered to the element in Example 1.5 at t = 5 ms if the current remains the same but the voltage is: (a) v = 2iV

(B) $v = (10 + 5\int_0^t i\,dt\,)v$

Sol : (a) $v(t) = 2\,i(t) = 2(5\cos 60\pi t) = 10\cos 60\pi t$

Jaith Abdulsalam Awwal

Practice problem 1.5 :

$P(t) = v(t) \times i(t) = (10\cos 60\pi t) \times (5\cos 60\pi t) =$
$50\,(\cos 60\pi t)^2 \rightarrow p(5)=17.27w$

(b) $V(t) = (10+5\int_0^t i\,dt\,) = 10 + \int_0^t \cos 60\pi t\,dt = 10 + 5(\frac{\sin 60\pi t}{12\pi})v$

$P(t)= v(t)\, i(t) = (50\cos 60\pi t + \frac{25}{24\pi}\sin 120\pi t)w \quad \rightarrow \quad p(5) = 29.7w$

Example 1.6 How much energy does a 100-W electric bulb consume in two hour ?

Sol : w = pt = 100 (W) × 2 (h) × 60 (min/h) × 60 (s/min)= 720,000 J =720kJ

This is the same as

w = pt = 100 W × 2 h = 200 Wh

Practice problem 1.6 A home electric heater dra ws 10 A when connected to a 115 V outlet . How much energy is consumed by the heater over a period of 6 hour ?

Sol : p=Vi =115×10=1150w

W=pt=1150×6 = 6.9kwh W=1150×6×60×60=24840kJ

Example 1.7 Calculate the power supplied or absorbed by each element in Fig. 1.15.

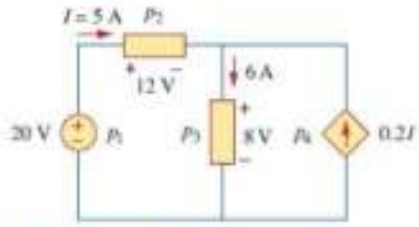


Figure 1.15
For Example 1.7.

Solution:
We apply the sign convention for power shown in Figs. 1.8 and 1.9. For p_1, the 5-A current is out of the positive terminal (or into the negative terminal); hence,

$$p_1 = 20(-5) = -100\text{ W} \quad \text{Supplied power}$$

For p_2 and p_3, the current flows into the positive terminal of the element in each case.

$$p_2 = 12(5) = 60\text{ W} \quad \text{Absorbed power}$$
$$p_3 = 8(6) = 48\text{ W} \quad \text{Absorbed power}$$

For p_4, we should note that the voltage is 8 V (positive at the top), the same as the voltage for p_3, since both the passive element and the dependent source are connected to the same terminals. (Remember that voltage is always measured across an element in a circuit.) Since the current flows out of the positive terminal,

p_4=-(0.2I)×8=-1×8 = -8 w produce , supplied power

يوجد في الكتاب مثال على هذا صفحة 16 لتمكن من المادة اكثر حل اسئلة البروبليمز صفحة 24 بالتوفيق للجمي

Salah Awad
Spark POLYTECHNIC

Chapter 2 (basic laws) القوانين الاساسية

❖Resistance المقاومة

- All materials resist the flow of current
- Resistance is usually represented by the variable R
- Depends on geometry and resistivity of the material
  - Copper 1.673e-8 ohm-meters
  - Lead 20.648e-8 ohm-meters
- Ohms per square for sheet.

A cylinder of length ℓ and cross-sectional area A has a resistance:

$$R = \rho \frac{\ell}{A}$$

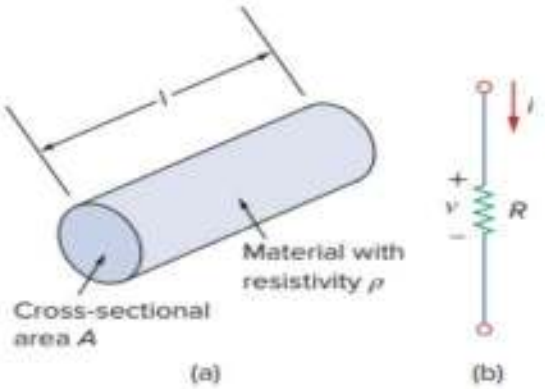
where

$R =$ resistance of an element in ohms (Ω)
$\rho =$ resistivity of the material in ohm-meters
$\ell =$ length of cylindrical material in meters
$A =$ Cross sectional area of material in meters^2

المقاومة هي خاصية فيزيائية تمتاز بها الموصلات المعدنية في الدوائر الكهربائية , وتعرف على انها قابلية المواد لمرور التيار الكهربائي فيها . جميع المواد تقاوم تدفق التيار الكهربائي
عند مرور التيار الكهربائي في الموصل ذو مقطع متجانس , وفي درجة حرارة معينة , يمكن لنا قياس مقاومة المقطع الكهربائية بدلالة نوع المادة التي صنع منها وبمعرفة ابعاده

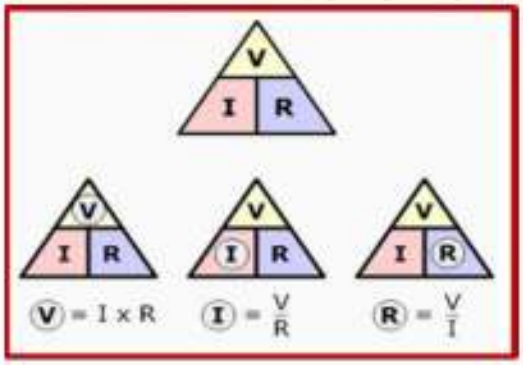
❖ Ohm's law قانون اوم

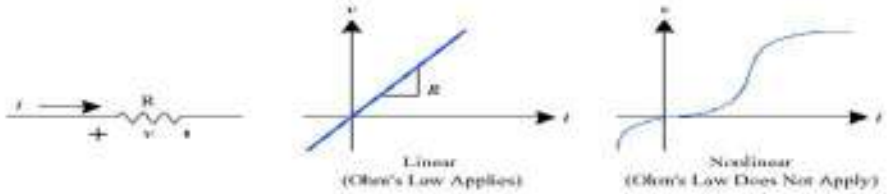
قانون اوم هو مبرأ اساسي في الكهرباء اطلق عليه هذا الاسم نسبة الى العالم الالماني "جورج سيمون اوم" ينص هذا القانون على ان فرق الجهد الكهربائي بين طرفي ناقل معدني يتناسب طردياً مع شدة التيار الكهربائي المار فيه
وبتم تعريف النسبة الثابتة بين فرق الجهد وشدة التيا ر الكهربائي بالمقاومة الكهربائية ويلاحظ ايضاً ان المقاومةلناقل ما هي الا قيمة ثابتة ولا تتغير بتغير فرق الجهد بين طرفيه ويعبر فرق الجهد على شدة التيار بقانون اوم .



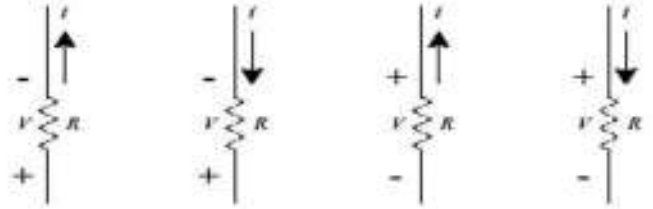
المقاومة لا يمكنها انتاج الطاقة ,لذا فان الطاقة التي تمتصها المقاومة تكون موجبة دائماً

Resistor connot produce power so the power absorbed by resistor will always be positive

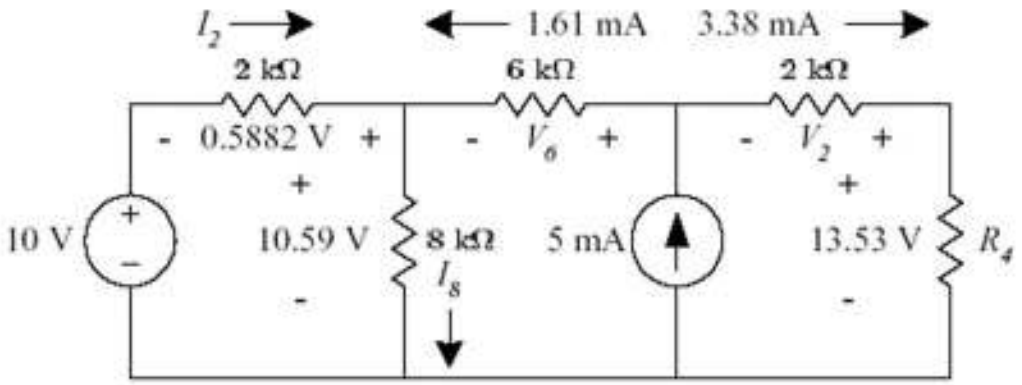




- As with all circuit elements, we need to know how the current through and voltage across the device are related
- Many materials have a complicated nonlinear relationship (including light bulbs): $v = \pm f(i)$
- Materials with a linear relationship satisfy Ohm's law: $v = \pm mi$
- The slope, m, is equal to the resistance of the element
- Ohm's Law: $v = \pm iR$
- Sign, $\pm$, is determined by the passive sign convention (PSC)



- Recall that relationships between current and voltage are sign sensitive
- Passive Sign Convention: Current enters the positive terminal of an element
  - If PSC satisfied: $v = iR$
  - If PSC not satisfied: $v = -iR$



I_2 = -0.5882/2×10³=-29.41mA

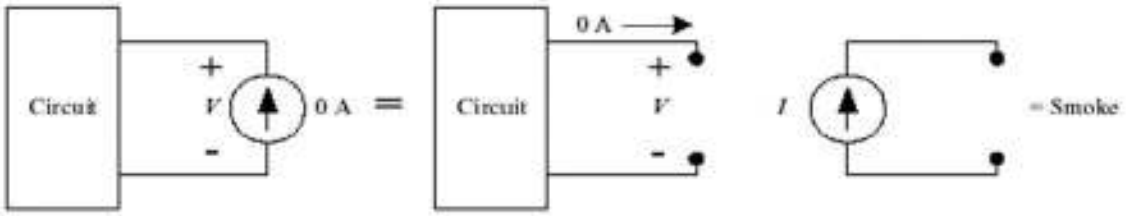
V_6 = 6×10³×1.61×10⁻³=9.6 v

R_4 = 13.53/3.38mA =4k ohm

V_2 =3.38mA×2kohm=-6.76v

I_8 =

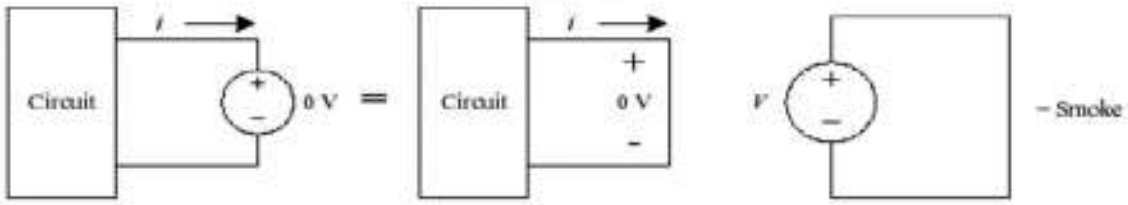
Short circuit and open circuit



- An ideal current source $I = 0$ A is also equivalent to an open circuit
- Could draw a source with $I = 0$ A, but is not done in practice
- Cannot connect a current source to an open circuit (spark coil)
- Irresistible force meets immovable object
- In practice, you blow the current source (if not protected)
- The insulator (air) usually wins. Else, sparks fly.

open circuit

هي دارة ذات مقاومة عالية جداً ك مقاومة الهواء وفي حالة الدارة المفتوحة لنظام الكهربائي التيار يأخذ قيمة صفر



- An ideal voltage source $V_s = 0$ V is also equivalent to a short circuit
- Since $v = iR$ and $R = 0$, $v = 0$ regardless of i
- Could draw a source with $V_s = 0$ V, but is not done in practice
- Can not connect a voltage source to a short circuit
- Irresistible force meets immovable object
- In practice, the wire usually wins and the voltage source melts (if not protected)

short circuit

هي دارة ذات مقاومة صغيرة جداً تقترب الى الصفر , وفي حالة قصر الدارة لنظام الكهربائي التيار يأخذ اقصى قيمة ويسمى بتيار قصر الدارة

Spark POLYTECHNIC

Conductance (الموصلية)

- Sometimes conductance is specified instead of resistance
- Inverse of resistance
- $G = \frac{1}{R} = \frac{i}{v}$
- Units: siemens (S) or mhos (℧)
- $1\ \mathrm{S} = 1\ ℧ = 1\ \mathrm{A/V}$
- In words: the ability of an element to conduct electric current

$$v = Ri \qquad p = vi = i^2R = \frac{v^2}{R}$$
$$i = Gv \qquad p = vi = v^2G = \frac{i^2}{G}$$

الموصلية وهي الدرجة التي يوصل بها الجسم الكهرباء , محسوبة على انها نسبة التيار الذي يتدفق الى فرق الجهد , وهو مقلوب المقاومة ويقاس ب السيمنز او moh's

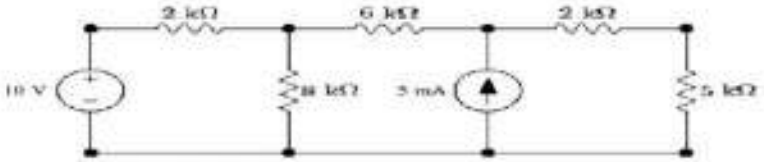
Circuit building blocks لبنات بناء الدوائر الكهربائية

- Before we can begin analysis, we need a common language and framework for describing circuits
- For this course, networks and circuits are the same
- Networks are composed of nodes, branches, and loops

قبل ان نبدأ في تحليل الدوائر الكهربائية نحتاج الى لغة مشتركة وأطار عمل لوصف هذه الدوائر . بالنسبة لهذه الدوائر , فأن الشبكات والدوائر هي نفس الشبكات التي تتكون من العقد والفروع والحلقات

Branches الفروع

Spark POLYTECHNIC



هي العناصر الموجودة في الدارة

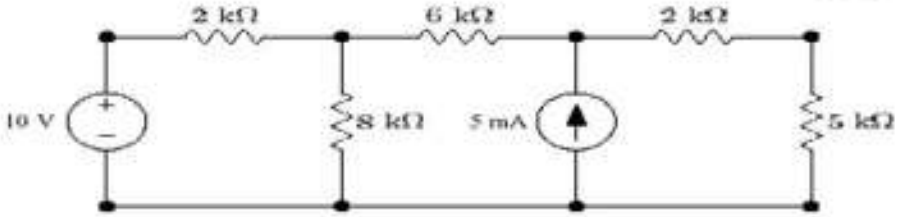
Example: How many branches?

- Branch: a single two-terminal element in a circuit
- Segments of wire are not counted as elements
- Examples: voltage source, resistor, current source

عدد العناصر 7

Nodes

(العقد)



Example: How many nodes? How many essential nodes?

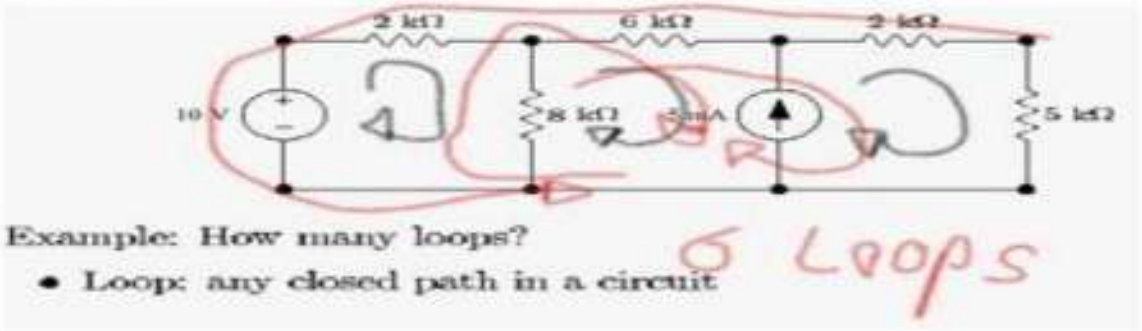
- Node: the point of connection between two or more branches
- May include a portion of the circuit (more than a single point)
- Essential Node: the point of connection between three or more branches

العقد او العقدة تشير الى اي نقطة في الدارة الكهربائية حيث يتجمع عنصرين او اكثر بها وبين كل عقدتين يجب ان يكون جهد كهربائي مختلف

وفي الشكل التالي يوجد لدينا 5 عقد السؤال الذي يسأل عنه معظم الطلاب لماذا 5 عقد وليس 8 ؟؟؟ الجواب لانو العقدة تربط بين عنصرين وا اكثر وفي الشكل التالي العقدة في الاسفل تربط بين مقاومة خمسة وا 8 اوم وا مصدر تيار ومصدر الجهد .

يوجد لدينا نوعان من العقد عقدة عادية : هي نقطة توصل بين عنصرين وا العقدة الاساسية : هي التي يتم توصيل من خلالها بين اكثر من ثلاث عناصر او اكثر

Loop and Meshes



Example: How many loops?

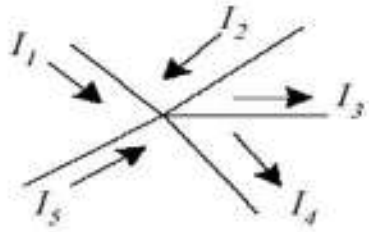
- Loop: any closed path in a circuit

Mesh : is loop which dosen't contin any other loop

لوب هو مسار مغلق نسلكه في الدارة الكهربائية اما المش ف هو ايضاً مسار كهربائي مغلق في الدارة الكهربائيو لكن لا يوجد بداخله مسارات اخرى اي في الدارة التالية يوجد 6 لوب ولكن يوجد 3 مش

Kirchhoff's laws (قوانين كيرشوف)

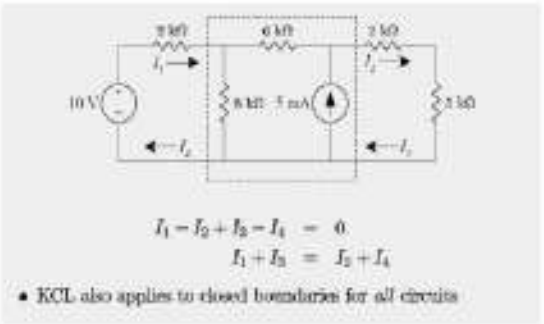
❖Kirchhoff's current law (KCL)

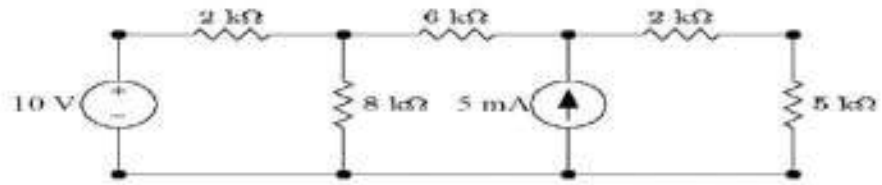


$$I_1 + I_2 - I_3 - I_4 + I_5 = 0$$
$$I_1 + I_2 + I_5 = I_3 + I_4$$

- Kirchhoff's Current Law (KCL): the algebraic sum of currents entering a node (or a closed boundary) is zero
- The sum of currents entering a node is equal to the sum of the currents leaving a node
- Common sense:
  - All of the electrons have to go somewhere
  - The current that goes in, has to come out some place
- Based on law of conservation of charge and $\nabla j = \frac{\partial \rho}{\partial t} = 0$

ينص قانون كيرشوف ان المجموع الجبري لتيارات التي تدخل في العقدة او حد مغلق هو صفر وان مجموع التيارات الداخلة في العقدة تساوي مجموع التيارات الخارجة من العقدة وا التيار الذي يدخل يجب ان يخرج في مكان ما وقانون كيرشوف مبني استناداً الى قانون حفظ الشحنة





Apply KCL to each essential node in the circuit.

Essential Node 1:

Essential Node 2:

Essential Node 3:

Essential Node 1 : I1-I2-I3=0

Essential Node 2 : I2 +5mA –I4=0

Essential Node 3 : -I1+I3-5mA+I4=0

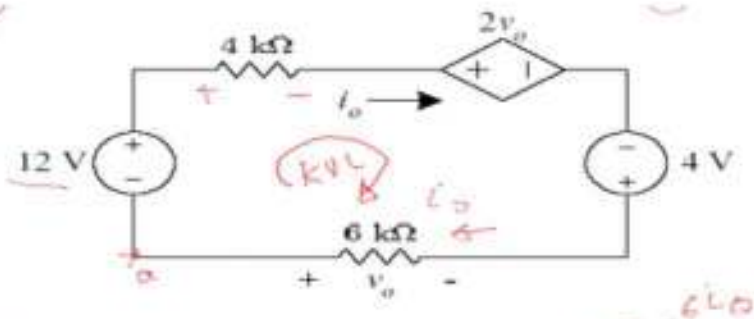
Kirchhoff's Voltage law (KVL)

$$\sum_{m=1}^{M} V_m = 0$$

- Kirchhoff's Voltage Law (KVL): the algebraic sum of voltages around a closed path (ie loop) is zero

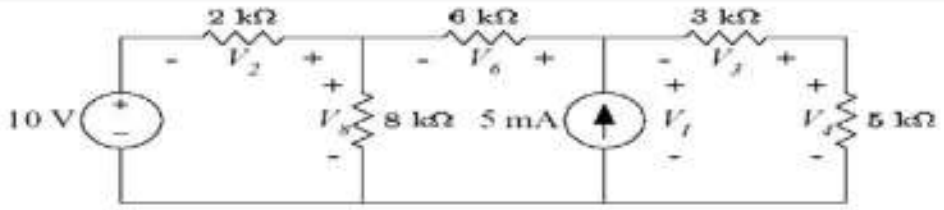
kvl:هو المجموع الجبري للجهود حول مسار مغلق اي حلقة هو صفر

Examples :



Find i_o and v_o.

Laith Abu-Awwad

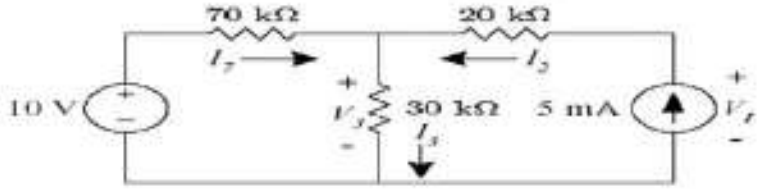


Apply KVL to each loop in the circuit.

Loop 1:
Loop 2:
Loop 3:
Loop 4:
Loop 5:
Loop 6:

sol : $12-4i_0-2v_0+4+v_0=0$ ---- 2 $v_0=6i_0$ ----- 1

$12-4i_0+12i_0+4-6i_0=0 \rightarrow i_0=-8A \rightarrow v_0=48v$



Find I_7, I_3, I_2, V_3, and V_1.

KCL at node, $I_3=I_2+I_7$ ———1

KVL at loop 1, $-10+(70K)(I_7)+(30k)(I_3)=0$ ———2

KVL at loop 2, $V_1+(20K)(I_2)+(30K)(I_3)=0$ ———3

لاحظ ان I2= 5mA لانو تيار المصدر .

$V_1=100+(30K)(I_7)$, $I_3=5mA+I_7$

$\rightarrow I_7=-1.4\ A$,,, $I_3=-1.395A$,,, $V_3=-41.85kv$,,, $V_1=-41.750\ kv$

Laith Abdul Awwal

Overview of kirchhoff's laws ملخص قانون كيرشوف

- The foundation of circuit analysis is
  - The defining equations for circuit elements (e.g. Ohm's law)
  - Kirchhoff's current law (KCL)
  - Kirchhoff's voltage law (KVL)
- The defining equations tell us how the voltage and current within a circuit element are related
- Kirchhoff's laws tell us how the voltages and currents in different branches are related

اساس تحليل المعدلات هو 1- المعدلات المحددة للعناصر الدارة قانون اوم 2- Kcl 3-Kvl

قانون كيرشوف يخبرنا القيم المحددة وكيف يرتبط الجهد والتيار داخل عنصر الدارة . وتخبرنا قوانين كيرشوف كيف يتم الربط بين الجهد والتيارات في الفروع المختلفة وقانون كيرشوف مبني على قانون حفظ الشحنة والطاقة .

Ideal current source : series

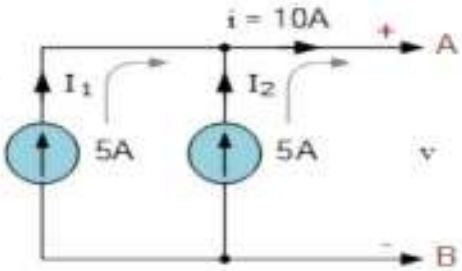


- Ideal current sources *cannot* be connected in series
- Recall: ideal current sources guarantee the current flowing through source is at specified value
- Recall: the current entering a circuit element must equal the current leaving a circuit element, $I_{in} = I_{out}$
- Could easily cause component failure (smoke)
- Ideal sources do not exist
- Technically allowed if $I_1 = I_2$, but is a bad iden

الشرح : لا يمكن توصيل مصادر التيار على التوالي لانها 1)تضمن المصادر الحالية المثالية ان التيار الكهربائي المتدفق عبر المصدر بقيمة محددة 2) يجب ان يساوي التيار الداخل لعنصر الدارة التيار الذي يخرج من عنصر الدارة يعني Iout=Iin يمكن ان تسبب في فشل المكون وينتج عنها دخان والمصادر المثالية غير موجودة , ومسموح تقنياً اذا كانت I1=I2 لكنها فكرة سيئة .

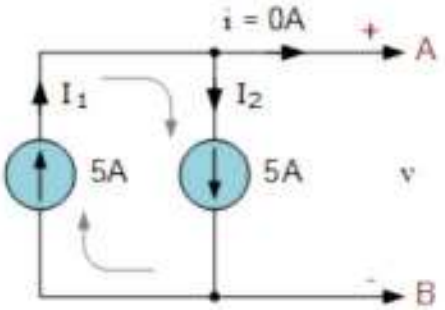
Ideal current source : parallel

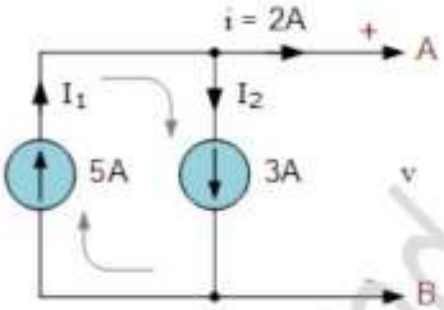
Connecting two or more current sources in parallel is equivalent to one current source whose total current output is given as the algebraic addition of the individual source currents. Here in this example, two 5 amp current sources are combined to produce 10 amps as IT = I1 + I2.

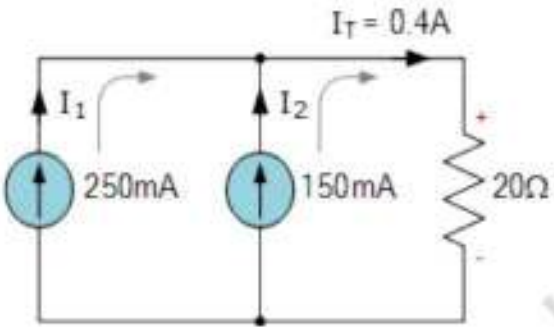


عند توصيل مصدرين او اكثر من المصادر الحالية بالتوازي معادل لمصدر تيار واحد يتم تقديم اجمالي انتاجه الحالي كاضافة جبرية لتيارات المصدر الفردية IT=I1+I2

EXAMPLES





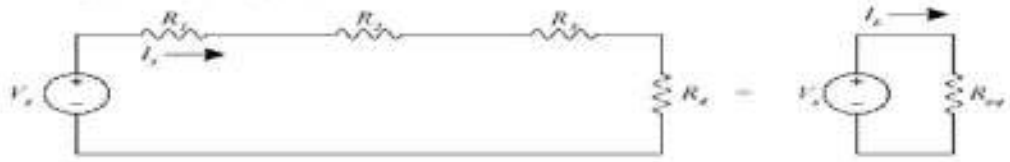


$I_T = I_1 + I_2 = 0.25 + 0.15 = 0.4A$

$V_R = I_T \times R = 0.4 \times 20 = 8V$

$P_R = I_T^2 \times R = 0.4^2 \times 20 = 3.2W$

Resistors in Series



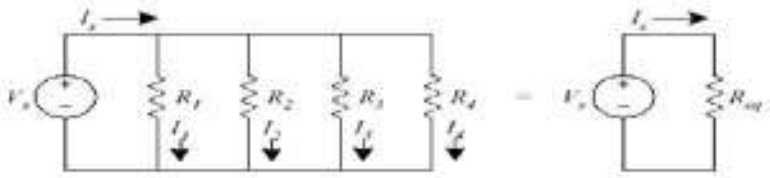
By KVL

$$\begin{aligned} V_s &= R_1 I_s + R_2 I_s + R_3 I_s + R_4 I_s \\ &= I_s (R_1 + R_2 + R_3 + R_4) \\ &= R_{eq} I_s \\ R_{eq} &= R_1 + R_2 + R_2 + R_4 \end{aligned}$$

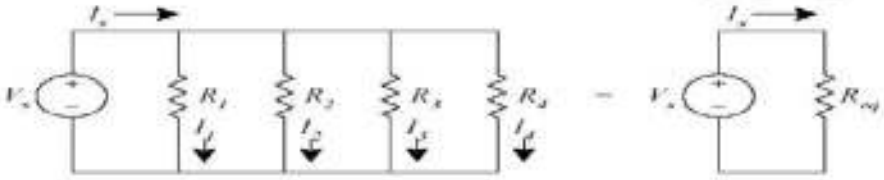
- Resistors in series add

Resistors in paralle

Spark POLYTECHNIC



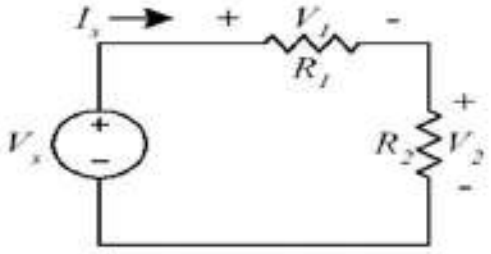
$$
\begin{aligned}
I_s &= I_1 + I_2 + I_3 + I_4 \\
&= V_s/R_1 + V_s/R_2 + V_s/R_3 + V_s/R_4 \\
&= V_s\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}\right) \\
&= \frac{V_s}{R_{eq}} \\
\frac{1}{R_{eq}} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \\
R_{eq} &= \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}}
\end{aligned}
$$



$$
\begin{aligned}
\frac{1}{R_{eq}} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \\
G_{eq} &= G_1 + G_2 + G_3 + G_4
\end{aligned}
$$

- Resistors in parallel have a more complicated relationship
- Easier to express in terms of conductance
- For two resistors: $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

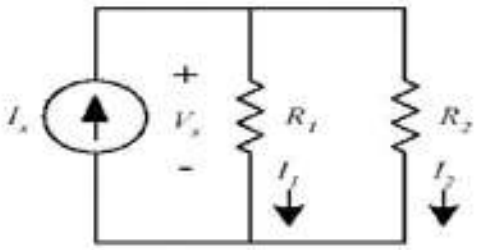
Voltage Divider



$$
\begin{aligned}
R_{eq} &= R_1 + R_2 \\
I_s &= \frac{V_s}{R_{eq}} \\
&= \frac{V_s}{R_1 + R_2} \\
V_2 &= I_s R_2 \\
&= V_s \frac{R_2}{R_1 + R_2} \\
V_1 &= I_s R_1 \\
&= V_s \frac{R_1}{R_1 + R_2}
\end{aligned}
$$

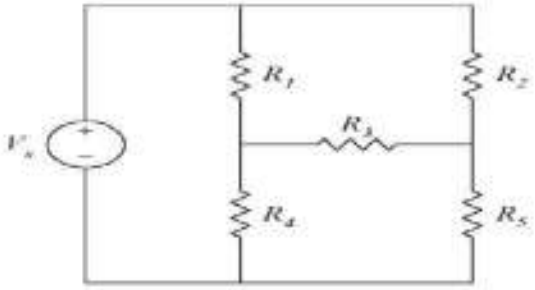
Current Divider

Spark POLYTECHNIC



$$
\begin{aligned}
V_s &= I_s \frac{R_1 R_2}{R_1 + R_2} \\
I_2 &= \frac{V_s}{R_2} \\
&= I_s \frac{R_1}{R_1 + R_2} \\
I_1 &= \frac{V_s}{R_1} \\
&= I_s \frac{R_2}{R_1 + R_2}
\end{aligned}
$$

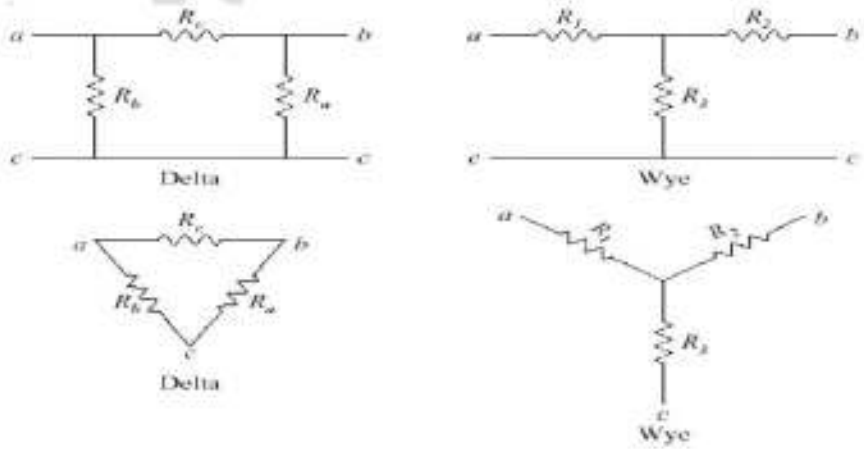
Resistor Network-Comments



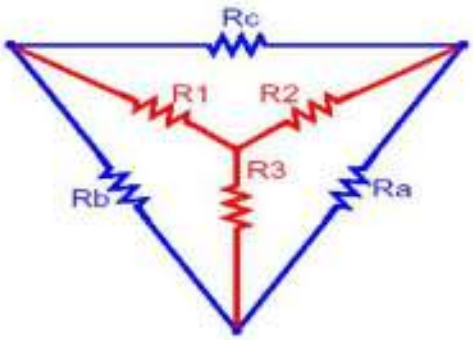
- Knowing the equivalent and parallel equivalents of resistors is not quite adequate
- There are some configurations that require one more tool

معرفة المكافئات والمتوازية للمقاومات ليست كافية تماماً , فهناك بعض التكوينات التي تتطلب أداة أخرى , في هذا الدارة لكي نجد المكافئة نستخدم طريقة التحويل من ستار الى دلتا لحل وايجاد المكافئة.

Delta ↔Wye Transformations



- Every Delta network is functionally equivalent to a Wye network (and vice versa).

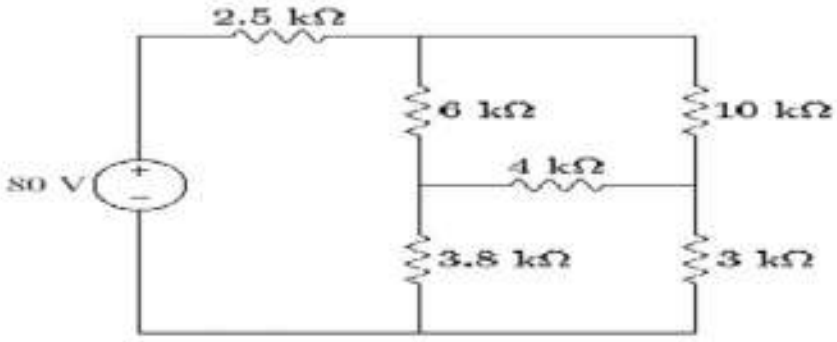


The following must be satisfied for equivalence

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1} \qquad R_1 = \frac{R_bR_c}{R_a + R_b + R_c}$$

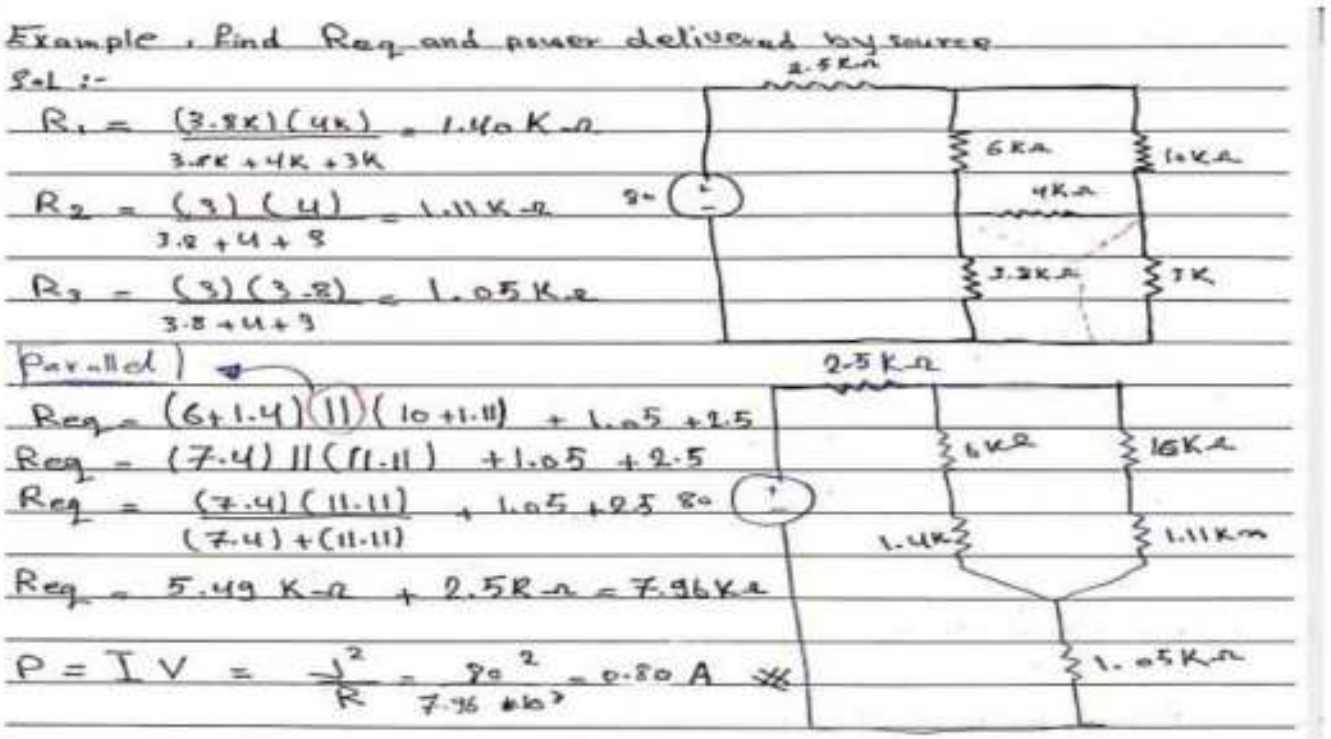
$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2} \qquad R_2 = \frac{R_cR_a}{R_a + R_b + R_c}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3} \qquad R_3 = \frac{R_aR_b}{R_a + R_b + R_c}$$



Find R_{eq} and Power delivered by source

Laith Abu



Exampel 2.1 An electric iron draws 2 A at 120 V. Find its resistance

$$R=\frac{V}{I}=\frac{120}{2}=60\Omega$$

Practice problem 1.2 The essential component of a toaster is an electrical element (a resistor) that converts electrical energy to heat energy. How much current is drawn by a toaster with resistance 15 Ω at 110 V ? Answer: 7.333 A.

Example 2.2 In the circuit shown in Fig. 2.8, calculate the current i, the conductance G, and the power p ?

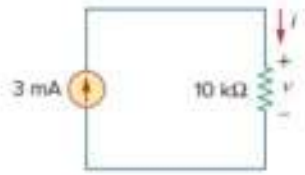


$$i=\frac{V}{R}=\frac{30}{5\times10^{3}}=6mA$$

$$G=\frac{1}{R}=0.2\ \mho$$

$$p=v\times i=(6\times10^{-3})\times(30)=0.18W=180mW$$

Practice Problem 2.2 For the circuit shown in Fig. 2.9, calculate the voltage v, the conductance G, and the power p ? Answer: 30 V, 100 μS, 90 mW.



Spark POLYTECHNIC

Example 2.3 A voltage source of 20 sin πt V is connected across a 5-kΩ resistor. Find the current through the resistor and the power dissipated.

Sol : $i = \frac{V}{R} = \frac{20 \sin \pi t}{5 \times 10^3} = 4 \sin \pi t \ \ mA$

Practice Problem 2.3 A resistor absorbs an instantaneous power of 30 $\cos^2$ t mW when connected to a voltage source v = 15 cos t V . Find i and R.

Sol :$p = i \times v \ \rightarrow i = \frac{30 \cos^2 t \ (mW)}{15 \cos t \ (V)} = 2 \cos t \ \ mA$

$$\rightarrow R = \frac{15 \cos t \ (v)}{2 \cos t \ (mA)} = 7.5 \ K\Omega$$

Example 2.4 Determine the number of branches and nodes in the circuit shown in Fig. 2.12. Identify which elements are in series and which are in parallel



Figure 2.12
For Example 2.4.

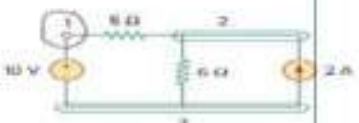


Figure 2.13
The three nodes in the circuit of Fig. 2.12.

Sol: number of branches = 4

Vs=10v,5Ω,6Ω,is=2A

Number of Node = 3

Practice Problem 2.4 How many branches and nodes does the circuit in Fig. 2.14 have? Identify Practice Problem 2.4the elements that are in series and in parallel

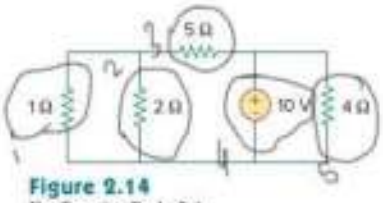


Figure 2.14
For Practice Prob. 2.4.

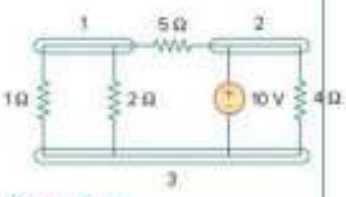


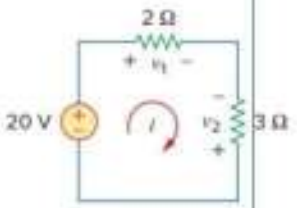
Figure 2.15
Answer for Practice Prob. 2.4.

Sol: number of branches = 4

Vs=10v,4,5,2,1Ω

Number of nodes =3

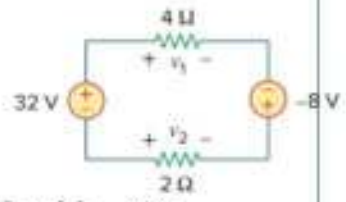
Example 2.5 For the circuit in Fig. 2.21(a), find voltages v_1 and v_2



$-20+v_1-v_2=0 \rightarrow -20+2i+3i=0 \rightarrow i=4A \rightarrow V_1=8v \ , V_2= -12v$

Spark POLYTECHNIC

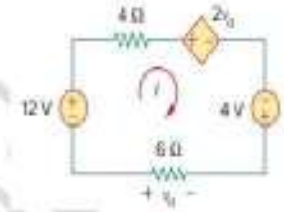
Practice Problem 2.5 Find v1 and v2 in the circuit of Fig. 2.22.



Sol: Applying KVL around the loop gives

$-32 + V_1 - - 8 - V_1 = 0 \rightarrow -32 + 4i + 8 + 2i = 0 \rightarrow i = 4A$, $V_1 = 16v$, $V_2 = -8v$

Example 2.6 Determine v_o and i in the circuit shown

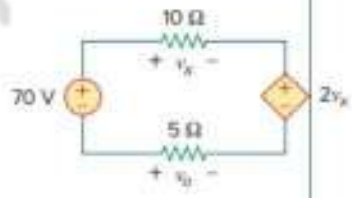


Sol : Applying KVL around the Loop

$-12+4i+2V_o-4+6i=0 \rightarrow -12+4i+-12i-4+6i=0 \rightarrow i = -8A$

$PSC = -6i \rightarrow V_o = -6i = 48v$

Practice Problem 2.6 Find v_x and v_o in the circuit

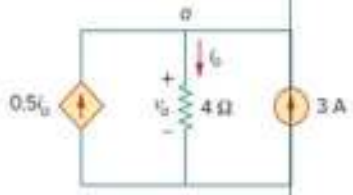


Sol : Applying KVL around the loop

$-70 + V_X + 2V_X - V_o = 0 \rightarrow -70 + 10i + 20i + 5i = 0 \rightarrow i = 2A$

$PSC\ V_o = -5i$,,,,, $V_X = 10i \rightarrow V_o = -10V$, $V_X = 20V$

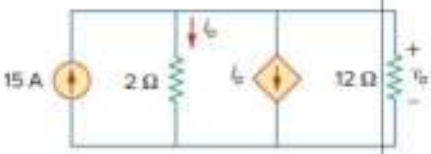
Example 2.7 Find current i_o and voltage v_o in the circuit



Sol : Applying KCL to node a

$3 + 0.5i_o = i_o \rightarrow i_o = 6A$ ohm's law $V_o = i_o \times 4 = 24v$

Practice Problem 2.7 Find v_o and i_o in the circuit



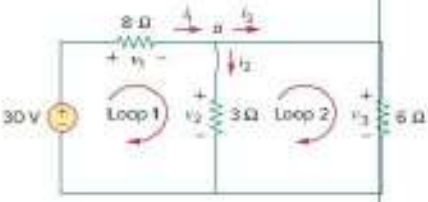
Sol : Applying KCL At node A

$-15 + i_o + i_o + \frac{v}{12} = 0 \rightarrow 2i_o + \frac{v_o}{12} = 15$

شرح جهد مقاومة 12اوموالمقاومة 2 اوم على التوازي وبالتالي لهم نفس الجهد

$V_o = 2i_o \rightarrow 2i_o + \frac{2i_o}{12} = 15$ $i_o = 6.923A$,,, $V_o = 13.846v$

Example 2.8 Find currents and voltages in the circuit shown



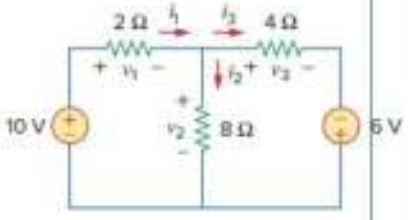
Sol : KVL loop 1 $-30 + V_1 + V_2 = 0$ —1

KVL loop 2 $-V_2 + V_3 = 0$ ——2 At node a $i_1 - i_2 - i_3 = 0$ –3

$V_1 = 8i_1$, $v_2 = 3i_2$, $v_3 = 6i_3$ $i_1 = \frac{30 - 3i_2}{8}$ —4 ,,, $i_3 = \frac{i_2}{2}$ ——5

Spark POLYTECHNIC

I_2= 2A , I_1=3A , I_3=1A , V_1=24v , V_2=6v , V_3=6v



Practice Problem 2.8 Find the currents and voltages in the circuit shown in Fig. 2.28.

Sol : KVL loop $-10+2i_1+8i_2=0$ ——1

KVL loop $-6-8i_2+4i_3=0$ ——2

KVL loop $i_1-i_3-i_2=0$ ——3

بالتعويض معادلة 3 في المعادلة 1

$-10+10i_2+2i_3=0$ ——4

I_2= 0.5A , I_3=2.5A , I_1= 3A , $V_1=2i_1$=6v , $V_2=8i_2$=4v , $V_3=4i_3$=10v

---

Example 2.9 Find Req for the circuit shown

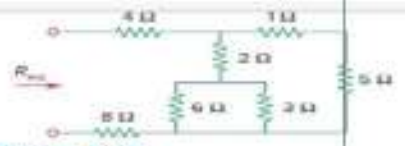
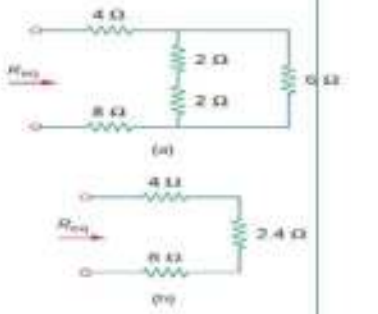


Figure 2.34
For Example 2.9.

Sol :

عندك المقاومة 3, 6 على التوازي

عندك المقاومة 1 , 5 على التوالي

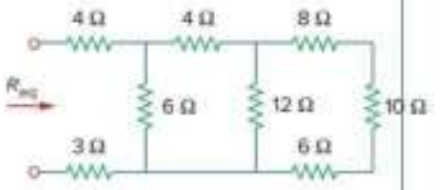


$R1=\frac{6\times3}{6+3}=2\Omega$, $R2=1+5=6\Omega$

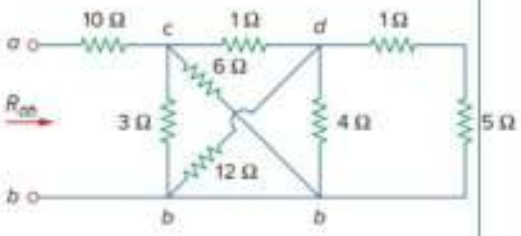
R1+2=2+2=4Ω ,, R3= $\frac{4\times6}{4+6}=2.4\Omega$

4+2.4+8=14.4Ω

Practice problem 2,9 By combining the resistors in Fig. 2.36, find Req answer: 11 Ω



Example 2.10 Calculate the equivalent resistance Rab in the circuit in Fig. 2.37



Lalit Abdulsalam Awad

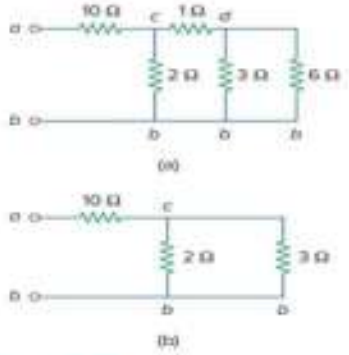


Figure 2.38
Equivalent circuits for Example 2.10.

Solution:
The 3-Ω and 6-Ω resistors are in parallel because they are connected to the same two nodes c and b. Their combined resistance is

$$3\ \Omega \parallel 6\ \Omega = \frac{3 \times 6}{3 + 6} = 2\ \Omega \qquad (2.10.1)$$

Similarly, the 12-Ω and 4-Ω resistors are in parallel since the y are connected to the same two nodes d and b. Hence

$$12\ \Omega \parallel 4\ \Omega = \frac{12 \times 4}{12 + 4} = 3\ \Omega \qquad (2.10.2)$$

Also the 1-Ω and 5-Ω resistors are in series; hence, their equivalent re - sistance is

$$1\ \Omega + 5\ \Omega = 6\ \Omega \qquad (2.10.3)$$

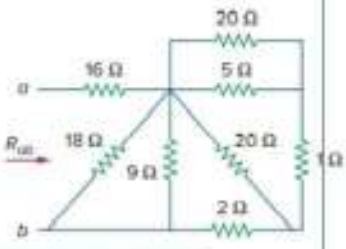
With these three combinations, we can replace the circuit in Fig. 2.37 with that in Fig. 2.38(a). In Fig. 2.38(a), 3-Ω in parallel with 6-Ω gives 2-Ω, as calculated in Eq. (2.10.1). This 2-Ω equivalent resistance is now in series with the 1-Ω resistance to give a combined resistance of $1\ \Omega + 2\ \Omega = 3\ \Omega$. Thus, we replace the circuit in Fig. 2.38(a) with that in Fig. 2.38(b). In Fig. 2.38(b), we combine the 2-Ω and 3-Ω resistors in parallel to get

$$2\ \Omega \parallel 3\ \Omega = \frac{2 \times 3}{2 + 3} = 1.2\ \Omega$$

This 1.2-Ω resistor is in series with the 10-Ω resistor, so that

$$R_{ab} = 10 + 1.2 = 11.2\ \Omega$$

Practice problem 2.10 Find Rab for the circuit



Answer: 19 Ω.

Example 2.14

Convert the Δ network in Fig. 2.50(a) to an equivalent Y network.

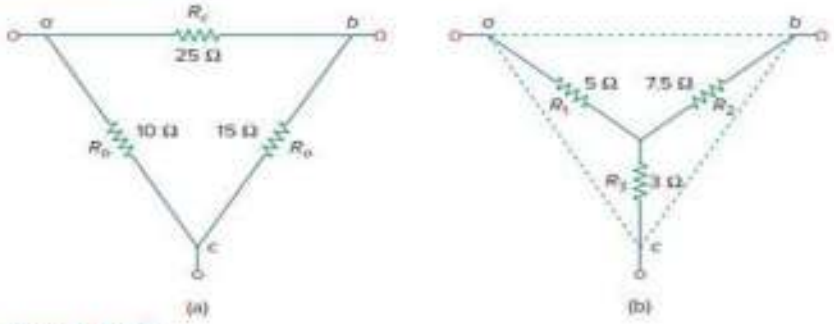


Figure 2.50
For Example 2.14: (a) original Δ network, (b) Y equivalent network.

Solution:
Using Eqs. (2.49) to (2.51), we obtain

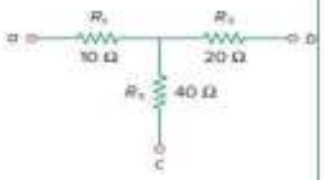
$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{10 \times 25}{15 + 10 + 25} = \frac{250}{50} = 5\ \Omega$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} = \frac{25 \times 15}{50} = 7.5\ \Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{15 \times 10}{50} = 3\ \Omega$$

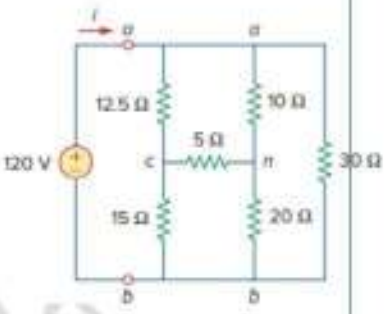
The equivalent Y network is shown in Fig. 2.50(b).

practice problem 2.14 Transform the wye network in Fig. 2.51 to a delta network.



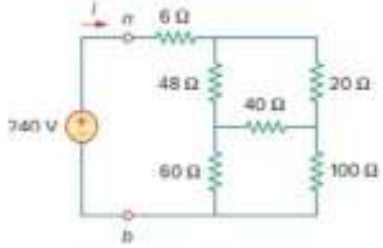
Answer: Ra = 140 Ω, Rb = 70 Ω, Rc = 35 Ω

Example 2.15 Obtain the equivalent resistance Rab for the circuit in Fig. 2.52 and use it to find current i.



R ab = 9.631 Ω ,,, i = 12.46 A

Practice problem : For the bridge network in Fig. 2.54, find Rab and i.



Answer: 60 Ω, 4 A

خلص شابتر 2 لتمكن من المادة اكثر وا اكثر حل اسئلة
البروبليمز صفحة 240 وعليك بالسنوات بالتوفيق للجميع

Laith Abdulsalam AlWa

Chapter 3 Mthods of Analysis (طرق التحليل)

❖ Nodel Analysis (التحليل العقدي)

في تحليل الدوائر الكهربائية , هناك طريقة التحليل العقدي او تحليل العقد للفولتية وهي طريقة لحساب قيمة الجهد بين العقد وفق شروط مرور التيارات داخل الدارة الكهربائية.

➢ Step to Determine Node Voltages :

- Select a node as the reference node. Assign voltage v_1, v_2, ...v_{n-1} to the remaining n-1 nodes. The voltages are referenced with respect to the reference node.

  حددعقدة كعقدة مرجعية ك V_1 V_2 ... V_{n-1} الى العقد المتبقية.

- Apply KCL to each of the n-1 nonreference nodes. Use Ohm's law to express the branch currents in terms of node voltages.

  نقوم بتطبيق KCL على كل من العقد واستخدام قانون اوم لتعبير عن تيارات الفرع في شروط الفولتية العقدة.

- Solve the resulting simultaneous equations to obtain the unknown node voltages.

  حل المعادلات الناتجة للحصول على فولتية العقدة الغير معروفة.

(بكل بساطة عندي عقدة مثل هيك رح نطبق عليها KCL مجموع التيارت الداخلة = الخارجة وبستخدام قانون اوم $i = \frac{\Delta v}{R}$, تبسط المعادلات وتجد الفولتية عند العقدة)

Common symbols for indicating a reference node, (a) common ground, (b) ground, (c) chassis

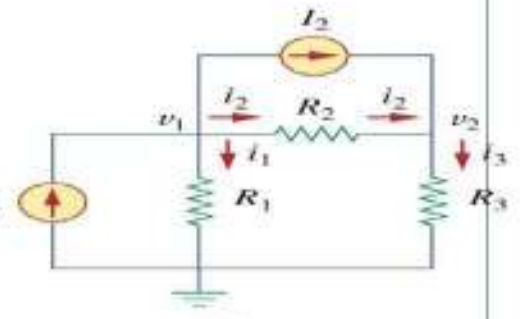


Typical circuit for nodal analysis

Applying KCL at $V_1 \rightarrow I_1=I_2+i_1+i_2$

Applying KCL at $V_2 \rightarrow I_2+i_2=i_3$

Ohm's law $i = \frac{v_{higher} - v_{lower}}{R}$



(b)

$$i_1 = \frac{v_1 - 0}{R_1} \text{ or } i_1 = G_1 v_1 \qquad \Rightarrow I_1 = I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2}$$

$$i_2 = \frac{v_1 - v_2}{R_2} \text{ or } i_2 = G_2(v_1 - v_2) \qquad I_2 + \frac{v_1 - v_2}{R_2} = \frac{v_2}{R_3}$$

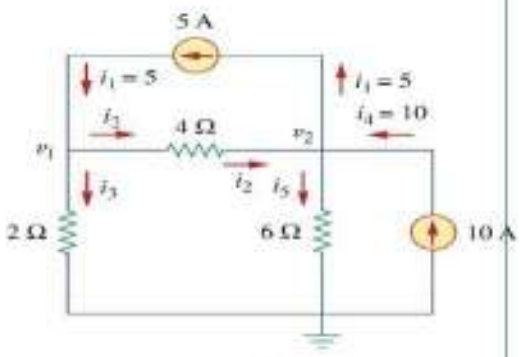
$$i_3 = \frac{v_2 - 0}{R_3} \text{ or } i_3 = G_3 v_2 \qquad \Rightarrow I_1 - I_2 = G_1 v_1 + G_2(v_1 - v_2)$$

$$I_2 = -G_2(v_1 - v_2) + G_3 v_2$$

$$\Rightarrow \begin{bmatrix} G_1 + G_2 & -G_2 \\ -G_2 & G_2 + G_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} I_1 - I_2 \\ I_2 \end{bmatrix}$$

Eam Al Qudah Wad

Example 3.1 Calculate the node voltage in the circuit shown in Fig. 3.3(a)



(b)

Sol :

نفرض اتجاه التيارات وتطبق KCL على العقد V_1 , V_2

At node V_1 $\quad i_1 = i_2 + i_3$

$$\Rightarrow 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$

At node V_2 $\quad i_2 + i_4 = i_1 + i_5$

$$\Rightarrow 5 = \frac{v_2 - v_1}{4} + \frac{v_2 - 0}{6}$$

At node $V_1 \rightarrow 20 = 3V_1 - V_2$ ——————(1)

At node $V_2 \rightarrow 30 = -1.5V_1 + 2.5\ V_2$ ———(2)

باستخدام الالة الحاسبة $V_1 = 13.33v$, $V_2 = 20v$

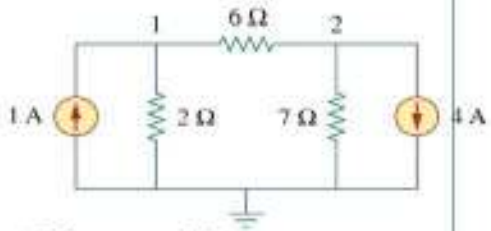
ممكن يضيف شغلة في الامتحان وهو انك تحسب i_1 , i_2 , i_3

$$i_1 = \frac{v_1 - v_2}{4} = \frac{20 - 13.33}{4} = 1.6675\ A$$

$$i_2 = \frac{v_1 - 0}{2} = \frac{13.33}{2} = 6.665A$$

$$i_3 = \frac{v_2 - 0}{6} = \frac{20}{6} = 3.333A$$

Practice Problem 3.1 Obtain the node voltages in the circuit



At node $V_1 \rightarrow 1 = i_2 + i_1$ $\quad 1 = \frac{v_1}{2} + \frac{v_1 - v_2}{6} \rightarrow 6 = 4V_1 - V_2$

At node $V_2 \rightarrow i_1 = i_3 + 4$ $\quad \frac{v_1 - v_2}{6} = \frac{v_2}{7} + 4 \rightarrow 28 = 1.16V_1 - 2.16V_2$ ——(2)

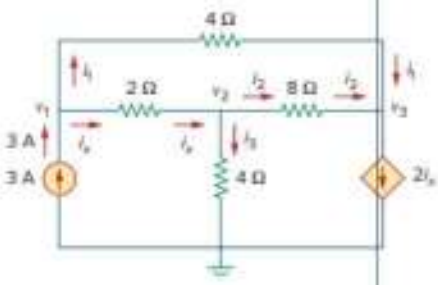
استخدام الالة الحاسبة معادلة رقم 1 وا معادلة رقم 2 نجد

$V_1 = -2.010$ v

$V_2 = -14.04$ v

Spark POLYTECHNIC

Example 3.2 Determine the voltages at the nodes



Sol : at node 1 $3=i_1+i_x \rightarrow 3 = \frac{v_1-v_3}{4} + \frac{v_1-v_2}{2}$

At node 2 $i_x=i_3+i_2 \rightarrow \frac{v_1-v_2}{2} = \frac{v_2}{4} + \frac{v_2-v_3}{8}$

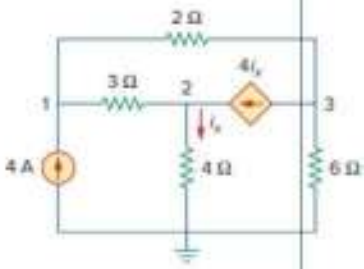
At node 3 $i_1+i_2=2i_x \rightarrow \frac{v_1-v_3}{4} + \frac{v_2-v_3}{8} = 2\frac{v_1-v_2}{2}$

At node 1 → $3v_1-2v_2-v_3=12$ $v_1=4.8v$

At node 2 → $4v_1-7v_2+v_3=0$ $v_2= 2.4v$

At node 3 → $-6v_1 + 9v_2 - 3v_3=0$ $v_3= -2.4v$

Practice Problem 3.2 Find the voltages at the three nonreference nodes in the circuit



Sol :

At node 1 $4 - \frac{v_1-v_2}{3} - \frac{v_1-v_2}{2} = 0$

At node 2 $\frac{v_2}{4} = \frac{v_1-v_2}{3} + \frac{4v_2}{4}$

At node 3 $\frac{v_3}{6} + \frac{4v_2}{4} - \frac{v_1-v_2}{2} = 0$

→ $5v_1-2v_2-3v_3=24$ ———(1)

→ $4v_1+5v_2=0$ ————(2)

→ $6v_1 - 12v_2 - 8v_3=0$ ———(3)

استخدم الالة الحاسبة او الماتركس لايجاد الجهود

$V_1=32v$

$V_2= -25.6v$

$V_3= 62.4v$

$I_x=v_2/4 = -6.4A$

Spark POLYTECHNIC

❖ Nodal Analysis with voltage source (مع مصادر جهد)

سابقاً لقد قمنا في عملية التحليل وايجاد الجهود عند العقد ولكن هناك كانت الدارت الكهربائية يوجد فقط بها مصادر تيار الان سوف نتعرف كيف نتعامل مع الدارة عندما يكون فيها مصادر جهد , لتحليل العقدي مع مصادر جهد يوجد حالتين .

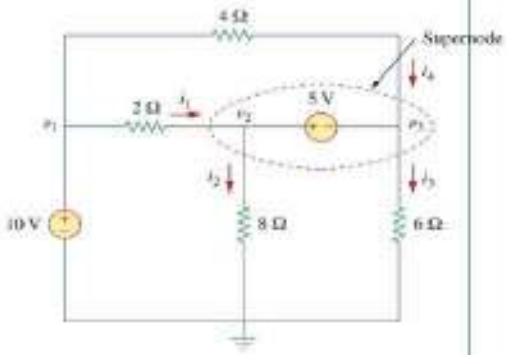
- Case 1: The voltage source is connected between a nonreference node and the reference node: The nonreference node voltage is equal to the magnitude of voltage source and the number of unknown nonreference nodes is reduced by one.

الحالة 1: يتم توصيل مصدر الجهد بين عقدة غير مرجعية والعقدة المرجعية : جهد العقدة المرجعية يساوي حجم مصدر الجهد ويتم تقليل عدد العقد غير المرجعية بمقدار واحد

- Case 2: The voltage source is connected between two nonreferenced nodes: a generalized node (supernode) is formed.

الحالة 2: يتم توصيل مصدر الجهد بين عقدتين غير مرجعيتين : يتم تشكيل عقدة معممة او ما يسمى ب (super node).

- A supernode is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.

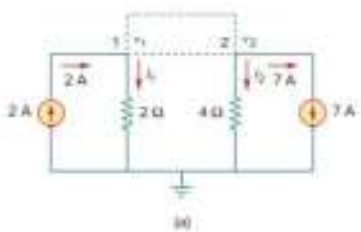


يتم تكوين *supernode* من خلال احاطة مصدر الجهد ان كان معتمد او غير معتمد متصل بين عقدتين غير مرجعيتين واي عناصر متصلة بالتوازي معها .

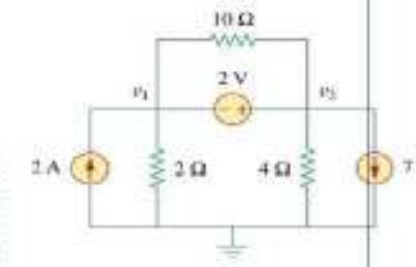
- The required two equations for regulating the two nonreference node voltages are obtained by the KCL of the supernode and the relationship of node voltages due to the voltage source.

يتم الحصول على معادلتين المطلوبتين لتنظيم جهدي العقدة الغير مرجعية بواسطة *KCL* لل *Supernode* وعلاقة جهد العقدة بسبب مصدر الجهد .

Example 3.3 For the circuit shown in Fig. 3.9, find the node voltages.







Lait Alhassan Awad

Example 3.3

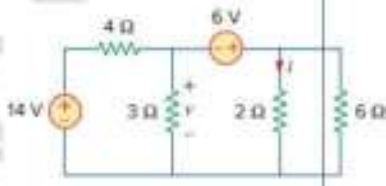
في المثال هذا يوجد عنا super node (اي عنصر متصل مع مصدر جهد على توازي
مثال هذا المثال نقوم باستبدالها بشورت سيركت يعني بنحذف المقاومة معها)

$2-7-i1-i2=0$

$2-7-\frac{v_1}{2}-\frac{v_2}{4}=0$ → v_2= -20 -2v_1 ——(1) v_1= -7.333v

$v_1-v_2=-2$ → v_2=v_1+2 ————(2) v_2 = -5.333v

Practice Problem 3.3 Find v and i in the circuit



Sol : V_0=14 حسب الحالة 1

Apply KCL at Node 1,2 Super node

$I_1=I_2+i+I_3 \rightarrow \frac{14-V_1}{4}=\frac{V_1}{3}+\frac{V_2}{6}+\frac{V_2}{2}$ →

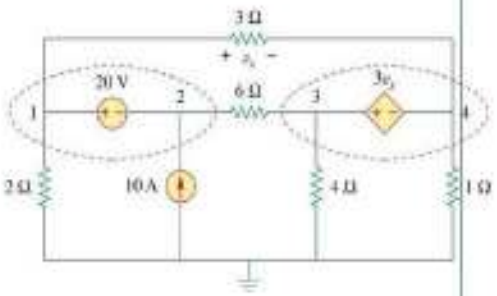
0.583V_1+0.666V_2=3.5 ——(1)

Apply KVL at super node $V_2-V_1=6$ ——(2)

From equation 1 , 2 V_1= - 0.4 ,,, V_2=5.6V

$i=\frac{v_2-0}{2}=\frac{5.2}{2}=2.8A$ V = V_1 = -0.4v

Example 3.4 Find the node voltages in the circuit



At super node 1 to 2

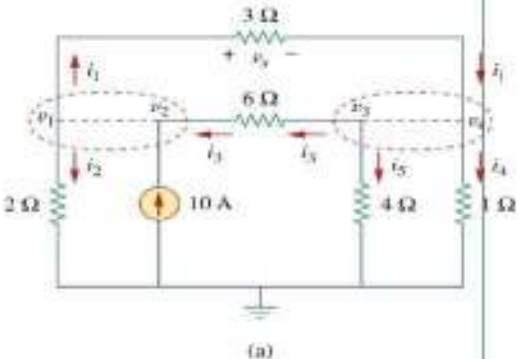
$$\frac{v_3-v_2}{6}+10=\frac{v_1-v_4}{3}+\frac{v_1}{2}$$

$$v_1-v_2=20$$

At super node 3 to 4

$$\frac{v_1-v_4}{3}=\frac{v_3-v_2}{6}+\frac{v_4}{1}+\frac{v_3}{4}$$

$$v_3-v_4=3(v_1-v_4)$$



(a)

Laith Abdulsalam Awni

Example 3.4

$5V_1+V_2-V_3-2V_4=60$ ——(1)

$V_1-V_2=20$————(2)

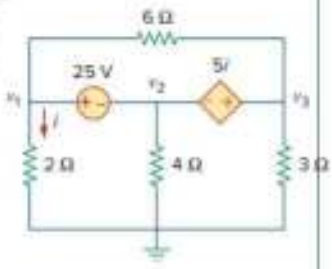
$4V_1+2V_2-5V_3-16V_4=0$ ———(3)

$3V_1-V_3-2V_4=0$ ————(4)

استخدام Matrix Equation

$V_1=26.67v$, $V_2= 6.67v$, $V_3= 173.33v$, $V_4=-46.67v$

Practice Problem 3.4 Find v1, v2, and v3 in the circuit using nodal analysis



Sol :

At super node 1 $V_1-V_2=25$ —(1)

At super node 2 $V_3-V_2=5i \rightarrow V_3-V_2=2.5V_1$ ———(2) ($i=V_1/2$)

Applying KCL to outer path of sopes super node

$$\frac{v_1}{2}+\frac{v_2}{4}+\frac{v_3}{3}+\frac{v_1-v_3}{6}+\frac{v_3-v_1}{6}=0$$

$V_1-V_2=25$ ——(1)

$-2.5V_1-V_2+V_3=0$ ———(2)

$0.5V_1+0.25V_2+0.33V_3=0$ ———(3)

$V_1=7.611V$,,, $V_2=-17.38V$,,, $V_3=1.64V$

❖ Mesh Analysis

- Mesh analysis: another procedure for analyzing circuits, applicable to planar circuit.

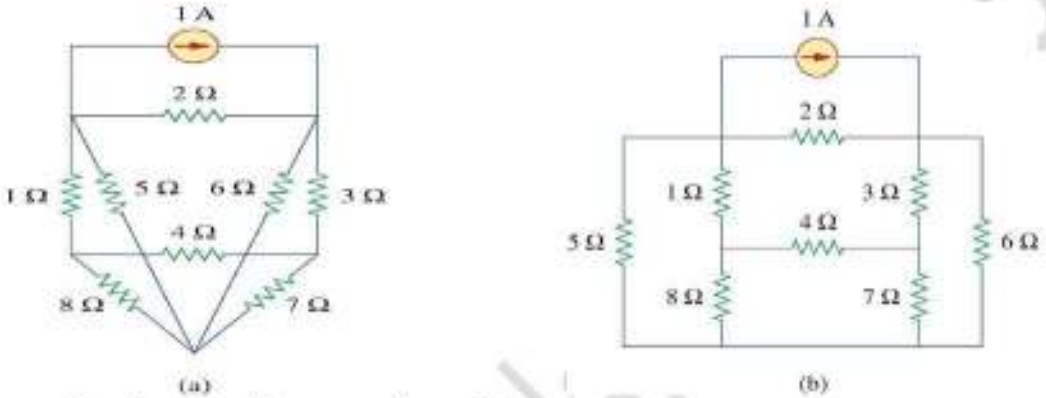
Mesh analysis : هو طريقة اخرى لتحليل الدارت الكهربائية , وينطبق mesh فقط على الدارت المستوية .

Latifa Abdulsalam Awad

- A Mesh is a loop which does not contain any other loops within it

Mesh : هي حلقة لا تحتوي على اي حلقات اخرى بداخلها (loop) كما تم شرحها سابقاً .

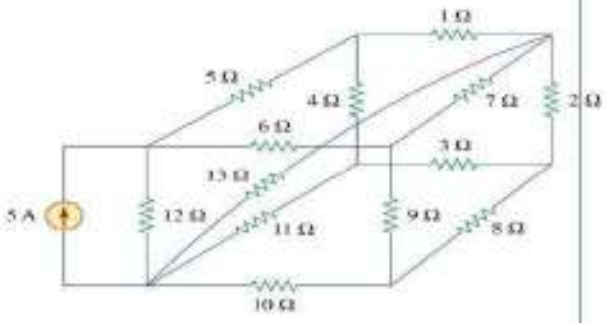
- a) A Planar circuit with crossing branches,
- (b) The same circuit redrawn with no crossing branches



- A nonplanar circuit

Planar(مستوية) يقال عن الدارة مستوية اذا كان من الممكن رسمها على سطح مستوآ بدون تقاطعات والدارتين مثال على هذا الكلام

اما الدارة الغير مستوية لايمكن رسمها على سطح بدون تقاطعات وهذا الدارت ما يزبط نطبق عليها mesh analysis ومثال على الدارت الغير مستوية الدارة على يمين الشرح .



Step to determine Mesh currents :

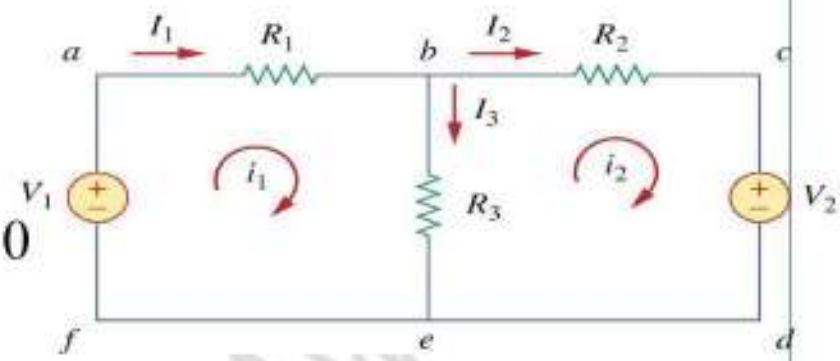
- Steps to Determine Mesh Currents:
- Assign mesh currents i_1, i_2, .., i_n to the n meshes.
- Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
- Solve the resulting n simultaneous equations to get the mesh currents.

Spark POLYTECHNIC

خطوات ايجاد او حساب تيارات mesh

1- تعين mesh في الدارة in.....,i2, i1 اي نفرض اتجاه التيارات في المسارات كلها اما مع عقارب الساعة او عكس عقارب الساعة
2- نأخذ KVL لكل مسار mesh ونستخدم قانون اوم لتعبير عن الفولتية من حيث التيارات
3- حل المعادلات الناتجة عن كل mesh للحصول على تيارات

A circuit with two meshes



Apply KVL to each mesh.
For mesh 1,

$$-V_1 + R_1 i_1 + R_3(i_1 - i_2) = 0$$

$$(R_1 + R_3)i_1 - R_3 i_2 = V_1$$

For mesh 2

$$R_2 i_2 + V_2 + R_3(i_2 - i_1) = 0$$

$$-R_3 i_1 + (R_2 + R_3)i_2 = -V_2$$

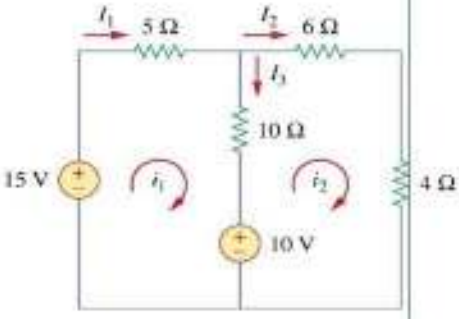
Solve for the mesh currents.

$$\begin{bmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix}$$

Use i for a mesh current and I for a branch current. It's evident

$$I_1 = i_1, \quad I_2 = i_2, \quad I_3 = i_1 - i_2$$

Example 3.5 Find the branch current I_1, I_2, and I_3 using mesh analysis.



Sol : For mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

$$3i_1 - 2i_2 = 1$$

For mesh 2,

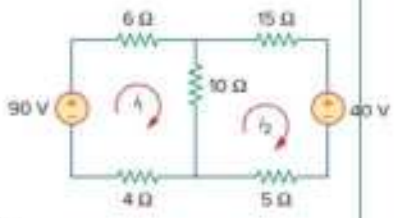
$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

$$i_1 = 2i_2 - 1$$

i_1=1A , i_2=1A

Spark POLYTECHNIC

Practice Problem 3.5 Calculate the mesh currents i1 and i2 of the circuit



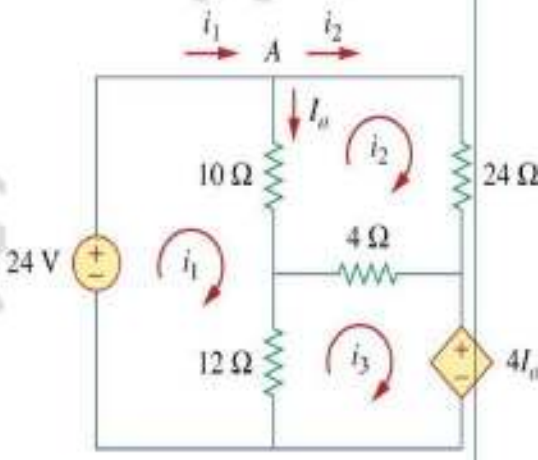
Sol : for mesh 1

$4i_1-90+6i_1+10(i_2-i_1)=0 \rightarrow 20i_1 - 10i_2 =90$ ——(1)

$40 +5i_2+10(i_2-i_1)+15i_2=0 \rightarrow -10i_1+30i_2=-40$ ——(2)

$i_1=4.6A$,,, $i_2=0.2$

Example 3.6 Use mesh analysis to find the current I_o in the circuit



Sol:

Apply KVL to each mesh. For mesh 1,

$$-24+10(i_1-i_2)+12(i_1-i_3)=0$$

$$11i_1-5i_2-6i_3=12$$

For mesh 2

$$24i_2+4(i_2-i_3)+10(i_2-i_1)=0$$

$$-5i_1+19i_2-2i_3=0$$

For mesh 3,

$$4I_o+12(i_3-i_1)+4(i_3-i_2)=0$$

At node A, $I_o=I_1-i_2$,

$$4(i_1-i_2)+12(i_3-i_1)+4(i_3-i_2)=0$$

$$-i_1-i_2+2i_3=0$$

In matrix from become

$$\begin{bmatrix}11 & -5 & -6\\ -5 & 19 & -2\\ -1 & -1 & 2\end{bmatrix}\begin{bmatrix}i_1\\ i_2\\ i_3\end{bmatrix}=\begin{bmatrix}12\\ 0\\ 0\end{bmatrix}$$

$i_1=2.25A$

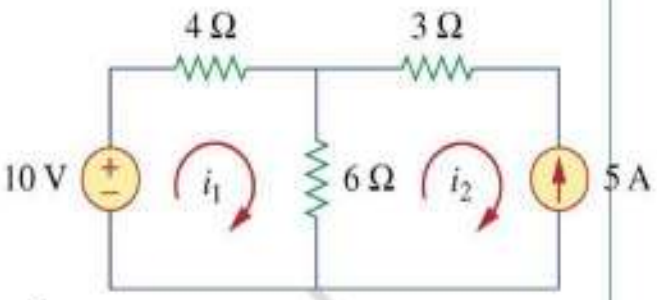
$i_2=0.75A$

$i_3= 1.5A$ $i_o=i_1-i_2=2.25-0.75=1.5A$

Spark POLYTECHNIC

❖ Mesh Analysis with current source (مع مصدر تيار)

A circuit with a current source. →



هناك حالتين خاصتين في Mesh analysis تيارات تحتوي على super mesh وتيارات تحتوي على مصادر معتمدة

- Case 1

  Current source exist only in one mesh

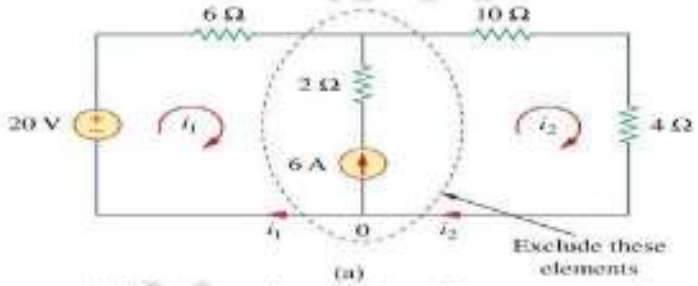
  One mesh variable is reduced

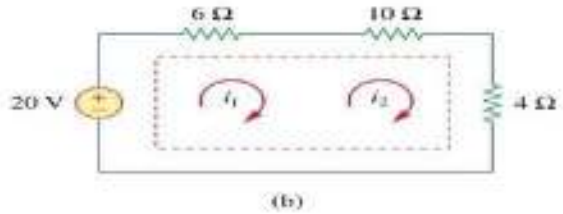
- Case 2
  Current source exists between two meshes, a super-mesh is obtained.

الحالة 1: عندما يوجد مصدر تيار في الشبكة واحدة فقط ضع في اعتبارك الدارة فوق , على سبيل المثال I2=-5A (لانه عكس اتجاه تيار المصدر) وهيك تم ايجاد تيار mesh2 ونكتب معادلة والشبكة الاخرى بالطريقة المعتادة

الحالة 2: يحدث super mesh عندما يتم احتواء مصدر تيار بين الشبكتين اساسيتين , يتم التعامل مع الدارة كما لو ان المصدر التيار غير موجود ويؤدي هذا الحكي الى معادلة واحدة تضمن تياراً

a supermesh results when two meshes have a (dependent , independent) current source in common





❖ Properties of a Supermesh

- The current is not completely ignored
  provides the constraint equation necessary to solve for the mesh current.
- A supermesh has no current of its own.
- Several current sources in adjacency form a bigger supermesh

Spark POLYTECHNIC

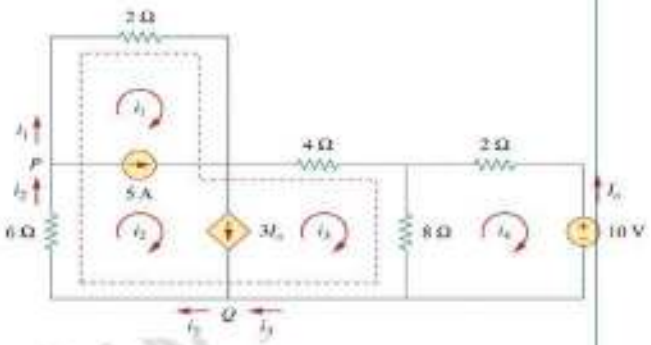
خصائص super mesh

1- لا يتم تجاهل التيار تماماً , يوفر معادلة القيد اللازم لحل شبكة التيار

2- Supermesh ليس لها تيار خاص فيها

3- العديد من مصادر التيار في الجوار تشكل شبكة عملاقة اكبر

Example 3.7 Use mesh analysis to find the current I_o in the circuit



Sol: two supermesh

$2i_1+4i_3+8(i_2-i_4)+6i_2=0$

$I_1+3i_2+6i_3-4i_4=0$ ——(1)

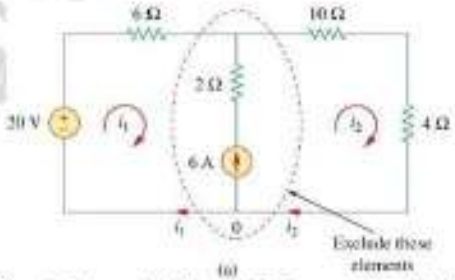
Apply KVL mesh 4 $5i_4-4i_3=-5$ ——(2)

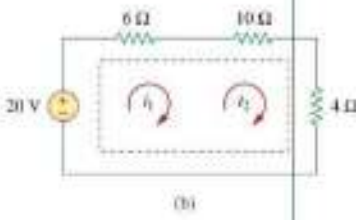
Apply KCL node P → $i_2=i_1+5$ ——(3)

Apply KCL node Q → $i_2=i_3+3I_o$ → $i_2=i_3-3i_4$ ——(4) but $i_o=-i_4$

i_1= -7.5A ,, i_2=-2.5A ,, i_3=3.93A ,, i_4=2.143A

Example 3.7 Use mesh analysis to find the current in the circuit





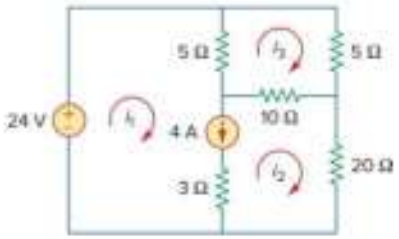
Sol :

$-20 + 6i_1 +10i_2 + 4i_2 = 0$ → $6i_1+14i_2=20$ ——(1)

We apply KCL to A node in the branch where the two meshes intersect $i_2=i_1+6$ ——(2)

Solving Eq (1) and (2) , we get i_1 = -3.2A ,, i_2 = 2.8 A

Practice Problem 3.7 Use mesh analysis to determine i_1, i_2, and i_3 in Fig. 3.25.



Answer: i_1 = 12.379 A, i_2 = 378.9 mA, i_3 = 3.284 A

Dr. Abdulsalam Awad

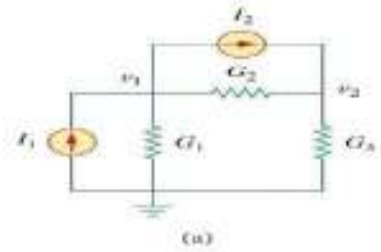
Spark POLYTECHNIC

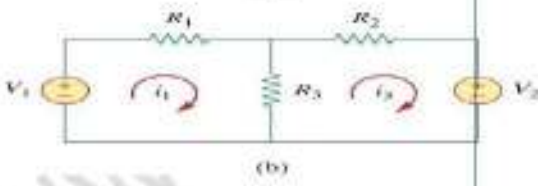
❖Nodal and Mesh Analysis by inspection

The analysis equations can be obtained by direct inspection

(a)For circuits with only resistors and independent current sources

(b) For planar circuits with only resistors and independent voltage sources





يمكن الحصول على معادلات التحليل عن طريق الفحص المباشر

(a) لدارات التي تحتوي فقط على مقاومات ومصادر تيار مستقلة
(b) لدارات المستوية ذات المقاومات ومصادر الجهد المستقلة فقط

In general, the node voltage equations in terms of the conductances is

or simply

$\mathbf{Gv} = \mathbf{i}$

$$\begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1N} \\ G_{21} & G_{22} & \cdots & G_{2N} \\ \vdots & \vdots & \vdots & \vdots \\ G_{N1} & G_{N2} & \cdots & G_{NN} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix}$$

where G : the conductance matrix,
v : the output vector, i : the input vector

In general, if the circuit has N meshes, the mesh-current equations as the resistances term is

or simply

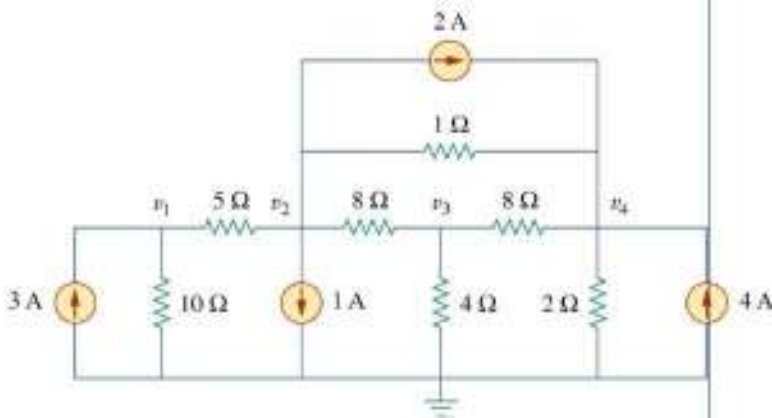
$\mathbf{Rv} = \mathbf{i}$

$$\begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1N} \\ R_{21} & R_{22} & \cdots & R_{2N} \\ \vdots & \vdots & \vdots & \vdots \\ R_{N1} & R_{N2} & \cdots & R_{NN} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

where R : the resistance matrix,
i : the output vector, v : the input vector

Spark POLYTECHNIC

Example 3.9 Write the node voltage matrix equations in Fig.3.27



Sol :

The circuit has 4 nonreference nodes, so

$$G_{11}=\frac{1}{5}+\frac{1}{10}=0.3,\quad G_{22}=\frac{1}{5}+\frac{1}{8}+\frac{1}{1}=1.325$$

$$G_{33}=\frac{1}{8}+\frac{1}{8}+\frac{1}{4}=0.5,\quad G_{44}=\frac{1}{8}+\frac{1}{2}+\frac{1}{1}=1.625$$

The off-diagonal terms are

$$G_{12}=-\frac{1}{5}=-0.2,\quad G_{13}=G_{14}=0$$

$$G_{21}=-0.2,\ G_{23}=-\frac{1}{8}=-0.125,\ G_{24}=-\frac{1}{1}=-1$$

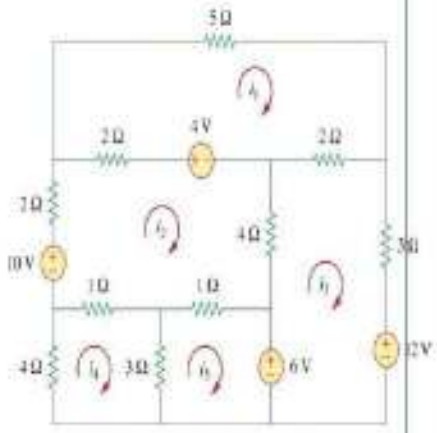
$$G_{31}=0,\ G_{32}=-0.125,\ G_{34}=-0.125$$

$$G_{41}=0,\ G_{42}=-1,\ G_{43}=-0.125$$

The node-voltage equations are

$$\begin{bmatrix} 0.3 & -0.2 & 0 & 0 \\ -0.2 & 1.325 & -0.125 & -1 \\ 0 & -0.125 & 0.5 & -0.125 \\ 0 & -1 & -0.125 & 1.625 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}=\begin{bmatrix} 3 \\ -3 \\ 0 \\ 6 \end{bmatrix}$$

Example 3.9 Write the mesh current equations in Fig.3.27



Latif Abdulsalam Awwad

The input voltage vector v in volts

$$v_1 = 4,\ v_2 = 10 - 4 = 6,$$
$$v_3 = -12 + 6 = -6,\ v_4 = 0,\ v_5 = -6$$

The mesh-current equations are

$$\begin{bmatrix} 9 & -2 & -2 & 0 & 0 \\ -2 & 10 & -4 & -1 & -1 \\ -2 & -4 & 9 & 0 & 0 \\ 0 & -1 & 0 & 8 & -3 \\ 0 & -1 & 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -6 \\ 0 \\ -6 \end{bmatrix}$$

❖ Nodal Versus Mesh Analysis

- Both nodal and mesh analyses provide a systematic way of analyzing a complex network.
- The choice of the better method dictated by two factors.

1-First factor : nature of the particular network. The key is to select the method that results in the smaller number of equations.

2-Second factor : information required.

توفر كل من التحليلات العقدية والشبكات طريقة منهجية لتحليل شبكة معقدة.

اختيار الطريقة الافضل يميله عاملين

العامل الاول : طبيعة الشبكة المعينة , وهو المفتاح لتحديد الطريقة التي ينتج عنها عدد اقل من المعادلات

العامل الثاني : المعلومة المطلوبة , مثلاً اذا طلب node voltage رح استخدم nodal
اما اذا طلب تيار mesh رح استخدم Mesh

خلص الشابتر ..... لتمكن من المادة اكثر واكثر حل اسئلة proplem اخر الشابتر صفحة 112 وا عليك بالسنوات (هذا الشابتر كثير مهم رح يتكرر في سيركت 2) .

Laith Abdulsalam Awwad

Spark POLYTECHNIC

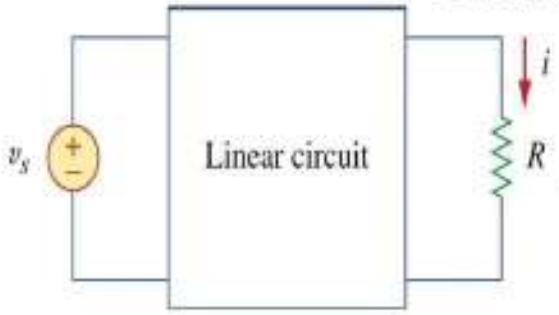
❖ Chapter 4 circuit theorem (نظريات في الدارات الكهربائية)

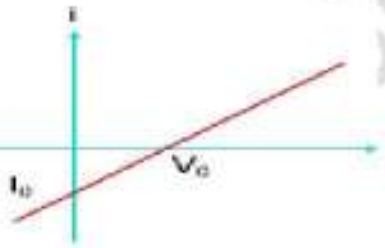
الهدف من هذه النظريات هي تبسيط او تحليل الدارة من دارة كبيرة ومعقدة الى دارات بسيطة من خلال هذه النظريات .

❖ Linearity Property

A linear circuit is one whose output is linearly related (or directly proportional) to its input

الدارة الخطية هي الدارة التي يرتبط ناتجها خطياً (اي يتناسب طردياً او عكسي) مع مدخلاتها . مثال



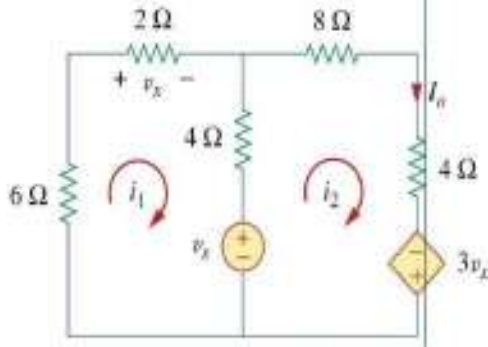


Linear circuit consist of
- linear elements
- linear dependent sources
- independent sources

$v_s = 10\text{V} \rightarrow i = 2\text{A}$
$v_s = 1\text{V} \rightarrow i = 0.2\text{A}$
$v_s = 5\text{mV} \leftarrow i = 1\text{mA}$

$$p = i^2 R = \frac{v^2}{R} : nonlinear$$

Example 4.1 For the circuit in fig 4.2 find I_o when v_s=12V and v_s=24V.



Sol : KVL

$12i_1 - 4i_2 + v_s = 0$ ——(1)

$-4i_1 + 16i_2 - 3v_x - v_s = 0$,,, $v_x = 2i_1$

$-10i_1 + 16i_2 - v_s = 0$ ——(2)

$2i_1 + 12i_2 = 0 \rightarrow i_1 = -6i_2$ $\quad -76i_2 = v_s = 0 \Rightarrow i_2 = \frac{v_s}{76}$

When $v_s = 12\text{V}$ when $v_s = 24\text{V}$ $I_o = i_2 = \frac{24}{76}\text{A}$

$I_o = i_2 = \frac{12}{76}\text{A}$

Spark POLYTECHNIC

❖ Superposition

The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone

ينص superposition على ان الجهد او التيار عبر عنصر في دارة خطية هو المجموع الجبري للجهد او التيارت عبر هذا العنصر بسبب كل مصدر مستقل يعمل بمفرده .

- Turn off, killed, inactive source:
  1- independent voltage source: 0 V (short circuit)
  2- independent current source: 0 A (open circuit)
- Dependent sources are left intact.

نقوم بايقاف او حذف مصدر غير معتمد لكي يصبح غير نشط

1- مصدر جهد غير معتمد(مستقل) : يصير short circuit 0V
2- مصدر تيار غير معتمد (مستقل) : يصير 0A open circuit

وتترك المصادر المعتمدة (غير مستقلة) سليمة ما بنحذفها

Steps to apply superposition principle:

- Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using nodal or mesh analysis.
- Repeat step 1 for each of the other independent sources.
- Find the total contribution by adding algebraically all the contributions due to the independent sources

1- نقوم بأيقاف تشغيل جميع المصادر المستقلة بأستثناء مصدر واحد وابحث عن الجهود او التيارات المطلوبة عند عنصر ما بسبب هذا العنصر النشط باستخدام قانون اوم او التحليل العقدي او MESH

Spark POLYTECHNIC

2- نكرر الخطوة رقم 1 لكل من المصادر المستقلة الاخرى
3- اوجد اجمالي المساهمة باضافة جميع المساهمالت جبرياً بسبب المصار المستقلة

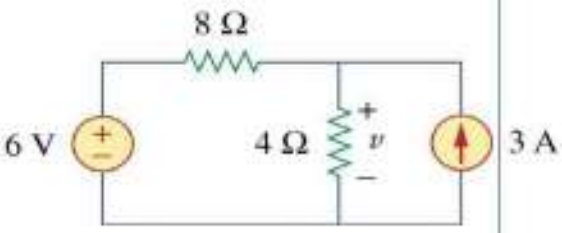
How to turn off independent sources

- Turn off voltages sources = short voltage sources; make it equal to zero voltage
- Turn off current sources = open current sources; make it equal to zero current
  - Superposition involves more work but simpler circuits.
  - Superposition is not applicable to the effect on power.

يتضمن Superposition مزيداً من العمل ولكن دارت ابسط

لا ينطبق superposition على تأثير power.

Example 4.3 Use the superposition theorem to find in the circuit in Fig.4.6.

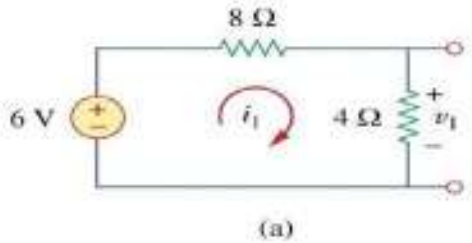


Sol : Since there are two sources,

Let $V = V_1 + V_2$

Voltage division to get

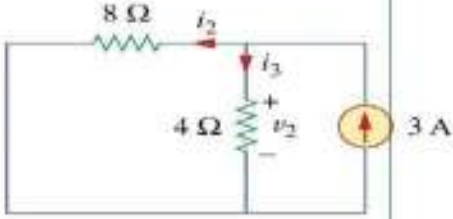
$$V_1 = \frac{4}{4+8}(6) = 2\text{V}$$



(a)

Current division, to get

$$i_3 = \frac{8}{4+8}(3) = 2\text{A}$$



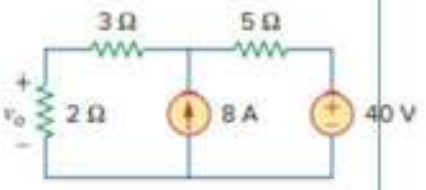
(b)

Hence $v_2 = 4i_3 = 8\text{V}$

$v = v_1 + v_2 = 2 + 8 = 10\text{V}$

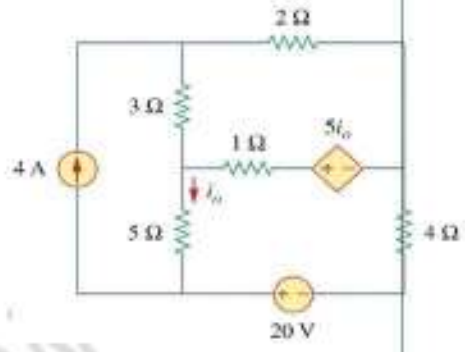
Laith Abdulsalam Awad

Practice Problem 4.3 Using the superposition theorem, find v_o in the circuit of Fig. 4.8.



Answer: 16 V

Example 4.4 Find I_o in the circuit in Fig.4.9 using superposition



Sol :

$I_o = I'_o + I''_o \rightarrow 3.0528 - 3.529 = -0.4706A$

For loop 3 $-5i_1 - i_2 + 10i_3 + 5i'_o = 0$

At node $i_3 = i_1 - i'_o = 4 - i'_o$

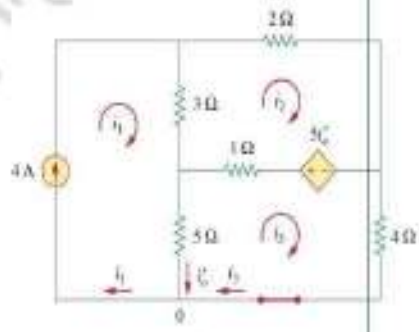
$3i_2 - 2i'_o = 8$, $i_2 + 5i'_o = 20 \rightarrow i'_o = 3.058$ A

For loop 4 $6i_4 - i_5 - 5i''_o = 0$

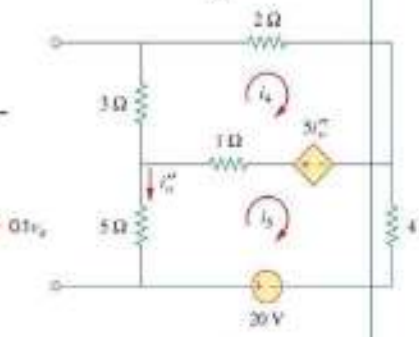
$i_5 = -i''_o$

$6i_4 - 4i''_o = 0 \rightarrow i''_o = -60/17$ A $= -3.529$A

$i_4 + 5i''_o = -20 \rightarrow$

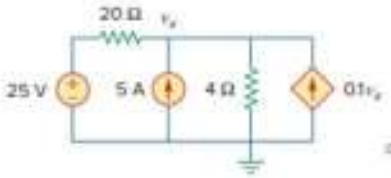


(a)



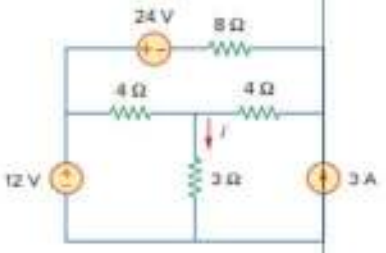
(b)

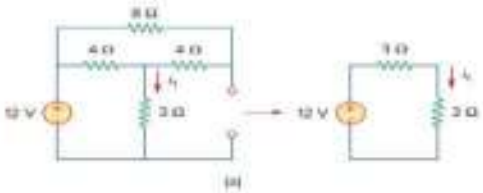
Practice Problem 4.4 Use superposition to find v_x in the circuit .



Answer: $v_x = 31.25$ V

Example 4.5 For the circuit in Fig. 4.12, use the superposition principle to find i.

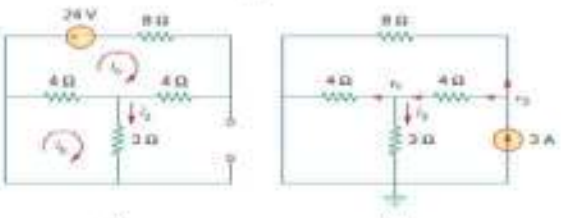




(a)

i=2A

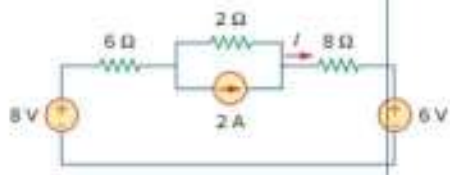
$i = i_1 + i_2 + i_3$



(b) (c)

Spark POLYTECHNIC

Practice Problem 4.5 Find I in the circuit of Fig. 4.14 using the superposition principle.

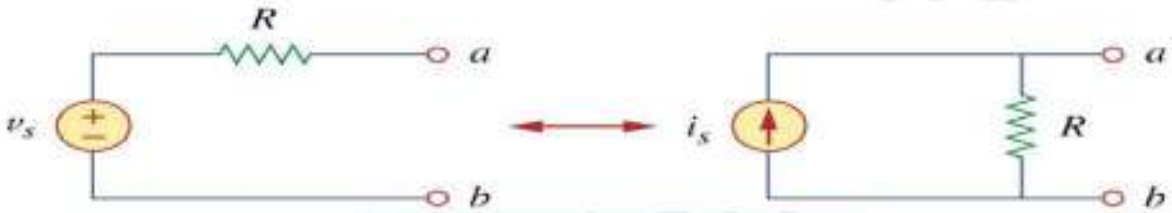


Answer: 375 mA

❖Source Transformation (تحويل المصادر)

A source transformation is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R, or vice versa

تحويل المصدر وهو هملية استبدال مصدر جهد موصول على التوالي مع مقاومة R بمصدر تيار موصول على التوازي مع مقاومة R , او العكس .



$$v_s = i_s R \quad \text{or} \quad i_s = \frac{v_s}{R}$$

Arrow of the current source positive terminal of voltage source

سهم مصدر التيار يكون باتجاه القطب الموجب لمصدر الجهد

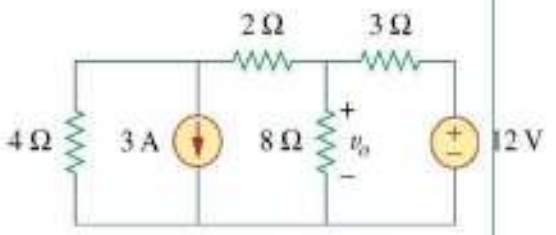
Impossible source Transformation

ideal voltage source (R = 0)

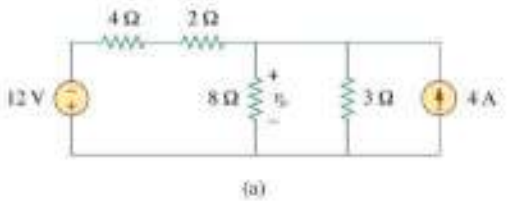
ideal current source (R=∞)

مصادر لا يمكن تحويلها مصدر جهد مربوط مع مقاومة قيمتها صفر
مصدر تيار مربوط مع مقاومة قيمتها ∞

Example 4.6 Use source transformation to find v_o in the circuit in Fig 4.17.



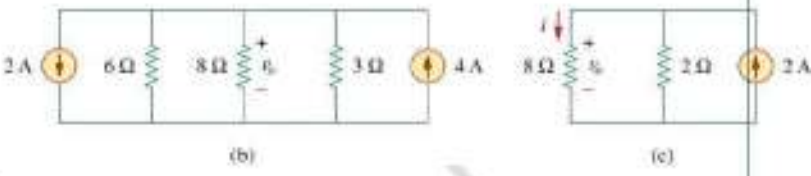
Example 4.6



Sol : $i_s = \frac{12}{6} = 2A$,,,

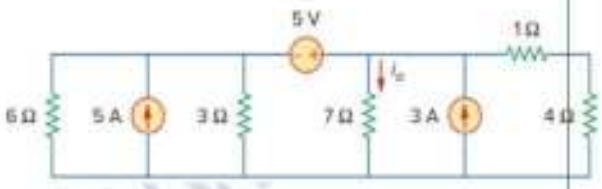
$$i = \frac{2}{2+8} \times 2 = 0.4\ A$$

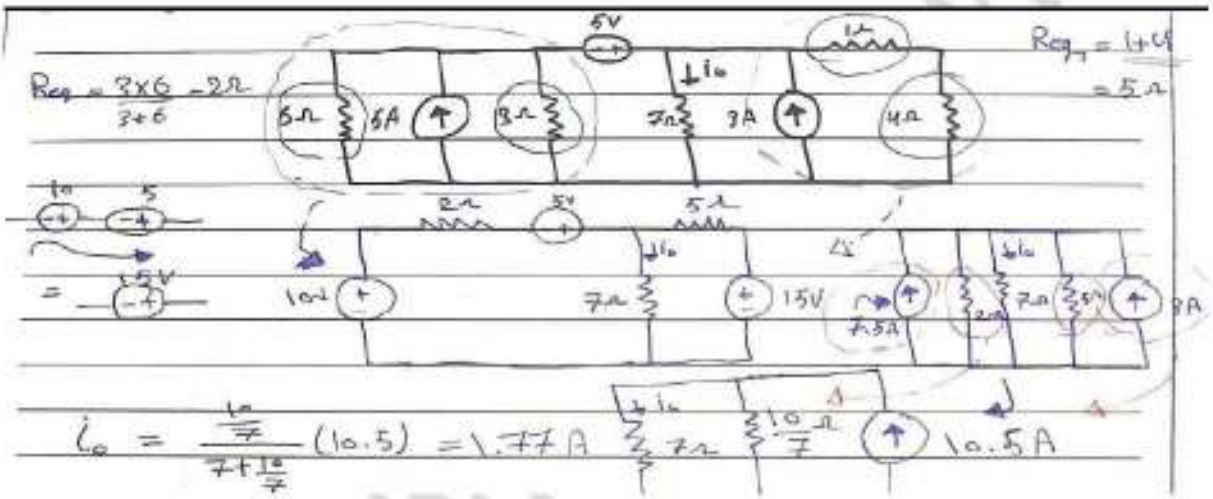
V_o=(0.4)(8)= 3.2v



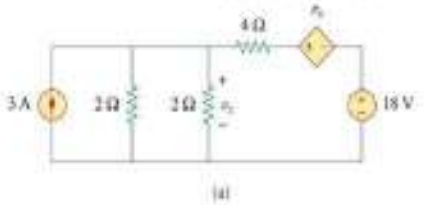
Practice problem 4.6

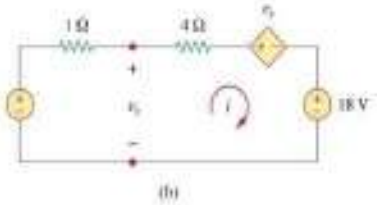
Sol :

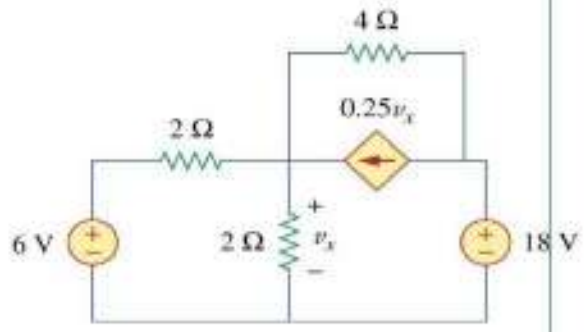




Example 4.7 Find v_x using source transformation sol :







Applying KVL around the loop in Fig 4.21(b) gives

$$-3+5i+v_x+18=0 \quad \textbf{(4.7.1)}$$

Appling KVL to the loop containing only the 3V voltage source, the 1 Ωresistor, and v_x yields

$$-3+1i+v_x=0\Rightarrow v_x=3-i \quad (4.7.2)$$

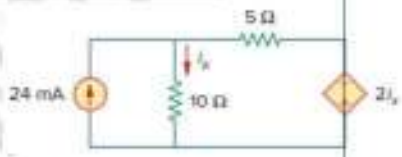
Substituting this into Eq.(4.7.1), we obtain

$$15+5i+3=0\Rightarrow i=-4.5\text{A}$$

Alternatively

$$-v_x+4i+v_x+18=0\Rightarrow i=-4.5\text{A} \quad \text{Thus} \quad v_x=3-i=7.5\text{V}$$

Practice Problem 4.7 Use source transformation to find i_x in the circuit

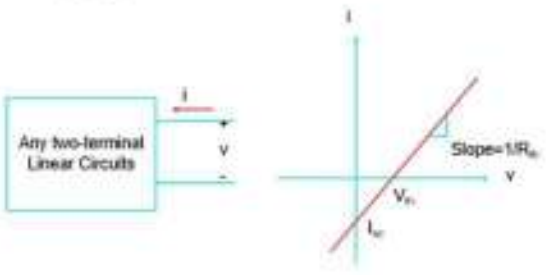


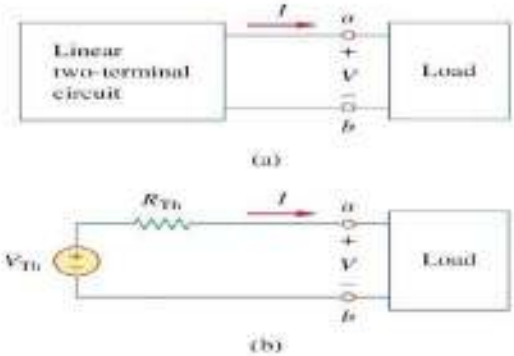
Answer: 7.059 mA.

❖Thevenin's Theorem

Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} where V_{Th} is the open circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent source are turn off.

تنص نظرية ثفينين على انه يمكن استبدال الدارة الخطية ثنائية الاطراف بدارة مكافئة تتكون من مصدر جهد Vth موصول على توالي Rth حيث Vth هو جهد الدارة المفتوحة وا Rth هي المدخلات او المقامة المكافئة للعناصر عندما يتم حذف او الغاء جميع المصادر المستقلة (غير المعتمدة) بحيث اذا كان مصدر تيار open circuit واذا كان مصدر الجهد short circuit





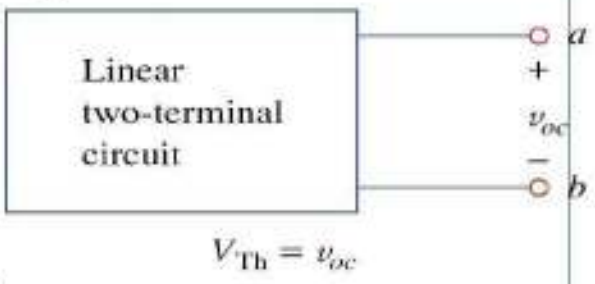
How to Find Thevenin's Voltage

Spark POLYTECHNIC

How to Find Thevenin's Voltage

Equivalent circuit: same voltage-current relation at the terminals.

$$V_{\text{Th}} = v_{oc} : \text{ open circuit voltage at } a-b$$



$V_{\text{Th}} = v_{oc}$

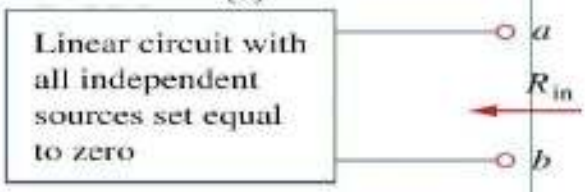
(a)

How to Find Thevenin's Resistance

$$R_{\text{Th}} = R_{\text{in}} :$$

$\text{input} - \text{resistance of the dead circuit at } a-b.$

- $a-b$ open circuited
- Turn off all independen t sources



$R_{\text{Th}} = R_{\text{in}}$

(b)

حذف جميع المصادر غير المعتمدة فقط

❖ CASE 1

If the network has no dependent sources:

- Turn off all independent source.
- RTH: can be obtained via simplification of either parallel or series connection seen from a-b

الحالة 2: اذا لم يكن في الدارة اي مصادر غير مستقلة , يعني كلها غير معتمدة نقوم بحذف (الغاء تنشيط) هذه المصادر المستقلة مصدر التيار اوبن سركت ومصدر الجهد شورت سيركت وبعد الغاء هذه العناصر نوجد Rth من خلال مكافئة التوالي والتوازي عبر (a – b) .

❖ CASE 2

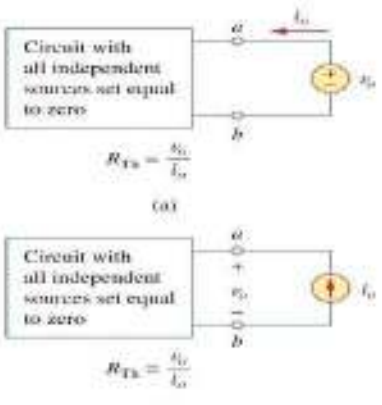
If the network has dependent sources

- Turn off all independent sources.
- Apply a voltage source v_o at a-b

$$R_{\text{Th}} = \frac{v_o}{i_o}$$

- Alternatively, apply a current source i_o at a-b

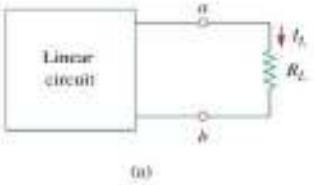
$$R_{Th} = \frac{v_o}{i_o}$$



The Thevenin's resistance may be negative, indicating that the circuit has ability providing power

قد تمون مقاومة thevenin سلبية , مما يشير الى ان الدارة لديها القدرة على توفير الطاقة .

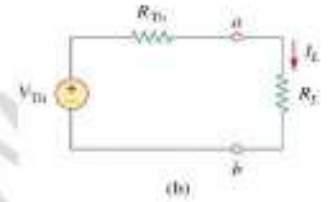
Simplified circuit



(a)

$$I_L = \frac{V_{\text{Th}}}{R_{\text{Th}} + R_L}$$

Voltage divider



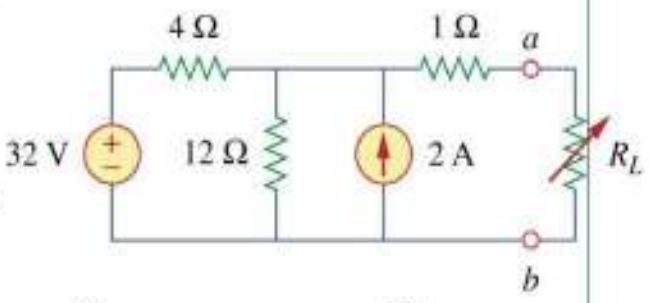
(b)

$$V_L = R_L I_L = \frac{R_L}{R_{\text{Th}} + R_L} V_{\text{Th}}$$

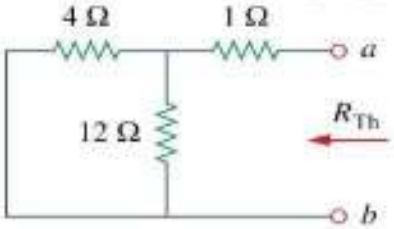
Example 4.8 Find the Thevenin's equivalent circuit of the circuit shown in Fig 4.27, to the left of the terminals a-b. Then find the current through RL = 6,16,and 36 Ω.

R_{Th} : 32V voltage source → short

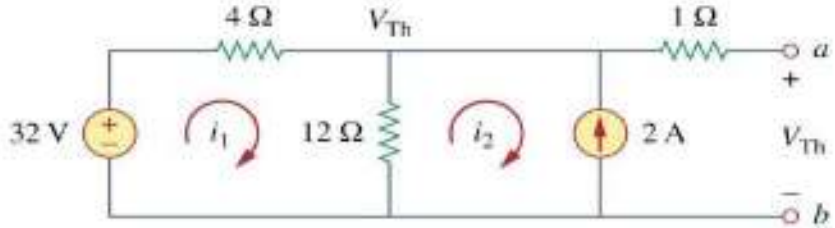
2A current source → open



$$R_{\text{Th}} = 4 \,\|\, 12 + 1 = \frac{4 \times 12}{16} + 1 = 4\Omega$$



(a)



(b)

V_{Th} :

Find Vth (1) Mesh analysis

$$-32 + 4i_1 + 12(i_1 - i_2) = 0 \,,\; i_2 = -2\text{A} \quad \therefore i_1 = 0.5\text{A}$$

$$V_{\text{Th}} = 12(i_1 - i_2) = 12(0.5 + 2.0) = 30\text{V}$$

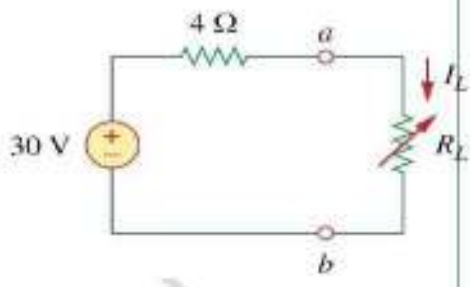
(2) Alternatively, Nodal Analysis

$$(32 - V_{\text{Th}})/4 + 2 = V_{\text{Th}}/12 \quad \therefore V_{\text{Th}} = 30\text{V}$$

Spark POLYTECHNIC

Example 4.8

(3) Alternatively, source transform



$$\frac{32 - V_{TH}}{4} + 2 = \frac{V_{TH}}{12}$$

$$96 - 3V_{TH} + 24 = V_{TH} \Rightarrow V_{TH} = 30\text{V}$$

$$\text{To get } i_L: \quad i_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

$$R_L = 6 \rightarrow I_L = 30/10 = 3\text{A}$$

$$R_L = 16 \rightarrow I_L = 30/20 = 1.5\text{A}$$

$$R_L = 36 \rightarrow I_L = 30/40 = 0.75\text{A}$$

Practice Problem 4.8 Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit of Fig. 4.30. Then find I.

Answer: $V_{Th} = 90$ V, $R_{Th} = 45\ \Omega$, $I = 1.5$ A.

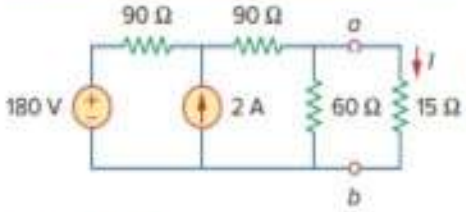
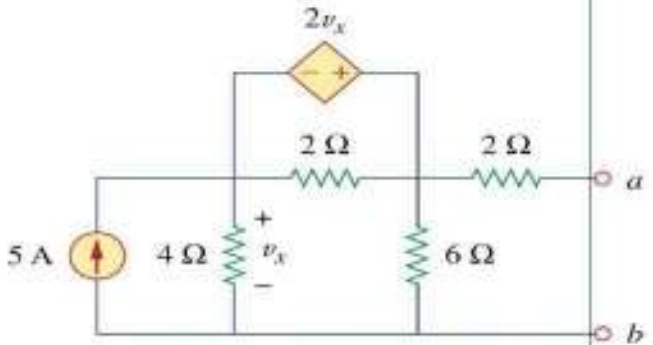


Figure 4.30
For Practice Prob. 4.8.

Example 4.9 Find the Thevenin's equivalent of the circuit in Fig. 4.31 at terminals a-b.



Sol

(independent + dependent source case)

$$\text{To find } R_{Th}: \text{ Fig(a)}$$

$$\text{independent source} \rightarrow 0$$

$$\text{dependent source} \rightarrow \text{intact}$$

$$v_o = 1\text{V}, \quad R_{Th} = \frac{v_o}{i_o} = \frac{1}{i_o}$$

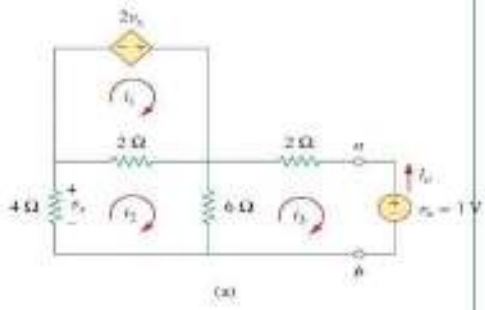
Spark POLYTECHNIC

For loop 1,

$$-2v_x + 2(i_1 - i_2) = 0 \text{ or } v_x = i_1 - i_2$$

$$\text{But } -4i = v_x = i_1 - i_2$$

$$\therefore i_1 = -3i_2$$



Loop 2 and 3 :

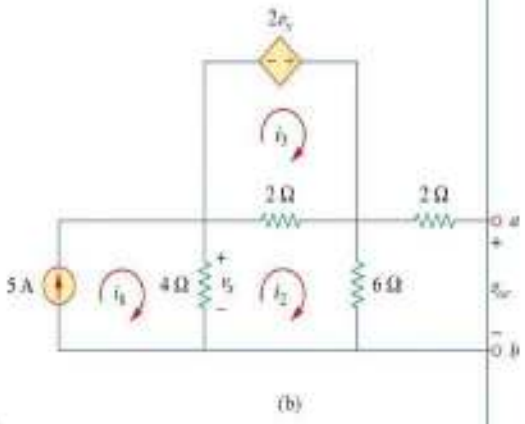
$$4i_2 + 2(i_2 - i_1) + 6(i_2 - i_3) = 0$$

$$6(i_3 - i_2) + 2i_3 + 1 = 0$$

Solving these equations gives

$$i_3 = -1/6\text{A}. \quad \text{But } i_o = -i_3 = \frac{1}{6}\text{A}$$

$$\therefore R_{\text{Th}} = \frac{1V}{i_o} = 6\Omega$$



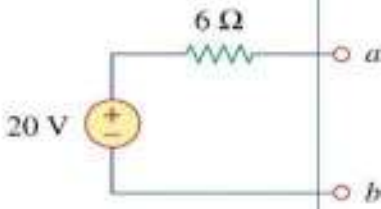
To get V_{Th} :Fig(b)Mesh analysis

$$i_1 = 5 \quad -2v_x + 2(i_3 - i_2) = 0 \Rightarrow v_x = i_3 - i_2$$

$$4(i_2 - i_1) + 2(i_2 - i_1) + 6i_2 = 0 \Rightarrow 12i_2 - 4i_1 - 2i_3 = 0$$

$$\text{But } 4(i_1 - i_2) = v_x \quad \therefore i_2 = 10/3.$$

$$V_{\text{Th}} = v_{oc} = 6i_2 = 20\text{V}$$



Practice Problem 4.9

Find the Thevenin equivalent circuit of the circuit in Fig. 4.34 to the left of the terminals.

Answer: $V_{\text{Th}} = 5.333$ V, $R_{\text{Th}} = 444.4$ mΩ.

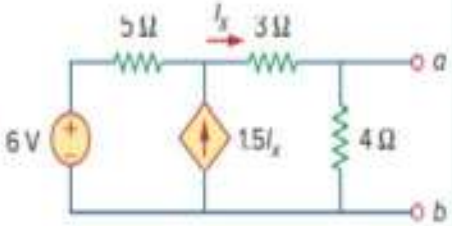
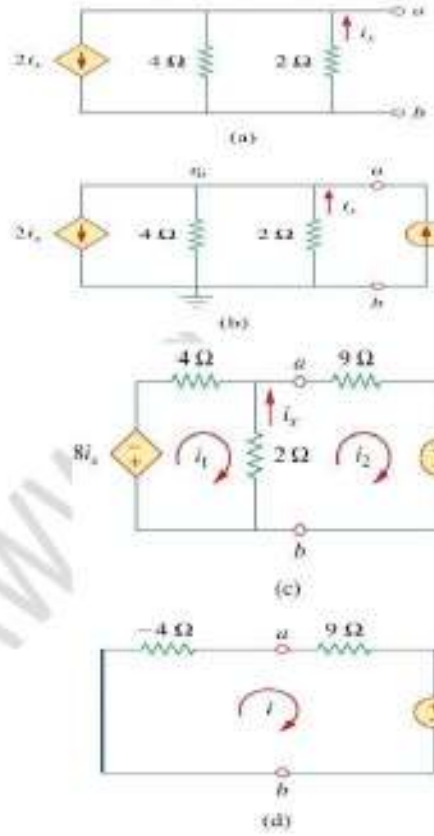


Figure 4.34
For Practice Prob. 4.9.

Example 4.10 Determine the Thevenin's equivalent circuit



Sol (dependent source only case)

$V_{Th} = 0 \quad R_{Th} = \frac{v_o}{i_o}$ ،

Nodal anaysis : $i_o + i_x = 2i_x + v_o/4$

But $i_x = \frac{0 - v_o}{2} = -\frac{v_o}{2}$

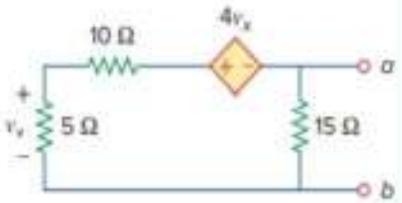
$i_o = i_x + \frac{v_o}{4} = -\frac{v_o}{2} + \frac{v_o}{4} = -\frac{v_o}{4}$ or $v_o = -4i_o$

Thus $R_{Th} = \frac{v_o}{i_o} = -4\Omega$: Supplying power

Practice Problem 4.10

Obtain the Thevenin equivalent of the circuit in Fig. 4.36.

Answer: $V_{Th} = 0$ V, $R_{Th} = -7.5\ \Omega$.



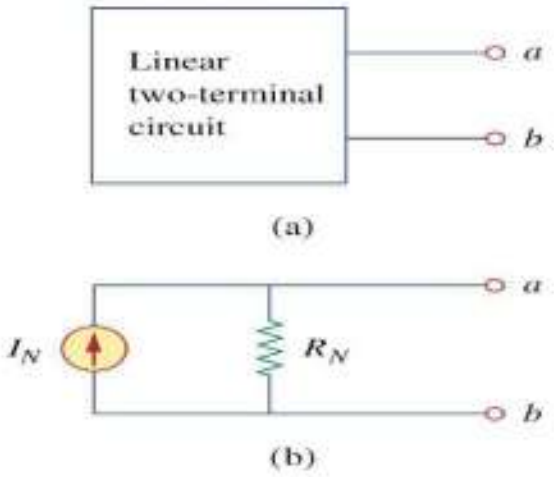
4.6 Norton's Theorem

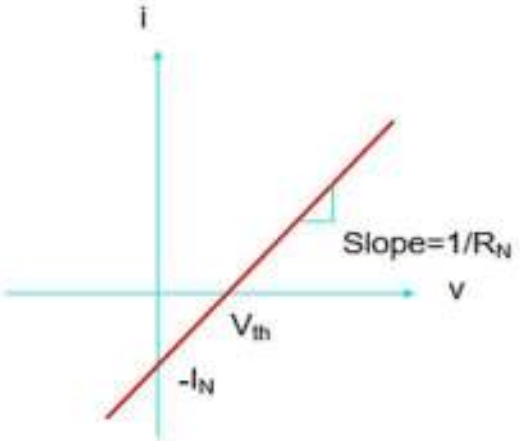
❖ Norton's Theorem

Norton's theorem states that a linear two-terminal circuit can be replaced by equivalent circuit consisting of a current source I_N in parallel with a resistor R_N where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent source are turn off.

تنص نظرية نورتون على انه يمكن استبدال الدارة الخطية الثنائية

الاطراف في دارة مكافئة تتكون من مصدر تيار على التوازي مع مقاومة R_N حيث يكون I_N هو تيار Short circuit وا R_N هو input او المقاومة المكافئة في الدارة عندما يتم ايقاف تشغيل المصدر المستقل غير المعتمد

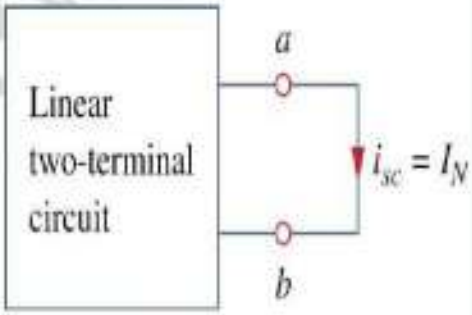




How to Find Norton Current

Thevenin and Norton resistances are equal:

$$R_N = R_{\text{Th}}$$



Short circuit current from a to b :

$$I_N = i_{sc} = \frac{V_{\text{Th}}}{R_{\text{Th}}}$$

Thevenin or Norton equivalent circuit :

- The open circuit voltage voc across terminals a and b
- The short circuit current isc at terminals a and b
- The equivalent or input resistance Rin at terminals a and b when all independent source are turn off.

الفرق بين دارة المكافئة ل Thevenin and Nortan

1- جهد الدارة المفتوحة voc بين العناصر a-b VTH = VO.C

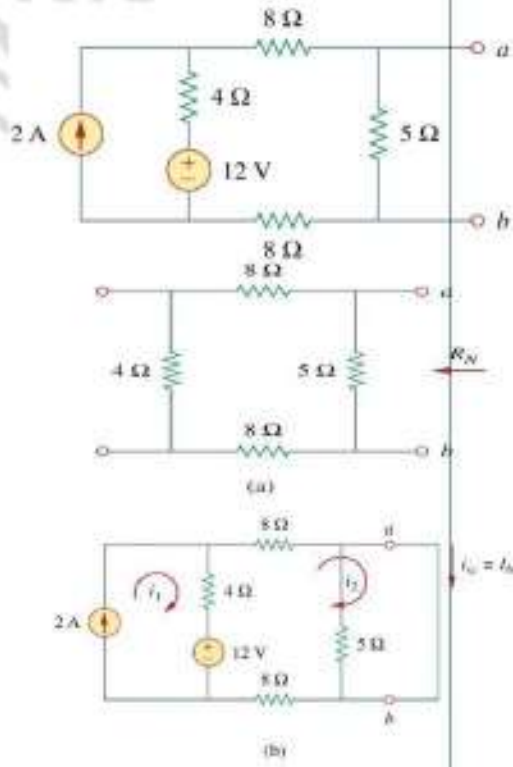
Spark POLYTECHNIC

2- تيار دارة القصر IS.C بين العناصر a - b isc=in

3- Rin المكافئة او المقاومة الداخلة بين العناصر a - b عند ايقاف تشغيل جميع المصادر المستقلة Rn=$R_{th}=\frac{vth}{in}$

- يمكن ان نقول نظرية نورتون ينطبق عليها كل ما جاء في نظرية thevenin
- يمكن ايجاد thevenin equivelent وتحويلها الى norton equivalent عن طريق source trasform
- ملاحظة سريعة عندما يكون سلك فارغ ومقاومة على توازي فاننا نقوم بالغاء المقاومة التي على توازي السبب لانه لايمر فيها تيار وسوف يختصر التيار الطريق ويمر عبر دارة القصر

Example 4.11 Find the Norton equivalent circuit of the circuit in Fig 4.39.



To find R_N Fig4.40(a):

$$R_N = 5 \,||\, (8+4+8)$$

$$= 5 \,||\, 20 = \frac{20\times 5}{25} = 4\Omega$$

To find i_N (Fig.4.40(b))

short – circuit terminals a and b.

Mesh : $i_1 = 2\text{A},\quad 20i_2 - 4i_1 - i_2 = 0$

$$i_2 = 1\text{A} = i_{sc} = I_N$$

Alternativ e method for I_N $\quad I_N = \frac{V_{Th}}{R_{Th}}$

V_{Th}: open – circuit voltage across terminals a and b (Fig4.40(c)):

Mesh analysis : $i_3 = 2A,\quad 25i_4 - 4i_3 - 12 = 0$

$$\therefore\ i_4 = 0.8\text{A} \quad \therefore v_{oc} = V_{Th} = 5i_4 = 4\text{V}$$

Hence, $\quad I_N = \frac{V_{Th}}{R_{Th}} = 4/4 = 1\text{A}$



Practice Problem 4.11

Find the Norton equivalent circuit for the circuit in Fig. 4.42, at terminals a-b.

Answer: $R_N = 90\ \Omega$, $I_N = 4.5$ A.

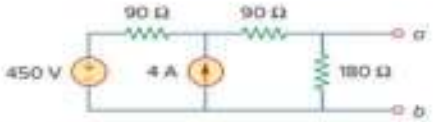
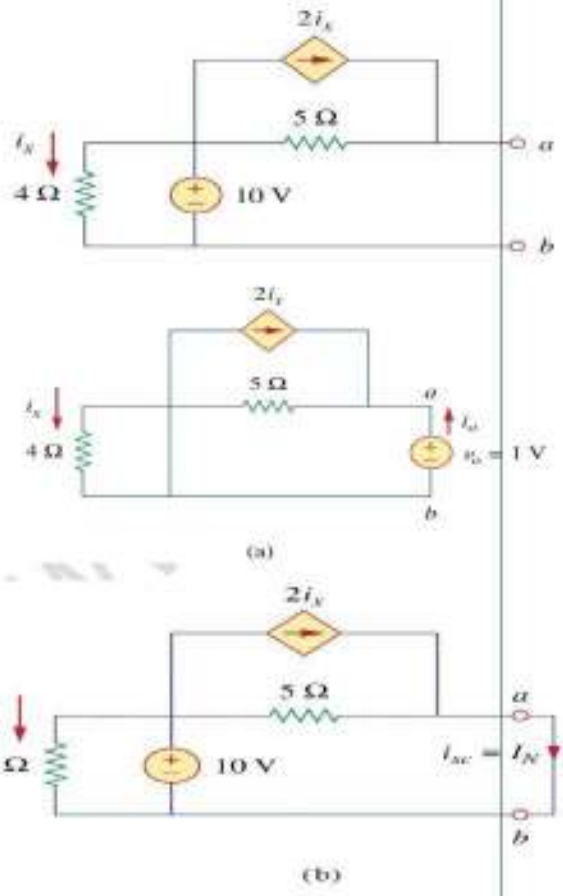


Figure 4.42
For Practice Prob. 4.11.

Latif Abdulsalam Ali

Example 4.12 Using Norton's theorem, find RN and IN of the circuit in Fig 4.43 at terminals a-b.



Sol

To find R_N *Fig.*4.44(*a*)

- 4Ω resistor shorted

$5\Omega \| v_o \| 2i_x$:Parallel

$$i_x = v_o/5 = 1/5 = 0.2$$

$$\therefore R_N = \frac{v_o}{i_o} = \frac{1}{0.2} = 5\Omega$$

To find I_N *Fig.*4.44(*b*)

- $4\Omega \| 10v \| 5\Omega \| 2i_x$:Parallel

$$i_x = \frac{10-0}{4} = 2.5\text{A},$$

$$i_{sc} = i_x + 2i_x = \frac{10}{5} + 2(2.5) = 7\text{ A} \therefore I_N = 7\text{A}$$

Practice Problem 4.12 Find the Norton equi valent circuit of the circuit in Fig. 4.45 at terminals *a-b*.

Answer: $R_N = 1\ \Omega$, $I_N = 10$ A.

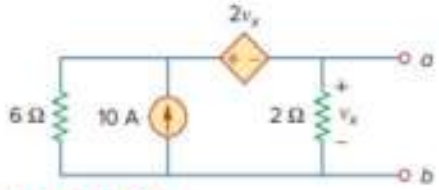
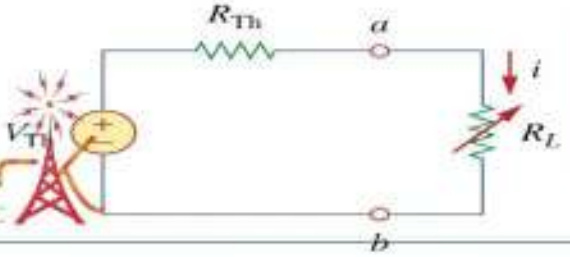


Figure 4.45
For Practice Prob. 4.12.

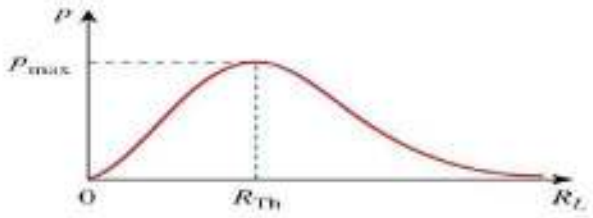
❖Maximum Power Transfer



$$p = i^2 R_L = \left(\frac{V_{TH}}{R_{TH} + R_L}\right)^2 R_L$$

Spar POLYTECHNIC

Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen the load ($R_L = R_{TH}$).



يتم نقل الطاقة القصوى الى الحمل
عندما تساوي $R_{th}=R_L$

رياضيا لايجاد P_{MAX} بنوخذ المشتقة
الاولى بنسبة ل R_L لانها هي المتغير

$$\frac{dp}{dR_L} = V_{TH}^2\left[\frac{(R_{TH}+R_L)^2 - 2R_L(R_{TH}+R_L)}{(R_{TH}+R_L)^4}\right]$$

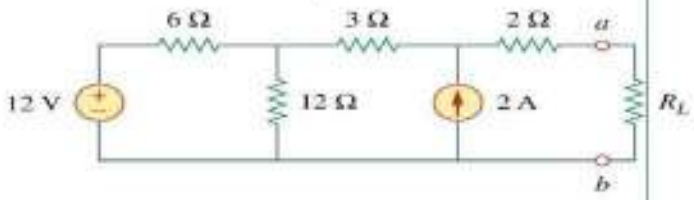
$$= V_{TH}^2\left[\frac{(R_{TH}+R_L-2R_L)}{(R_{TH}+R_L)^3}\right] = 0$$

$$0 = (R_{TH}+R_L-2R_L) = (R_{TH}-R_L)$$

$$R_L = R_{TH}$$

$$P_{\max} = \frac{V_{TH}^2}{4R_{TH}}$$

Example 4.13 Find the value of RL for maximum power transfer in the circuit of Fig. 4.50. Find the maximum power.



SOL

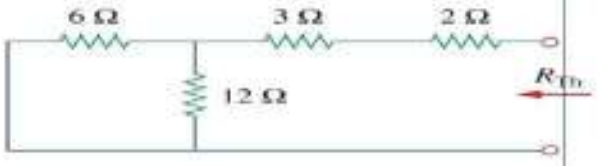
$$R_{TH} = 2+3+6\|12 = 5+\frac{6\times12}{18} = 9\Omega$$

$$-12+18i_1-12i_2,\quad i_2 = -2A$$

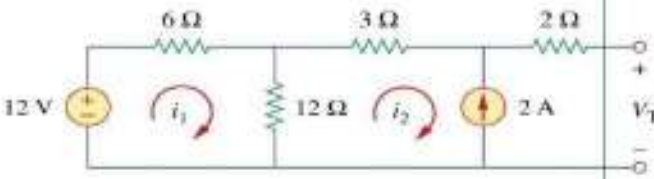
$$-12+6i_1+3i_2+2(0)+V_{TH} = 0 \Rightarrow V_{TH} = 22V$$

$$R_L = R_{TH} = 9\Omega$$

$$p_{\max} = \frac{V_{TH}^2}{4R_L} = \frac{22^2}{4\times9} = 13.44W$$



(a)



Practice Problem 4.13 Determine the value of R_L that will draw the maximum power from the rest of the circuit in Fig. 4.52. Calculate the maximum power.

Answer: 126.67 Ω, 96.71 mW.

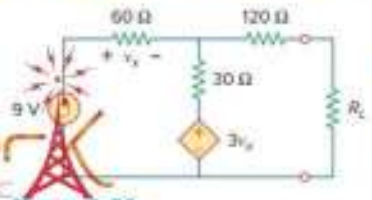


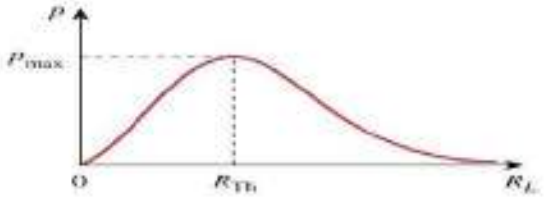
Figure 4.52
For Practice Prob. 4.13.

4.9 Verifying Circuit Theorems with *PSpice*

In this section, we learn how to use *PSpice* to verify the theorems covered in this chapter. Specifically, we will consider using DC Sweep analysis to

Spark POLYTECHNIC
Laith Abdulsalam Awad

Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen the load ($R_L = R_{TH}$).



يتم نقل الطاقة القصوى الى الحمل
عندما تساوى $R_{th}=R_L$

رياضيا لايجاد P_{MAX} بنوخذ المشتقة
الاولى بنسبة ل R_L لانها هي المتغير

$$\frac{dp}{dR_L} = V_{TH}^2\left[\frac{(R_{TH}+R_L)^2 - 2R_L(R_{TH}+R_L)}{(R_{TH}+R_L)^4}\right]$$

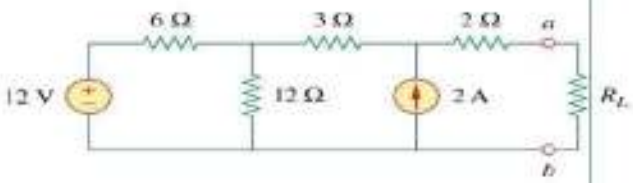
$$= V_{TH}^2\left[\frac{(R_{TH}+R_L-2R_L)}{(R_{TH}+R_L)^3}\right] = 0$$

$$0 = (R_{TH}+R_L-2R_L) = (R_{TH}-R_L)$$

$$R_L = R_{TH}$$

$$p_{\max} = \frac{V_{TH}^2}{4R_{TH}}$$

Example 4.13 Find the value of RL for maximum power transfer in the circuit of Fig. 4.50. Find the maximum power.



SOL

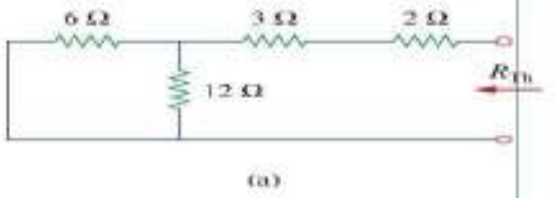
$$R_{TH} = 2+3+6\|12 = 5+\frac{6\times12}{18} = 9\Omega$$

$$-12+18i_1-12i_2,\quad i_2=-2A$$

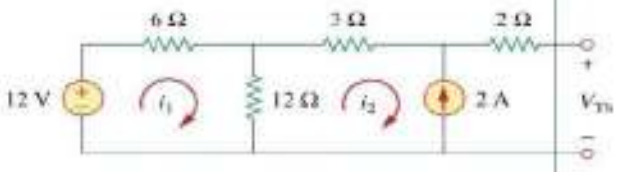
$$-12+6i_1+3i_2+2(0)+V_{TH}=0 \Rightarrow V_{TH}=22V$$

$$R_L = R_{TH} = 9\Omega$$

$$p_{\max} = \frac{V_{TH}^2}{4R_L} = \frac{22^2}{4\times9} = 13.44W$$



(a)



Practice Problem 4.13 Determine the value of R_L that will draw the maximum power from the rest of the circuit in Fig. 4.52. Calculate the maximum power.

Answer: 126.67 Ω, 96.71 mW.

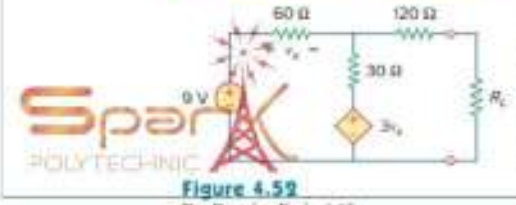


Figure 4.52
For Practice Prob. 4.13.

4.9 Verifying Circuit Theorems with *PSpice*

In this section, we learn how to use *PSpice* to verify the theorems covered in this chapter. Specifically, we will consider using DC Sweep analysis to

Spark POLYTECHNIC
Laith Abu Salah Awad

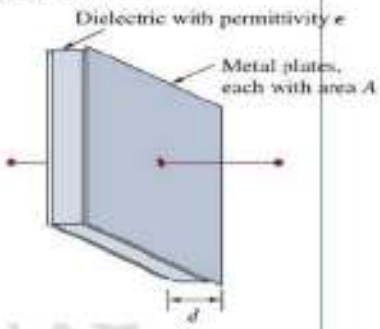
❖Chapter 6 Capacitors and Inductors

$$C=\frac{\varepsilon A}{d}$$

$$\varepsilon=\varepsilon_r\varepsilon_0$$

$$\varepsilon_0=8.854\times10^{-12}\ (\mathrm{F/m})$$

Capacitors A capacitor consists of two conducting plates separated by an insulator (or dielectric).



المواسع المكثف هو عبارة عن لوحين متوازين مشحون كل منهم بشحنة مختلفة عن الاخر وتفصل بينهم مسافة وظيفة هذا المواسع تخزين الطاقة الكهربائية

Three factors affecting the value of capacitance:

- Area: the larger the area, the greater the capacitance.
- Spacing between the plates: the smaller the spacing, the greater the capacitance.
- Material permittivity: the higher the permittivity, the greater the capacitance.

$$C=\frac{\varepsilon A}{d}$$

يوجد لدى المكثف ثلاث عوامل تؤثر على قيمة السعة

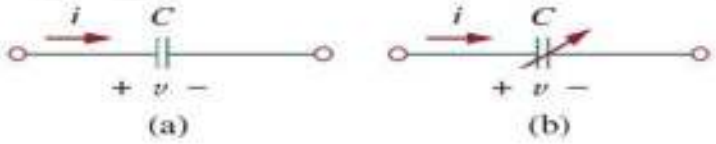
1- المساحة كلما كبرت المساحة ازدات السعة والعكس صحيح
2- التباعد بين الالواح المسافه بين اللوحين كلما كانت المسافه بين للوحين اصغر زادت السعة
3- سماحية المواد كلما زادت السماحية زادت السعة

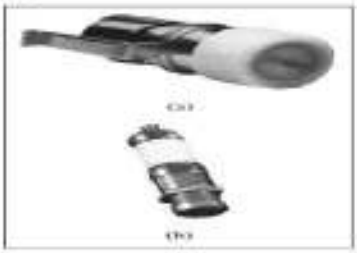


بعض انواع المكثفات

a) Polyester capacitor,
(b) Ceramic capacitor,
(c) Electrolytic capacitor

Variable capacitors

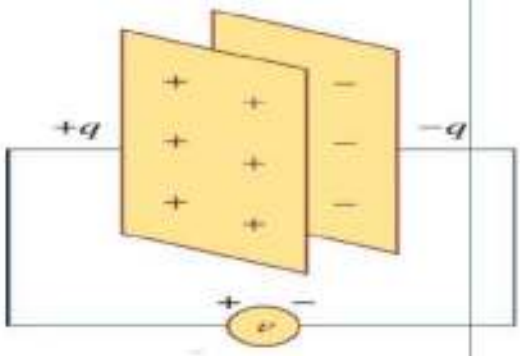




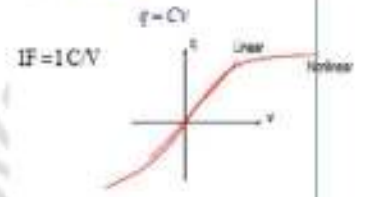
a- fired b- variable

شرح

اذا قمنا بزيادة الشحنة الى ضعف سوف يزداد الجهد الى ضعف ايضا لتبقى السعة ثابت اي ان الجهد والشحنة اذا حصل عليهم تغير طبقة المواسعة ثابتة slop= c



$$c = \left[\frac{c}{v}\right] = \left[\frac{A \times S}{V}\right] = \left[\frac{1}{\Omega} \times S\right] = [F]$$



Charge in Capacitors

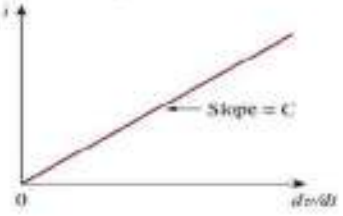
The relation between the charge in plates and the voltage across a capacitor is given below.

Voltage Limit on a Capacitor

Since q=Cv, the plate charge increases as the voltage increases. The electric field intensity between two plates increases. If the voltage across the capacitor is so large that the field intensity is large enough to break down the insulation of the dielectric, the capacitor is out of work. Hence, every practical capacitor has a maximum limit on its operating voltage

عندك q=cv تزداد شحنة اللوحة مع زيادة الجهد تزداد شدة المجال الكهربائي بين لوحين اذا كان الجهد عبر المكثف كبيرا جدا لدرجة ان شدة المجال كبيرة بما ينبغي لكسر عزل العازل الكهربائي فان المكثف يكون عاطلا عن العمل ومن ثم فان كل مكثف عملي له حد اقصى لجهد التشغيل الخاص به وجميع المكثفات مكتوب عليها rated voltage

I-V Relation of Capacitor



التيار يتناسب طرديا مع C

وايضا يتناسب طرديا مع معدل تغير الفولطية على اطراف المكثف

$$q = Cv, i = \frac{dq}{dt} = C\frac{dv}{dt}$$

Spark POLYTECHNIC

Physical Meaning

- when v is a constant voltage, then i=0; a constant voltage across a capacitor creates no current through the capacitor, the capacitor in this case is the same as an open circuit.
- If v is abruptly changed, then the current will have an infinite value that is practically impossible. Hence, a capacitor is impossible to have an abrupt change in its voltage except an infinite current is applied.

- عندما تكون V جهدا ثابتا فان i=0 لا ينتج الجهد الثابت عبر المكثف اي تيار من خلال المكثف والمكثف في هذه الحالة هو نفس الدارة المفتوحة
توضيح في حالة اذا كان مصدر الجهد ثابت سوف يشحن المكثف عندما يشحن يصبح نفس مصدر الجهد ف بتالي جهد المكثف يساوي نفس جهد المصدر ولا يمر تيار ف يصبح open circuit
- اذا تم تغيير الجهد فجاة فسيكون للتيار قيمة لا نهائية مستحيلة علميا ومن ثم من المستحيل ان يحدث تغير مفاجئ في جهد المكثف باستثناء تيار غير محدود
اي اذا تغيرت الفولطية بشكل لحظي اي انها تغيرت بزمن يقترب من الصفر t→0 i→∞ اذن الفولطية والشحنة لا يمكن تغيرها بشكل مفاجئ على اطراف المكثف

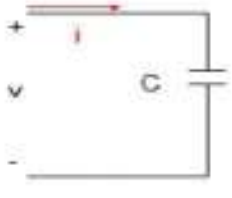
- A capacitor is an open circuit to dc.
- The voltage on a capacitor cannot change abruptly

المكثف يعمل عمل الدارة المفتوحة في dc circuit والفولطية على صفائح المكثف لا يمكن ان تتغير بشكل لحظي

Integral relation ships for capacitor

$$i = C\frac{dv}{dt} \qquad v(t) = \frac{1}{C}\int_{-\infty}^{t} i\,dt \qquad (v(-\infty) = 0)$$

$$v(t) = \frac{1}{C}\int_{t_o}^{t} i\,dt + v(t_o) \qquad (v(t_o) = q(t_o)/C)$$



The charge on a capacitor is an integration of current through the capacitor. Hence, the memory effect counts

Spark POLYTECHNIC

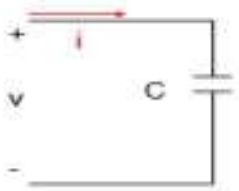
Energy Storing in Capacitor

$$p = vi = Cv\frac{dv}{dt}$$

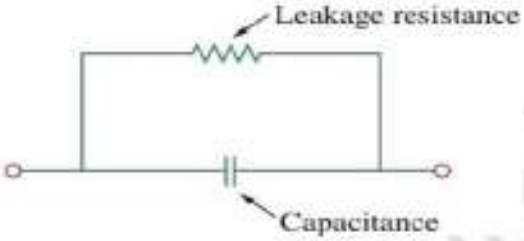
$$w = \int_{-\infty}^{t} p\,dt = C\int_{-\infty}^{t} v\frac{dv}{dt}dt = C\int_{v(-\infty)}^{v(t)} v\,dv = \frac{1}{2}Cv^2\Big|_{v(-\infty)}^{v(t)}$$

$$w(t) = \frac{1}{2}Cv^2(t) \qquad (v(-\infty) = 0)$$

$$w(t) = \frac{q^2(t)}{2C}$$



Model of Practical Capacitor



Example 6.1 (a) Calculate the charge stored on a 3-pF capacitor with 20 V across it.(b) Find the energy stored in the capacitor.

Sol :

(a) $q = Cv,\ q = 3\times10^{-12}\times20 = 60\text{pC}$

(b) $w = \frac{1}{2}Cv^2 = \frac{1}{2}\times3\times10^{-12}\times400 = 600\text{pJ}$

Practice Problem 6.1 What is the voltage across a 4.5- μF capacitor if the charge on one plate is 0.12 mC? How much energy is stored? Answer: 26.67 V, 1.6 mJ.

Lath Abdussalam AWS

Example 6.2 The voltage across a 5- μF capacitor is $v(t) = 10\cos 6000t$ V Calculate the current through it.

Sol : $i = C\frac{dv}{dt} = 5\times10^{-6}\ \frac{d}{dt}(10\cos 6000t)$

$= -5\times10^{-6}\times6000\times10\sin 6000t = -0.3\sin 6000t$ A

Practice Problem 6.2 If a 10-μF capacitor is connected to a voltage source with v(t) = 20 cos(200t) V determine the current through the capacitor.

Answer: −40 sin (200t) mA.

Example 6.3 Determine the voltage across a 2-μF capacitor if the current through it is i(t) = 6e−3000t mA Assume that the initial capacitor voltage is zero.

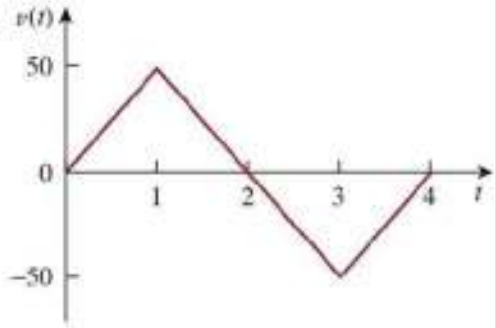
Sol : $v = \frac{1}{C}\int_0^t i dt + v(0) \text{and } v(0) = 0, v = \frac{1}{2\times10^{-6}}\int_0^t 6e^{-3000t}dt\cdot10^{-3}$

$= \frac{3\times10^3}{-3000}e^{-3000t}\Big|_0^t = (1-e^{-3000t})\text{V}$

Practice Problem 6.3 The current through a 100-μF capacitor is i(t) = 50 sin 120πt mA. Calculate the voltage across it at t = 1 ms and t = 5 ms. Take v(0) = 0.

Answer: 93.14 mV, 1.736 V.

Example 6.4 Determine the current through a 200- μF capacitor whose voltage is shown in Fig 6.9

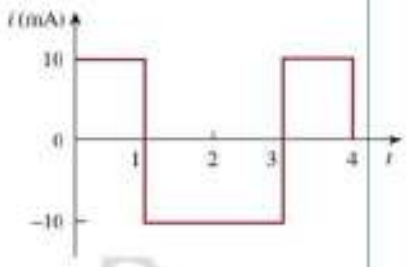


Spark POLYTECHNIC

Example 6.4

Sol : The voltage waveform can be described mathematically as

$$v(t) = \begin{cases} 50t \text{ V} & 0 < t < 1 \\ 100 - 50t \text{ V} & 1 < t < 3 \\ -200 + 50t \text{ V} & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

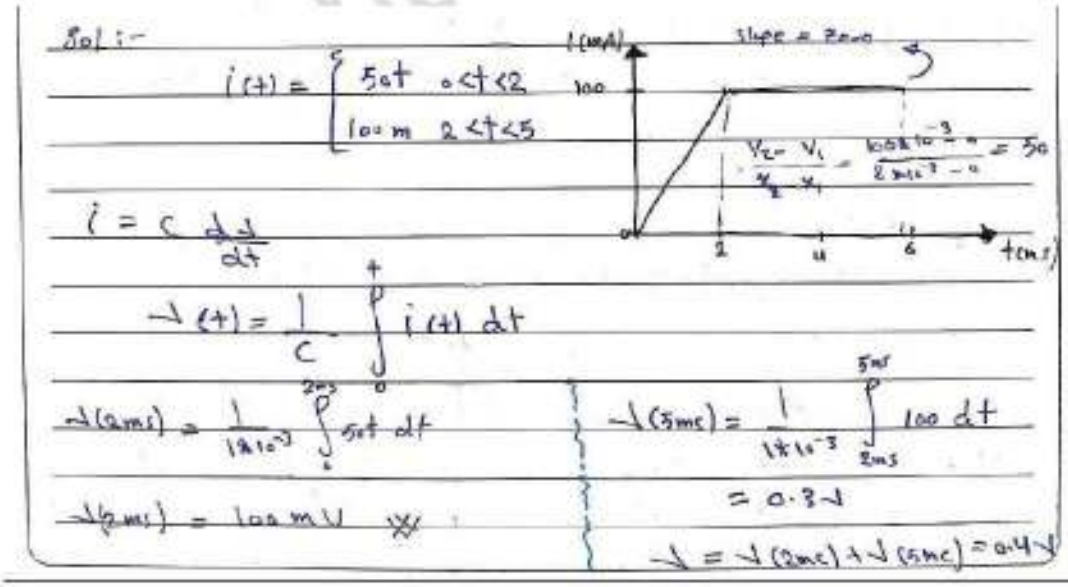


Since i = C dv/dt and C = 200 μF, we take the derivative of to obtain

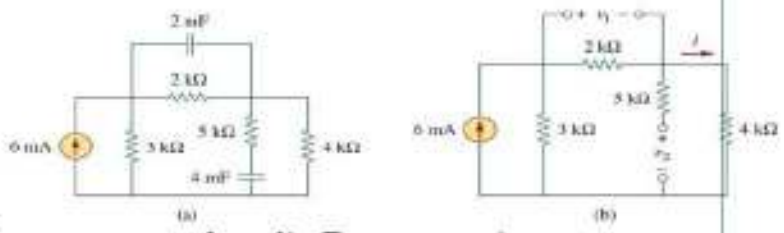
$$i(t) = 200\times10^{-6} \times \begin{cases} 50 & 0 < t < 1 \\ -50 & 1 < t < 3 \\ 50 & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 10\text{mA} & 0 < t < 1 \\ -10\text{mA} & 1 < t < 3 \\ 10\text{mA} & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

thus the current waveform is shown in Fig.6.10

Practice Problem 6.4 An initially uncharged 1-mF capacitor has the current shown in Fig. 6.11 across it. Calculate the voltage across it at t = 2 ms and t = 5 ms.



Example 6.5 Obtain the energy stored in each capacitor in Fig. 6.12(a) under dc condition



Sol :

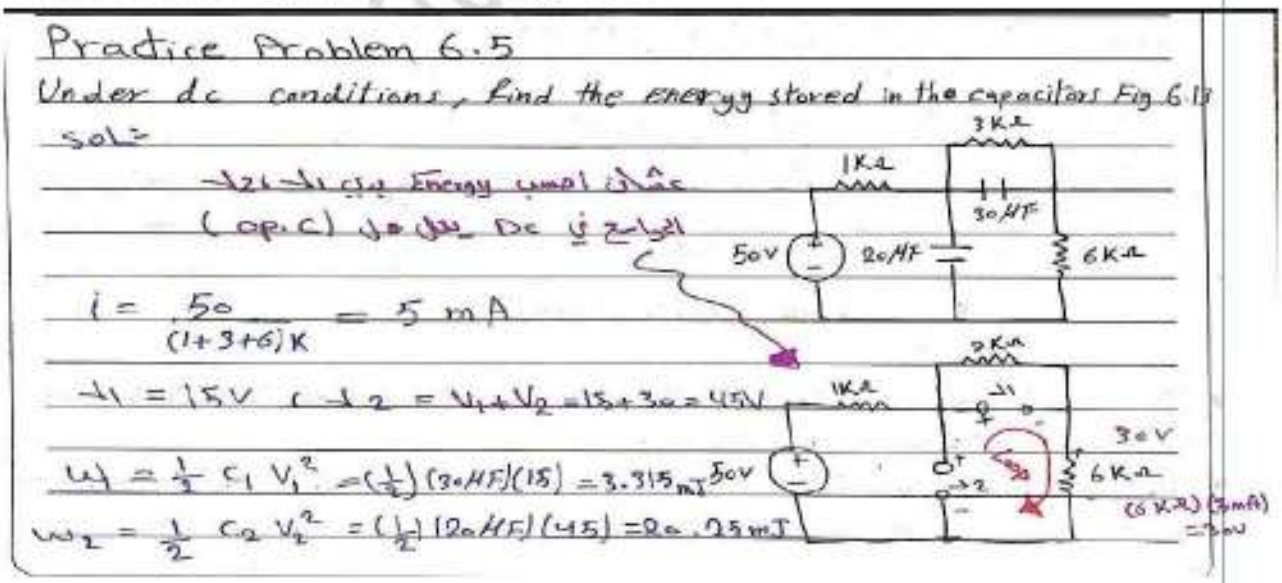
Under dc condition, we replace each capacitor with an open circuit. By current division

$$i = \frac{3}{3+2+4}(6\text{mA}) = 2\text{mA} \quad \therefore v_1 = 2000i = 4\text{V}, v_2 = 4000i = 8\text{V}$$

$$\therefore w_1 = \frac{1}{2}C_1 v_1^2 = \frac{1}{2}(2\times10^{-3})(4)^2 = 16\text{mJ}$$

$$w_2 = \frac{1}{2}C_2 v_2^2 = \frac{1}{2}(4\times10^{-3})(8)^2 = 128\text{mJ}$$

Practice Problem 6.5 Under dc conditions, find the energy stored in the capacitors in Fig. 6.13

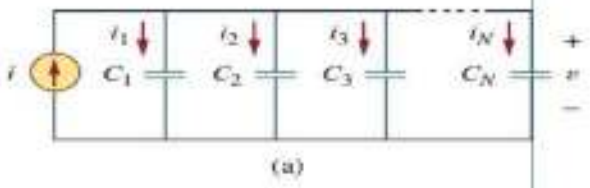


Abdalla Awwad

Spark POLYTECHNIC

Series and paralle capacitors

Parallel capacitor



(a)

$$i = i_1 + i_2 + i_3 + \ldots + i_N$$

$$i = C_1\frac{dv}{dt} + C_2\frac{dv}{dt} + C_3\frac{dv}{dt} + \ldots + C_N\frac{dv}{dt}$$

$$= \left(\sum_{k=1}^{N} C_K\right)\frac{dv}{dt} = C_{eq}\frac{dv}{dt}$$

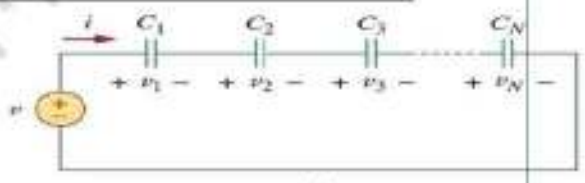


(b)

$$C_{eq} = C_1 + C_2 + C_3 + \ldots + C_N$$

❖ The equivalent capacitance of N parallel-connected capacitors is the sum of the individual capacitance.

Series capacitor

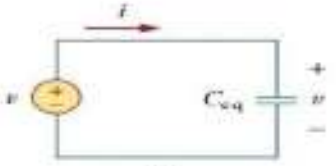


(a)

$$v(t) = v_1(t) + v_2(t) + \ldots + v_N(t)$$

$$\frac{1}{C_{eq}}\int_{-\infty}^{t} i\,d\tau = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots + \frac{1}{C_N}\right)\int_{-\infty}^{t} i\,d\tau$$

$$\frac{q(t)}{C_{eq}} = \frac{q(t)}{C_1} + \frac{q(t)}{C_2} + \cdots + \frac{q(t)}{C_N}$$



(b)

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \qquad \boxed{C_{eq} = \frac{C_1 C_2}{C_1 + C_2}}$$

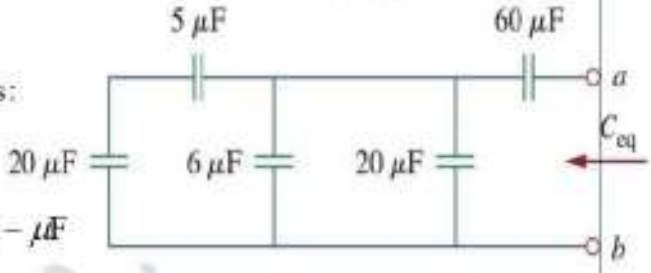
❖ The equivalent capacitance of series-connected capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances

Lain Abdulsalam Awwal

These results enable us to look the capacitor in this way: 1/C has the equivalent effect as the resistance. The equivalent capacitor of capacitors connected in parallel or series can be obtained via this point of view, so is the Y-Δ connection and its transformation

Example 6.7 Find the equivalent capacitance seen between terminals a and b of the circuit in Fig 6.16

Sol :



- $20-\mu F$ and $5-\mu F$ capacitors are in series:

$$\therefore \frac{20\times 5}{20+5}=4\mu F$$

- $4-\mu F$ capacitor is in parallel with the $6-\mu F$ and $20-\mu F$ capacitors:

$$\therefore 4+6+20=30\mu F$$

- $30-\mu F$ capacitor is in series with the $60-\mu F$ capacitor.

$$C_{eq}=\frac{30\times 60}{30+60}\mu F=20\mu F$$

Practice Problem 6.6

Find the equivalent capacitance seen at the terminals of the circuit in Fig. 6.17.

Answer: 40 μF.

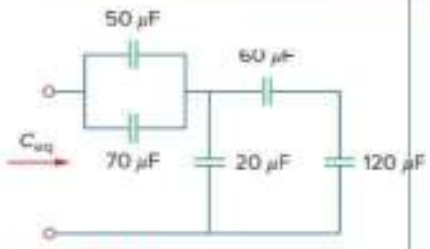
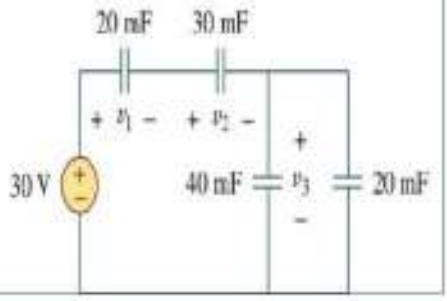


Figure 6.17
For Practice Prob. 6.6.

Example 6.7 For the circuit in Fig. 6.18, find the voltage across each capacitor.



Example 6.7

Sol : $\therefore C_{eq} = \frac{1}{\frac{1}{60}+\frac{1}{30}+\frac{1}{20}}\text{mF} = 10\text{mF}$

Total charge $q = C_{eq}v = 10\times10^{-3}\times30 = 0.3\text{C}$

This is the charge on the 20-mF and 30-mF capacitors, because they are in series with the 30-v source. (A crude way to see this is to imagine that charge acts like current, since i = dq/dt)

$$v_1 = \frac{q}{C_1} = \frac{0.3}{20\times10^{-3}} = 15\text{ V}, \quad v_2 = \frac{q}{C_2} = \frac{0.3}{30\times10^{-3}} = 10\text{V}$$

Having determined v1 and v2, we now use KVL to determine v3 by

$$v_3 = 30 - v_1 - v_2 = 5\text{V}$$

Alternatively, since the 40-mF and 20-mF capacitors are in parallel, they have the same voltage v3 and their combined capacitance is 40+20=60mF

$$\therefore v_3 = \frac{q}{60\text{mF}} = \frac{0.3}{60\times10^{-3}} = 5\text{V}$$

Practice Problem 6.7

Find the voltage across each of the capacitors in Fig. 6.20.

Answer: $v_1 = 75$ V, $v_2 = 75$ V, $v_3 = 25$ V, $v_4 = 50$ V.

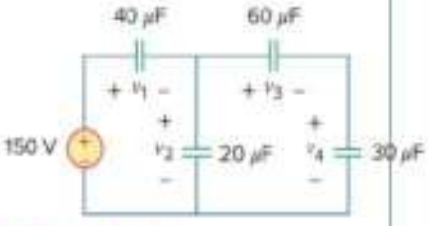


Figure 6.20
For Practice Prob. 6.7.

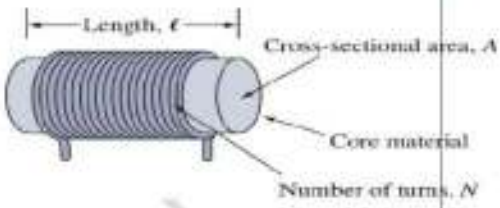
Spark POLYTECHNIC

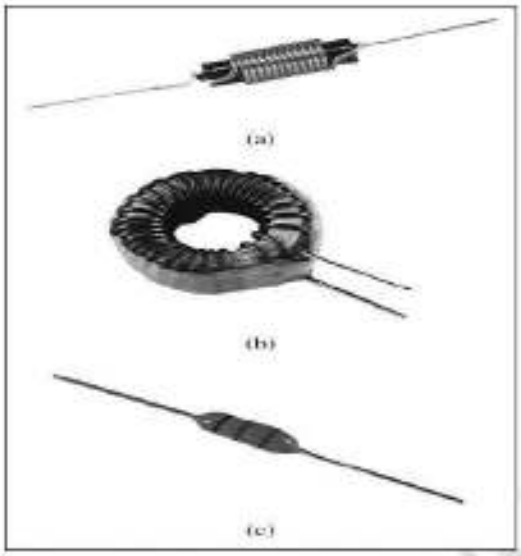
$$L = \frac{N^2 \mu A}{l}$$

- **Inductors المحاثات**

An inductor is made of a coil of conducting wire

المحث مصنوع من ملف من الاسلاك الموصلة (هذا الملف يصنع من مجموعة من الاسلاك المعزولة بمادة عازلة من الورنيش او مواد اخرى نقوم بلف هذه الاسلاك بعدد لفات معين)





$$L = \frac{N^2 \mu A}{l}$$

$$\mu = \mu_r \mu_0$$

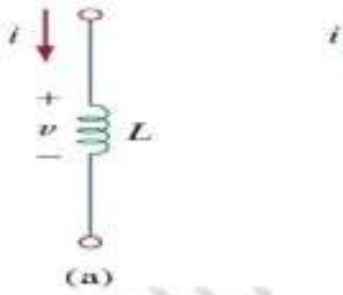
$$\mu_0 = 4\pi \times 10^{-7} \text{ (H/m)}$$

N : number of turns

L : length

A : cross-sectional area

μ : *permeability of the core*



(a) air-core

(b) iron-core

(c) variable iron-core

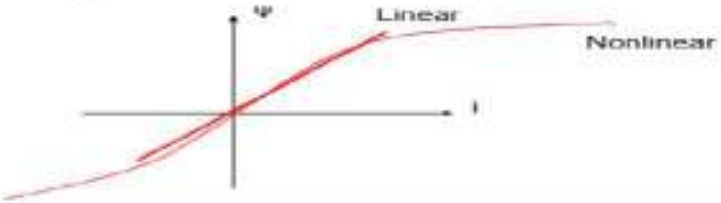
Flux in Inductors

The relation between the flux in inductor and the current through the inductor is given below

العلاقة بين التدفق في المحاثة والتيار عبر المحاثة مذكور ادناه

$$\phi = Li$$

1H = 1 Weber/A



Spark POLYTECHNIC

Energy Storage Form

An inductor is a passive element designed to store energy in the magnetic field while a capacitor stores energy in the electric field

الحث هو عنصر سلبي مصمم لتخزين الطاقة في مجال مغناطيسي بينما يقوم المكثف بتخزين الطاقة والمجال الكهربائي .

I-V Relation of Inductors

An inductor consists of a coil of conducting wire.

الفولطية على اي ملف تتناسب طردياً مع المحاثاة (او الحث ذاتي)

$$v = \frac{d\phi}{dt} = L\frac{di}{dt}$$



Physical Meaning

- When the current through an inductor is a constant, then the voltage across the inductor is zero, same as a short circuit.
- No abrupt change of the current through an inductor is possible except an infinite voltage across the inductor is applied.
- The inductor can be used to generate a high voltage, for example, used as an igniting element

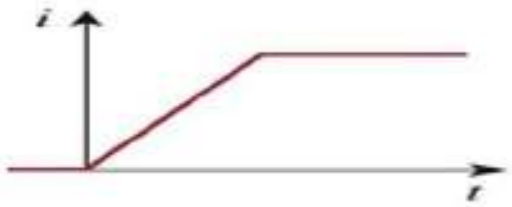
1- عندما يكون التيار عبر المحاثاة ثابتاً , فأن الجهد عبر المحاثاة يساوي صفر , مثل SHORT CIRCUIT

2- لا يمكن حدوث تغيير مفاجئ في التيار خلال محث بأستثناء استخدام جهد لا نهائي عبر المحاثاة

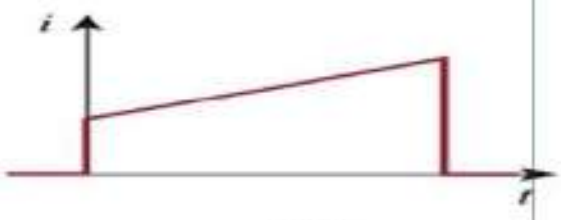
3- يمكن استخدام المحاثاة لتوليد جهد عالي , على سبيل المثال , ويستخدم كعنصر اشعال .

- An inductor are like a short circuit to dc.
- The current through an inductor cannot change instantaneously.

Spark POLYTECHNIC



(a)



(b)

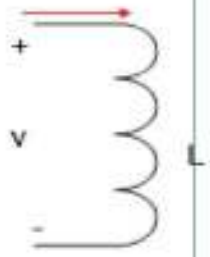
1- المحاثاة في CIRCUIT DC يصبح ك short circuit

2- التيار من خلال المحاثاة لا يمكن ان يتغير على الفور . (I and φ) لا يمكن ان يتغيرو بشكل لحظي على اطراف الملف .

Integral relation ships of the Inductor .

$$di = \frac{1}{L} v dt \qquad i = \frac{1}{L}\int_{-\infty}^{t} v(t)dt$$

$$i = \frac{1}{L}\int_{t_o}^{t} v(t)dt + i(t_o)$$

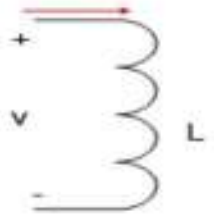


Energy Stored in an Inductor

$$P = vi = \left(L\frac{di}{dt}\right)i$$

$$w = \int_{-\infty}^{t} p dt = \int_{-\infty}^{t}\left(L\frac{di}{dt}\right)i dt$$

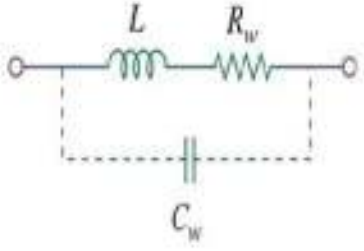
$$= L\int_{i(-\infty)}^{i(t)} i di = \frac{1}{2}Li^2(t) - \frac{1}{2}Li^2(-\infty) \quad i(-\infty) = 0$$



The energy stored in an inductor

$$w(t) = \frac{1}{2} L i^2(t)$$

Model of a Practical Inductor



Spark POLYTECHNIC

Example 6.8 The current through a 0.1-H inductor is $i(t)=10te^{-5t}$ A. Find the voltage across the inductor and the energy stored in it.

Sol : $\text{Since } v = L\frac{di}{dt} \text{ and } L = 0.1\text{H},$

$$v = 0.1\frac{d}{dt}(10te^{-5t}) = e^{-5t} + t(-5)e^{-5t} = e^{-5t}(1-5t)\text{V}$$

$$w = \frac{1}{2}Li^2 = \frac{1}{2}(0.1)100t^2e^{-10t} = 5t^2e^{-10t}\text{J}$$

Practice Problem 6.8 If the current through a 1-mH inductor is i(t) = 90 sin (200t) mA, find the terminal voltage and the energy stored. Answer: 18 cos (200t) mV, 4.05 $\sin^2(200t)$ μJ

Example 6.9 Find the current through a 5-H inductor if the voltage across it is

$$v(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t < 0 \end{cases}$$

Also find the energy stored within 0 < t < 5s. Assume i(0)=0.

Sol : $\text{Since } i = \frac{1}{L}\int_{t_0}^{t} v(t)dt + i(t_0) \text{and } L = 5\text{H}.$

$$i = \frac{1}{5}\int_0^t 30t^2dt + 0 = 6\times\frac{t^3}{3} = 2t^3 \text{ A}$$

The power $p = vi = 60t^5$, and the energy stored is then

$$w = \int pdt = \int_0^5 60t^5dt = 60\frac{t^6}{6}\Big|_0^5 = 156.25\text{kJ}$$

Spark POLYTECHNIC

Example 6.9

$$w(5) - w(0) = \frac{1}{2}Li^2(5) - \frac{1}{2}Li(0)$$

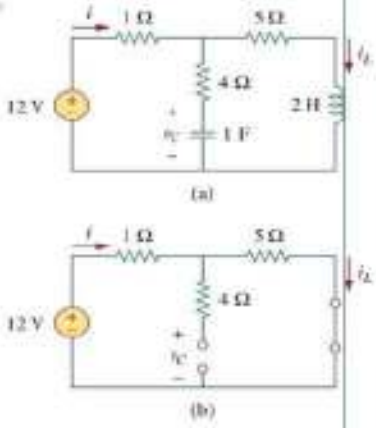
$$= \frac{1}{2}(5)(2\times 5^3)^2 - 0 = 156.25\text{kJ}$$

Practice Problem 6.9

The terminal voltage of a 2-H inductor is $v = 10(1 - t)$ V. Find the current flowing through it at $t = 4$ s and the energy stored in it at $t = 4$ s. Assume $i(0) = 2$ A.

Answer: −18 A, 324 J.

Example 6.10 Consider the circuit in Fig 6.27(a). Under dc conditions, find:



(a) i, v_C, and i_L.

(b) the energy stored in the capacitor and inductor.

(*a*) Under dc condition: capacitor → open circuit
inductor → short circuit

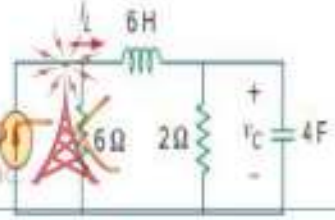
$$i = i_L = \frac{12}{1+5} = 2A,\ v_c = 5i = 10\text{ V}$$

(*b*) $$w_c = \frac{1}{2}Cv_c^2 = \frac{1}{2}(1)(10^2) = 50\text{J},$$

$$w_L = \frac{1}{2}Li^2 = \frac{1}{2}(2)(2^2) = 4\text{J}$$

Practice Problem 6.10

Determine v_C, i_L, and the energy stored in the capacitor and inductor in the circuit of Fig. 6.28 under dc conditions.



Answer: 15 V, 7.5 A, 450 J, 168.75 J.

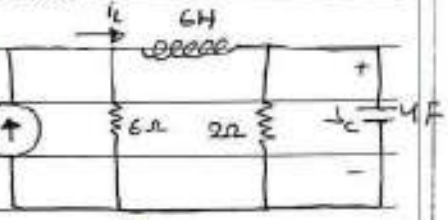
Practice Problem 6.10

Determine v_c, i_L and the energy stored in the capacitor and inductor in the circuit of Fig 6.28 under dc conditions.

Sol:- sh.c ... o.c ... في حالة الـ dc

Using current divider

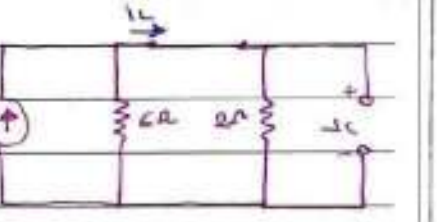
$$i_L = (10)\left(\frac{2}{2+6}\right) = 7.5\ A$$



$$v_c = (2\Omega)(7.5) = 15\ V$$

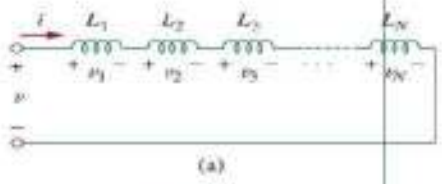
$$w_c = \frac{1}{2} C v_c^2 = \left(\frac{1}{2}\right)(4)(15V)^2 = 450\ J$$

$$w_L = \frac{1}{2} L i_L^2 = \left(\frac{1}{2}\right)(6)(7.5)^2 = 168.75\ J$$



Series and Parallel Inductors

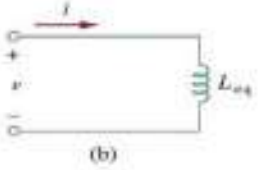
Inductor in series



Applying KVL to the loop

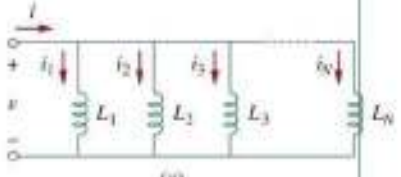
$$v = v_1 + v_2 + v_3 + \ldots + v_N$$

Substituting $v_K = L_K\, di/dt$ results in

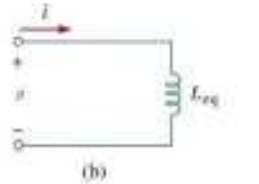


$$\begin{aligned} v &= L_1\frac{di}{dt} + L_2\frac{di}{dt} + L_3\frac{di}{dt} + \ldots + L_N\frac{di}{dt} \\ &= (L_1 + L_2 + L_3 + \ldots + L_N)\frac{di}{dt} \\ &= \left(\sum_{K=1}^{N} L_K\right)\frac{di}{dt} = L_{eq}\frac{di}{dt} \end{aligned}$$

$$L_{eq} = L_1 + L_2 + L_3 + \ldots + L_N$$



Parallel inductors



Abdulsalam

Using KCL

$$i = i_1 + i_2 + i_3 + \ldots + i_N$$

$$i_k = \frac{1}{L_k}\int_{t_0}^{t} v\,dt + i_k(t_0)$$

$$\therefore i = \frac{1}{L_k}\int_{t_0}^{t} v\,dt + i_1(t_0) + \frac{1}{L_2}\int_{t_0}^{t} v\,dt + i_2(t_0) + \ldots + \frac{1}{L_N}\int_{t_0}^{t} v\,dt + i_N(t_0)$$

$$= \left(\frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_N}\right)\int_{t_0}^{t} v\,dt + i_1(t_0) + i_2(t_0) + \ldots + i_N(t_0)$$

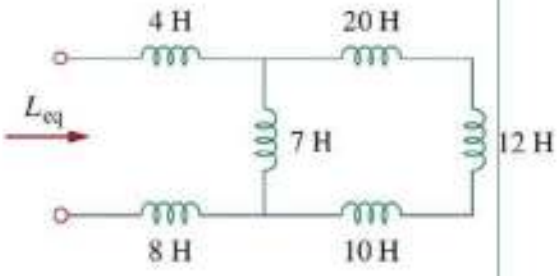
$$= \left(\sum_{k=1}^{N}\frac{1}{L_k}\right)\int_{t_0}^{t} v\,dt + \sum_{k=1}^{N} i_k(t_0) = \frac{1}{L_{eq}}\int_{t_0}^{t} v\,dt + i(t_0)$$

$$\frac{1}{Leq} = \frac{1}{L1} + \frac{1}{L2} + \frac{1}{L3} + \cdots + \frac{1}{Ln} \quad , \quad \frac{1}{Leq} = \frac{l1 \times l2}{l1 + l2}$$

The inductor in various connection has the same effect as the resistor. Hence, the Y-Δ transformation of inductors can be similarly derived

للمحث في اتصال مختلف نفس تأثير المقاومة ويمكن التحويل من ستار الى دلتا

Example 6.11 Find the equivalent inductance of the circuit shown in Fig. 6.31



Sol :

Laith Abdulsalam

$$\text{Series}: \quad 20\text{H}, 12\text{H}, 10\text{H}$$
$$\rightarrow 42\text{H}$$

$$\text{Parallel}: \quad \frac{7\times 42}{7+42} = 6\text{H}$$
$$\therefore L_{eq} = 4+6+8 = 18\text{H}$$

Practice Problem 6.11 Calculate the equivalent inductance for the inductive ladder network in Fig. 6.32.

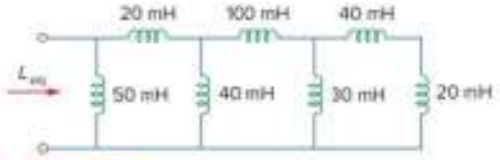


Figure 6.32
For Practice Prob. 6.11.

Answer: 25 mH.

Example 6.12 For the circuit in Fig. 6.33, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$

Sol :

Laith Abdulsalam

(a) $i(t) = 4(2 - e^{-10t})\text{mA} \rightarrow i(0) = 4(2-1) = 4\text{mA}.$

$\therefore\ i_1(0) = i(0) - i_2(0) = 4 - (-1) = 5\text{mA}$

(b) The equivalentinductanceis

$$L_{eq} = 2 + 4 \,||\, 12 = 2 + 3 = 5\text{H}$$

$$\therefore v(t) = L_{eq}\frac{di}{dt} = 5(4)(-1)(-10)e^{-10t}\text{mV} = 200e^{-10t}\text{mV}$$

$$v_1(t) = 2\frac{di}{dt} = 2(-4)(-10)e^{-10t}\text{mV} = 80e^{-10t}\text{mV}$$

$$\therefore v_2(t) = v(t) - v_1(t) = 120e^{-10t}\text{mV}$$

(c) $i = \frac{1}{L}\int_0^t v(t)\,dt + i(0) \Rightarrow$

$$i_1(t) = \frac{1}{4}\int_0^t v_2\,dt + i_1(0) = \frac{120}{4}\int_0^t e^{-10t}\,dt + 5\text{mA}$$

$$= -3e^{-10t}\Big|_0^t + 5\text{mA} = -3e^{-10t} + 3 + 5 = 8 - 3e^{-10t}\ \text{mA}$$

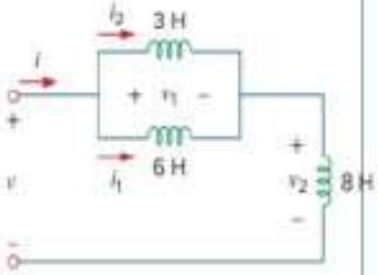
$$i_2(t) = \frac{1}{12}\int_0^t v_2\,dt + i_2(0) = \frac{120}{12}\int_0^t e^{-10t}\,dt - 1\text{mA}$$

$$= -e^{-10t}\Big|_0^t - 1\text{mA} = -e^{-10t} + 1 - 1 = -e^{-10t}\ \text{mA}$$

Note that $i_1(t) + i_2(t) = i(t)$

Practice Problem 6.12

In the circuit of Fig. 6.34, $i_1(t) = 3e^{-2t}$ A. If $i(0) = 7$ A, find:
(a) $i_2(0)$; (b) $i_2(t)$ and $i(t)$; (c) $v_1(t)$, $v_2(t)$, and $v(t)$.



Answer: (a) 4 A, (b) $(-2 + 6e^{-2t})$ A, $(-2 + 9e^{-2t})$ A, (c) $-36e^{-2t}$ V, $-144e^{-2t}$ V, $-180e^{-2t}$ V.

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v-i:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i\,dt + v(t_0)$	$v = L\frac{di}{dt}$
i-v:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} i\,dt + i(t_0)$
p or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

خلص شابتر شابتر خفيف لطيف لاتنسو تحلو الامثلة التي تم وضعها بين الشرح ولتمكن أكثر وأكثر من المادة حل اسئلة Problem صفحة 240 وعليك بالسنوات

تصبح الحياة ذات معنى حينما نجد شيئًا نكافح من أجله ونسعى بشغف للوصول إليه

Laith Abu Salah

- Chapter 7 First – order circuit

Spark POLYTECHNIC

Laith Abdulsalam Awwad

2020\2021

Circuit 1

laith abdulsalam

Eng . laith abdulsalam awwad

Spark polytechnic

2020\2021

- ## Chapter 7 first – order circuit
  - ❖ A first order circuit is characterized by a first order differential equation.
  - ❖ There are two types of first order circuits:
  - ❖ Resistive capacitive, called RC
  - ❖ Resistive inductive, called RL
  - ❖ There are also two ways to excite the circuits:
    - Initial conditions
    - Independent sources

❖ تتميز الدوائر من الدرجة الاولى بمعادلة تفاضلية من الدراجة الاولى

❖ هناك نوعان من الدوائر الكهربائية من الدرجة الاولى : RL , RC circuit

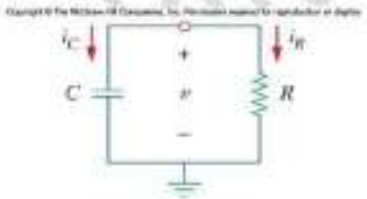
❖ هناك طريقتين لتغذية الدوائر : 1- Initial conditions (a) initial current Inductor

(b) initial voltage at the place of the capacitor (t=0)

2- Independent sources → step response

Source free Rc circuit

- ❖ A source free RC circuit occurs when its dc source is suddenly disconnected.
- ❖ The energy stored in the capacitor is released to the resistors.
- ❖ Consider a series combination of a resistor and a initially charged capacitor as shown



نحصل على (source free RC) عندما نقوم بفصل مصدر التيار المستمر الخاص بها فجأة .

ويتم تحرير الطاقة المخزنة في المكثف الى المقومات

الدارة هاي تسمى في source free RC circuit

وهذه الدارة بعد فصل المصدر عنها

Spark POLYTECHNIC

شو يعني (source free) ؟؟؟

Source free يعني بدي اوجد Response لهذه الدارة عندما لايكون هناك مصدر جهد او تيار يعني كان عندي في الدارة من قبل وفصلنا هذا المصدر, طيب كيف رح تتكون Response (الاستجابة) وانا ما عندي مصدر ؟؟؟؟ رح يتكون response (الاستجابة) عن او من خلال الطاقة المخزنة في المكثف او الملف وخلال فترة زمنية معنية ستنفذ هذه الطاقة المخزنة التي على شكل مجال كهربائي (للمكثف) او شكل مجال مغناطيسي (للملف) ويحرر بها الى المقاومة يعني رح يصير عندي فولطية تيار وطاقة وجميع الطاقة المخزنة في المكثف او الملف سوف تتحول الى حرارة

وهسا بما انو ما في عندي لا مصدر تيار ولا مصدر جهد رح نطلق على response (الاستجابة) اسم (Natural Response) الاستجابة الطبيعية لهذه الدارة

Since the capacitor was initially charged, we can assume at t=0 the initial voltages is $V(0)=V0=V(0^+)=V(0^-)$

الفولطية لا تتغير على اطراف المكثف بشكل مفاجئ

$V(0^-)$= the capacitor voltage just before switching

$V(0^+)$= the capacitor voltage just after switching

The energy stored at t=0 الطاقة المخزنة في المكثف في لحظة صفر

$$w(0) = \frac{1}{2} \times C \times v(0)^2$$

Applying KCL at the top node (a) :

$i_c + i_R = 0$ نقوم بايجاد قيمة التيار وقيمة تيار المقاومة

$$C\frac{dv}{dt} + \frac{v}{Rc} = 0 \rightarrow \frac{dv}{dt} + \frac{V}{Rc} = 0$$ **first order differential equation**

- **This is a first order differential equation , since only the first derivative of V is involved**

هذه المعادلة تفاضلية من الدرجة الاولى حيث ان المشتق الاول فقط من V متضمن

(تنبيه) اذا ما فهمت طريقة اشتقاق القانون التالي (لانه معتمد على مادة رياضيات هندسية (معادلات تفاضلية)) احنا بهمنا القانون الي رح نشتقو بالتوفيق للجميع

Spark POLYTECHNIC

اول خطوة نقوم بفصل المتغيرات V,t

$$\frac{dv}{v} = \frac{-1}{Rc}dt \rightarrow \int_{v(0)}^{v(t)} \frac{dv}{dy} = \frac{-1}{Rc}\int_0^t dt \rightarrow \ln v \mid_{v(0)}^{v(t)} = \frac{-1}{Rc}t$$

$$\ln v - \ln v0 = \ln \frac{v}{v0} = -\frac{1}{Rc}t \text{ taking power of e}$$

$$V(t) = V0\, e^{-\frac{1}{RC}t} = V0\, e^{-\frac{t}{\tau}}$$

the natural response of Rc circuit ((حفظ)) RC=τ

the Natural response of a circuit before to the behavior C in te terms of Voltage and currents of circuit it self with no external sources of excitation

الاستجابة الطبيعية لدارة ما قبل سلوك المكثف من حيث الفولطية وتيارات الدارة مع عدم وجود مصادر خارجية للاثارة

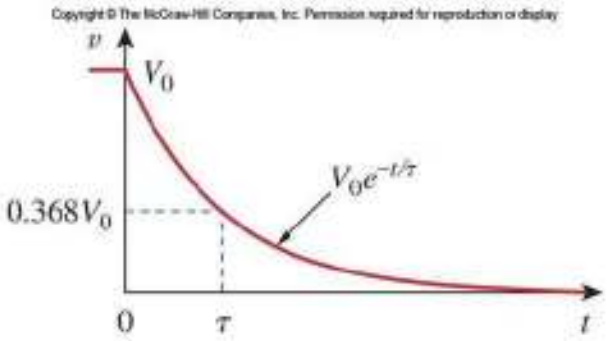
The result shows that the voltage response of the RC circuit is an exponential decay of the initial voltage.

Since this is the response of the circuit without any external applied voltage or current, the response is called the natural response.

تظهر النتيجة ان استجابة الجهد لدارة RC وهي على شكل دالة اسية للجهد الاول

ونظراً لان هذه اسجابة الدارة دون اي جهد خارجي او تيار خارجي تسمى هذه الاستجابة الطبيعية

عندما t=τ يكون V(t) = 0.368V0 وبعد خمس ثوابت زمنية انتهت



After 5τ – the transient process will die out

اذا نعتبر ان الحالة العابرة بنسبة لدوائر الكهربائية انتهت بعد خمس ثوابت زمنية اي ان المكثف فرغ جميع الطاقة المخزنة في داخله .

Spark POLYTECHNIC

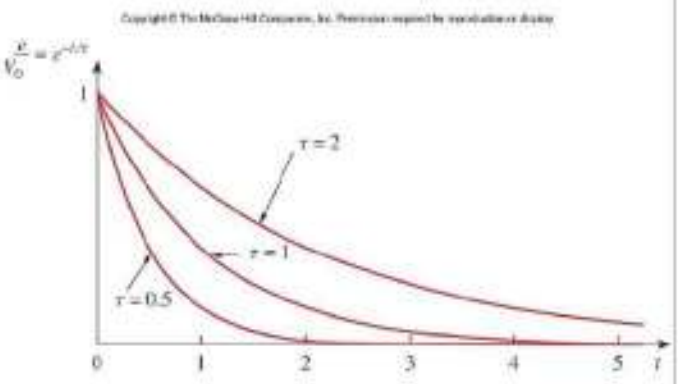
Time Constant ثابت الزمني

- **The speed at which the voltage decays can be characterized by how long it takes the voltage to drop to 1/e of the initial voltage.**
- **This is called the time constant and is represented by .**
- **By selecting 1/e as the reference voltage:** $\tau = Rc$
- **The voltage can thus be expressed as:** $v(t) = V_0 e^{-t/\tau}$

يمكن وصف السرعة التي يتحلل بها الجهد من خلال المدة التي يستغرقها الجهد للانخفاض الى 1/e من الجهد الاولي .

- **After five time constants the voltage on the capacitor is less than one percent.**
- **After five time constants a capacitor is considered to be either fully discharged or charged**
- **A circuit with a small time constant has a fast response and vice versa**

- بعد خمس ثوابت زمنية , يكون الجهد على المكثف اقل من واحد بالمائة
- بعد خمس ثوابت زمنية , يعتبر المكثف اما مفرغاً بالكامل او مشحوناً
- الدارة ذات ثابت زمني صغير لها ستجابة سريعة والعكس صحيح



طرق حل الاسئلة

- اوجد الجهد الاولي عبر المكثف V(0) وثابت الوقت τ
- وباستخدام الجهد الاولي وثابت الزمني يمكن معرفة الجهد كالدالة الوقت باستخدام القانون الذي تم اشتقاقه
- ومن الجهد , يمكن معرفة التيار عبر المقاومة باستخدام قانون اوم

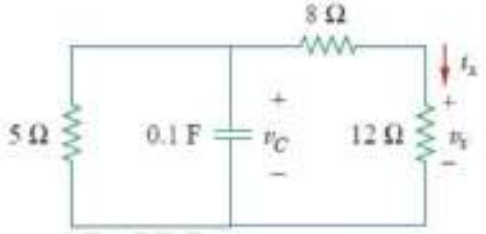
laith abdulsalam alwan

غالباً ما تكون مقاومة الذرة هي مكافئة Thevenin

Example 7.1

: In Fig. 7.5, let vC(0)= 15 V. Find vC , vx and ix for t>0.

Sol:



$$R_{eq} = \frac{20 \times 5}{20 + 5} = 4\ \Omega$$

$$\tau = R_{eq}C = 4(0.1) = 0.4\ \text{s}$$

$$v = v(0)e^{-t/\tau} = 15e^{-t/0.4}\ \text{V}, \qquad v_C = v = 15e^{-2.5t}\ \text{V}$$



نستخدم **voltage division** للحصول على**Vx**

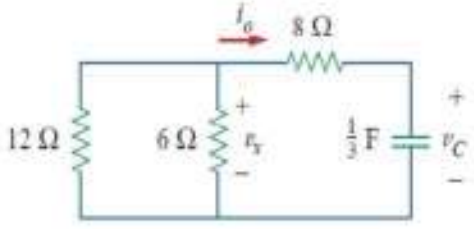
$$v_x = \frac{12}{12 + 8}v = 0.6(15e^{-2.5t}) = 9e^{-2.5t}\ \text{V}$$

$$i_x = \frac{v_x}{12} = 0.75e^{-2.5t}\ \text{A}$$

Practice problem 7.1

Refer to the circuit in Fig. 7.7. Let V_C (0) = 60 V. Determine V_C, V_X, and i_o for t ≥ 0.

SOL:



Consider Req

$$R_{eq} = \frac{12 \times 6}{18} + 8 = 12\ \Omega$$

- Time constant τ : $\tau = R_{eq}C = 12 \times \frac{1}{3} = 4\ s$
- Then: $v_C(t) = v_C(0)e^{\frac{-t}{4}} = 45e^{-0.25t}\ \text{V}$

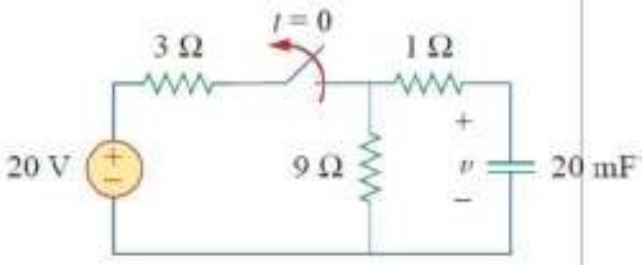
$$v_x(t) = \frac{4}{4 + 8}v_C(t) = \frac{1}{3}45e^{-0.25t} = 15e^{-0.25t}\ \text{V}$$

$$i_o(t) = \frac{v_x(t) - v_C(t)}{8} = \frac{15e^{-0.25t} - 45e^{-0.25t}}{8} = -3.75e^{-0.25t}\ \text{V}$$

Spark POLYTECHNIC

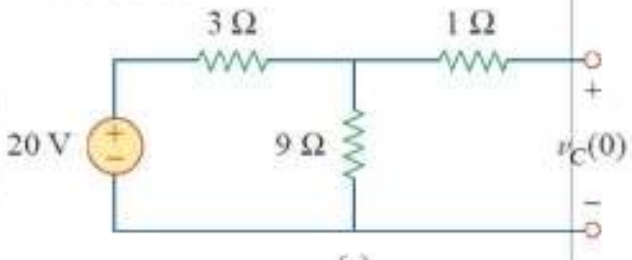
Example 7.2

Ex. 7.2: The switch in the circuit below has been closed for a long time, and it is opened at t= 0. Find v(t) for t≥0. Calculate the initial energy stored in the capacitor



Sol:

For t<0 the switch is closed; the capacitor is an open circuit to dc, as represented in (a)

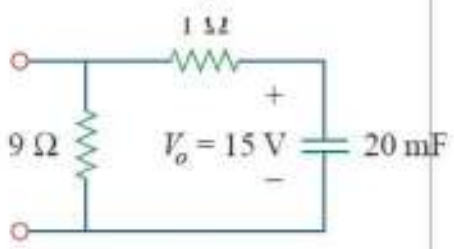


(a)

$$v_C(t) = \frac{9}{9+3}(20) = 15\,\text{V}, \qquad t < 0$$

$$v_C(0) = V_0 = 15\,\text{V}$$

For t>0 the switch is opened, and we have the RC circuit shown in (b).



(b)

$$R_{\text{eq}} = 1 + 9 = 10\,\Omega$$

$$\tau = R_{\text{eq}}C = 10 \times 20 \times 10^{-3} = 0.2\,\text{s}$$

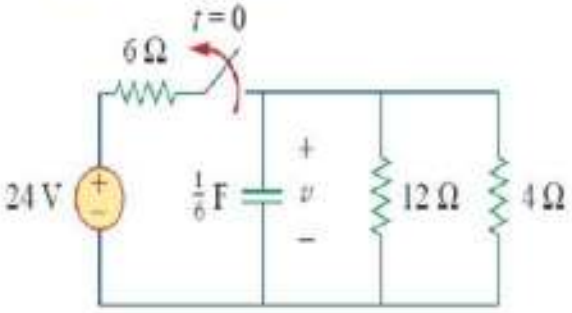
$$v(t) = v_C(0)e^{-t/\tau} = 15e^{-t/0.2}\,\text{V} = 15e^{-5t}\,\text{V}$$

$$w_C(0) = \frac{1}{2}Cv_C^2(0) = \frac{1}{2} \times 20 \times 10^{-3} \times 15^2 = 2.25\,\text{J}$$

Practice problem 7.4

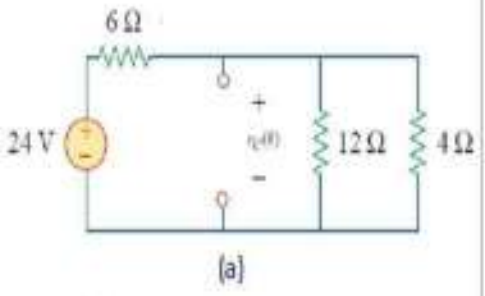
Spark POLYTECHNIC

Ex. : If the switch in Fig. below opens at $t = 0$, find $v(t)$ for $t \geq 0$ and $w_C(0)$.



Solution

- For $t<0$ the switch is closed; the capacitor is an **open circuit to dc** as shown in Fig. (a).

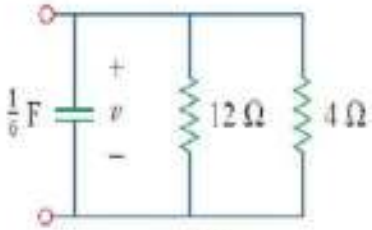


(a)

$$v_C(t) = \frac{3}{3+6}24 = 8 \text{ V} \quad for \quad t<0$$

$$v_C(0) = V_0 = 8 \text{ V}$$

- For $t>0$ the switch is opened, and we have the RC circuit shown in Fig. (b).



(b)

$$R_{eq} = \frac{12 \times 4}{16} = 3 \ \Omega$$

- Time constant τ : $\tau = R_{eq}C = 3 \times \frac{1}{6} = 0.5 \ s$

- Then: $v(t) = v_C(0)e^{\frac{-t}{0.5}} = 8e^{-2t} \text{ V}$

- The **initial energy** stored in the capacitor: $w_C(0) = \frac{1}{2}Cv_C^2(0) = \frac{1}{2} \times \frac{1}{6}(8)^2 = 5.33 \ J$

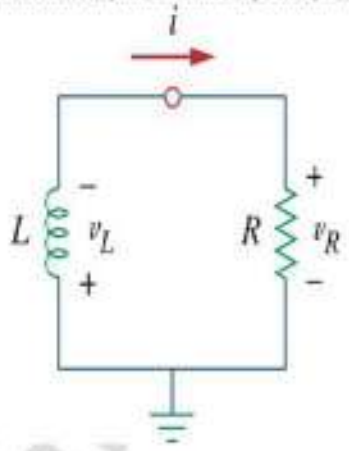
Source Free RL Circuit

- **A source free RL circuit occurs when it's dc source is suddenly disconnected**
- **The energy arready stored in the inductor is released to the resistors**

Spark POLYTECHNIC

نحصل على **source free RL** عندما نقوم بفصل مصدر التيار المستمر الخاص بها فجأة ويتم تحرير الطاقة المخزنة في الملف الى المقاومات .

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



- **Now lets consider the series connection of a resistor and inductor.**
- **In this case, the value of interest is the current through the inductor.**
- **Since the current cannot change instantaneously, we can determine its value as a function of time.**
- **Once again, we will start with an initial current passing through the inductor.**

الآن دعونا ننظر في الاتصال المتسلسل للمقاومة والمحث.

في هذه الحالة ،القيمة هي التيار عبر المحث.

نظرًا لأن التيار لا يمكن أن يتغير على الفور ، فيمكننا تحديد قيمته كدالة للوقت.

مرة أخرى ، سنبدأ بتيار أولي يمر عبر المحث.

We will take the initial current to be: $I_0 = I(0)$

Applying KVL around the loop: $V_L + V_R = 0$ **OR** $L\frac{di}{dt} + Ri = 0$

After integration: $i(t) = I_0 e^{-Rt/L}$

Once again, the natural response is an exponentially decaying current.

The time constant in this case is: $\tau = \frac{L}{R}$

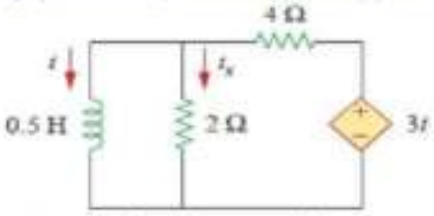
Spark POLYTECHNIC

The same principles as the RC circuit apply here.

source free RC طريقة الحل مثل

Example 7.3

Ex. 7.3: Assuming that $i(0)$ =10 A, calculate $i(t)$ and $i_x(t)$ in the circuit below.



Solution

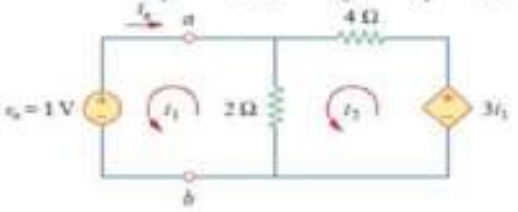
- Thevenin resistance at the inductor terminals. we insert a voltage source with v_o=1 V. Applying KVL to the two loops results

$$2(i_1 - i_2) + 1 = 0 \quad \Rightarrow \quad i_1 - i_2 = -\frac{1}{2} \quad (1)$$

$$6i_2 - 2i_1 - 3i_1 = 0 \quad \Rightarrow \quad i_2 = \frac{5}{6}i_1 \quad (2)$$

- Substituting Eq. (2) into Eq. (1) gives.

$$i_1 = -3\ \text{A}, \qquad i_o = -i_1 = 3\ \text{A}$$

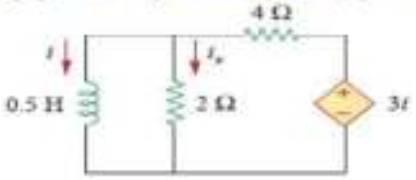


(a)



(b)

Ex. 7.3: Assuming that $i(0)$ =10 A, calculate $i(t)$ and $i_x(t)$ in the circuit below.



Solution

- Hence,

$$R_{eq} = R_{Th} = \frac{v_o}{i_o} = \frac{1}{3}\ \Omega$$

- Time constant is:

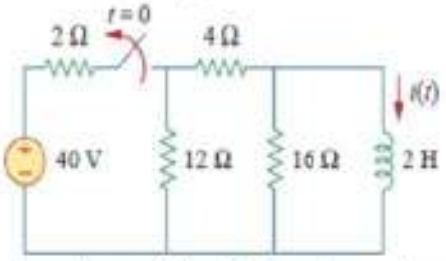
$$\tau = \frac{L}{R_{eq}} = \frac{\frac{1}{2}}{\frac{1}{3}} = \frac{3}{2}\ \text{s}$$

- Thus, the current through the inductor is: $\boxed{i(t) = i(0)e^{-t/\tau} = 10e^{-(2/3)t}\ \text{A}, \qquad t > 0}$

Spark POLYTECHNIC

Practice Problem 7.3

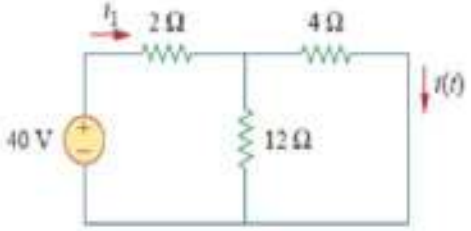
Ex. 7.4: The switch in the circuit below has been closed for a long time. At $t=0$ the switch is opened. Calculate $i(t)$ for $t>0$.



Solution

- When $t<0$ the switch is closed, and the inductor acts as a short circuit to dc,

$$\frac{4\times 12}{4+12} = 3\,\Omega \quad \Longrightarrow \quad i_1 = \frac{40}{2+3} = 8\,\text{A}$$



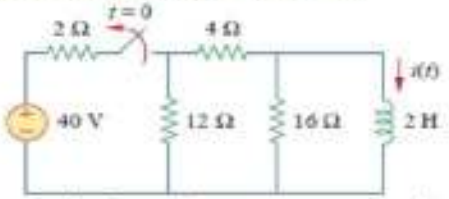
(a)

- Using current division: $i(t) = \dfrac{12}{12+4} i_1 = 6\,\text{A}, \qquad t < 0$

- Current through an inductor cannot change instantaneously,

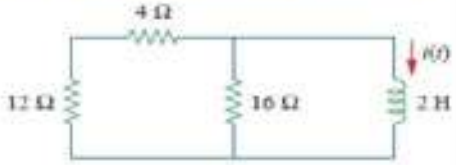
$$i(0) = i(0^-) = 6\,\text{A}$$

Ex. 7.4: The switch in the circuit below has been closed for a long time. At $t=0$ the switch is opened. Calculate $i(t)$ for $t>0$.



Solution

When $t>0$ the switch is open and the voltage source is disconnected. We now have the source-free RL circuit in Fig. (b).



(b)

$$R_{eq} = (12+4) \parallel 16 = 8\,\Omega$$

- The time constant is : $\tau = \dfrac{L}{R_{eq}} = \dfrac{2}{8} = \dfrac{1}{4}\,\text{s}$

- Thus,

$$i(t) = i(0)e^{-t/\tau} = 6e^{-4t}\,\text{A}$$

Spark POLYTECHNIC

Example 7.4

The switch in the circuit shown in Fig. 7.7 has been closed for a long time before it is opened at $t = 0$. Find

a) $i_L(t)$ for $t \geq 0$,

b) $i_o(t)$ for $t \geq 0^+$,

c) $v_o(t)$ for $t \geq 0^+$,

d) the percentage of the total energy stored in the 2 H inductor that is dissipated in the 10 Ω resistor.

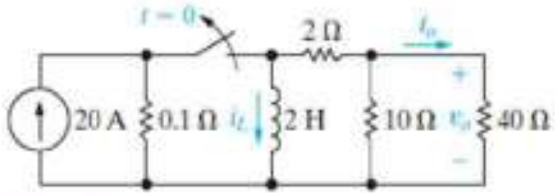


Figure 7.7 ▲ The circuit for Example 7.1.

Solution

a) The switch has been closed for a long time prior to $t = 0$, so we know the voltage across the inductor must be zero at $t = 0^-$. Therefore the initial current in the inductor is 20 A at $t = 0^-$. Hence, $i_L(0^+)$ also is 20 A, because an instantaneous change in the current cannot occur in an inductor. We replace the resistive circuit connected to the terminals of the inductor with a single resistor of 10 Ω:

$$R_{eq} = 2 + (40 \parallel 10) = 10\ \Omega.$$

The time constant of the circuit is L/R_{eq}, or 0.2 s, giving the expression for the inductor current as

$$i_L(t) = 20e^{-5t}\ \text{A}, \quad t \geq 0.$$

b) We find the current in the 40 Ω resistor most easily by using current division; that is,

$$i_o = -i_L\frac{10}{10 + 40}.$$

Note that this expression is valid for $t \geq 0^+$ because $i_o = 0$ at $t = 0^-$. The inductor behaves as a short circuit prior to the switch being opened, producing an instantaneous change in the current i_o. Then,

$$i_o(t) = -4e^{-5t}\ \text{A}, \quad t \geq 0^+.$$

c) We find the voltage v_o by direct application of Ohm's law:

$$v_o(t) = 40i_o = -160e^{-5t}\ \text{V}, \quad t \geq 0^+.$$

d) The power dissipated in the 10 Ω resistor is

$$p_{10\Omega}(t) = \frac{v_o^2}{10} = 2560e^{-10t}\ \text{W}, \quad t \geq 0^+.$$

The total energy dissipated in the 10 Ω resistor is

$$w_{10\Omega}(t) = \int_0^\infty 2560e^{-10t}\,dt = 256\ \text{J}.$$

The initial energy stored in the 2 H inductor is

$$w_L(0) = \frac{1}{2}Li^2(0) = \frac{1}{2}(2)(400) = 400\ \text{J}.$$

Therefore the percentage of energy dissipated in the 10 Ω resistor is

$$\frac{256}{400}(100) = 64\%.$$

Spark POLYTECHNIC

Example

The switch in the circuit shown in Fig. 7.13 has been in position x for a long time. At $t = 0$, the switch moves instantaneously to position y. Find

a) $v_C(t)$ for $t \geq 0$,

b) $v_o(t)$ for $t \geq 0^+$,

c) $i_o(t)$ for $t \geq 0^+$, and

d) the total energy dissipated in the 60 kΩ resistor.

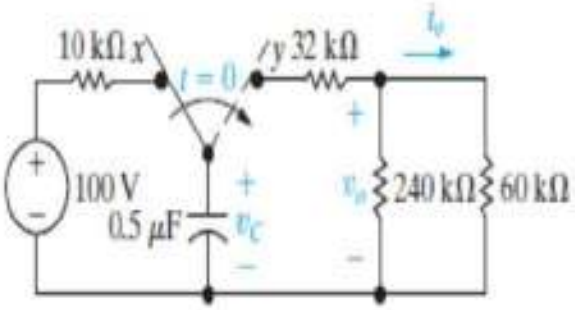


Figure 7.13 ▲ The circuit for Example 7.3.

Solution

a) Because the switch has been in position x for a long time, the 0.5 mF capacitor will charge to 100 V and be positive at the upper terminal. We can replace the resistive network connected to the capacitor at $t = 0^+$ with an equivalent resistance of 80 kΩ. Hence the time constant of the circuit is $(0.5 \times 10^{-6})(80 \times 10^3)$ or 40 ms. Then,

b) The easiest way to find $v_o(t)$ is to note that the resistive circuit forms a voltage divider across the terminals of the capacitor. Thus

$$v_o(t) = \frac{48}{80} v_C(t) = 60e^{-25t}\ \text{V}, \quad t \geq 0^+.$$

This expression for $v_o(t)$ is valid for $t \geq 0^+$ because $v_o(0^-)$ is zero. Thus we have an instantaneous change in the voltage across the 240 kΩ resistor.

c) We find the current $i_o(t)$ from Ohm's law:

$$i_o(t) = \frac{v_o(t)}{60 \times 10^3} = e^{-25t}\ \text{mA}, \quad t \geq 0^+.$$

d) The power dissipated in the 60 kΩ resistor is

$$p_{60\text{k}\Omega}(t) = i_o^2(t)(60 \times 10^3) = 60e^{-50t}\ \text{mW}, \quad t \geq 0^+.$$

The total energy dissipated is

$$w_{60\text{k}\Omega} = \int_0^\infty i_o^2(t)(60 \times 10^3)\,dt = 1.2\ \text{mJ}.$$

Spark POLYTECHNIC

Example 7.4

The switch in the circuit of Fig. 7.16 has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.

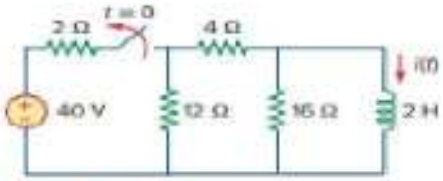


Figure 7.16
For Example 7.4.

Solution:
When $t < 0$, the switch is closed, and the inductor acts as a short circuit to dc. The 16-Ω resistor is short-circuited; the resulting circuit is shown in Fig. 7.17(a). To get i_1 in Fig. 7.17(a), we combine the 4- Ω and 12-Ω resistors in parallel to get

$$\frac{4 \times 12}{4 + 12} = 3\ \Omega$$

Hence,

$$i_1 = \frac{40}{2 + 3} = 8\ \text{A}$$

We obtain $i(t)$ from i_1 in Fig. 7.17(a) using current division, by writing

$$i(t) = \frac{12}{12 + 4} i_1 = 6\ \text{A}, \qquad t < 0$$

Since the current through an inductor cannot change instantaneously,

$$i(0) = i(0^-) = 6\ \text{A}$$

When $t > 0$, the switch is open and the voltage source is disconnected. We now have the source-free RL circuit in Fig. 7.17(b). Combining the resistors, we have

$$R_{eq} = (12 + 4) \parallel 16 = 8\ \Omega$$

The time constant is

$$\tau = \frac{L}{R_{eq}} = \frac{2}{8} = \frac{1}{4}\ \text{s}$$

Thus,

$$i(t) = i(0)e^{-t/\tau} = 6e^{-4t}\ \text{A}$$

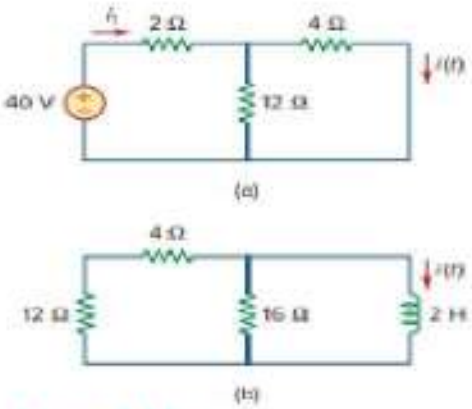


Figure 7.17
Solving the circuit of Fig. 7.16: (a) for $t < 0$, (b) for $t > 0$.

Practice Problem 7.4

For the circuit in Fig. 7.18, find $i(t)$ for $t > 0$.

Answer: $2e^{-2t}$ A, $t > 0$.

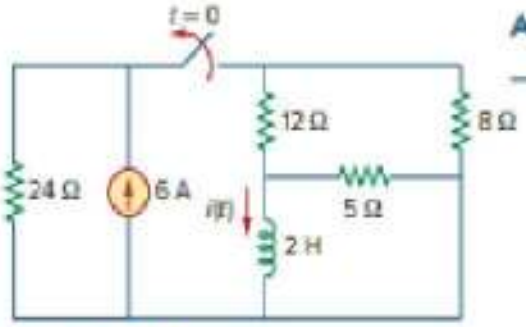


Figure 7.18
For Practice Prob. 7.4.

Spark POLYTECHNIC

Singularity Functions

- **Before we consider the response of a circuit to an external voltage, we need to cover some important mathematical functions.**
- **Singularity functions serve as good approximations to switching on or off a voltage.**
- **The three most common singularity functions are the unit step, unit impulse, and unit ramp.**

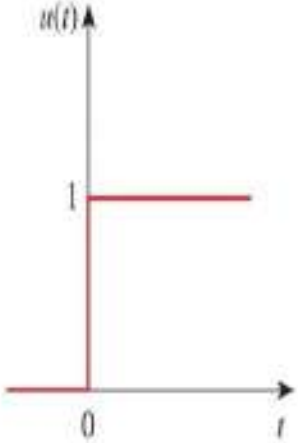
قبل أن نفكر في استجابة الدائرة لجهد خارجي ، نحتاج إلى تغطية بعض الوظائف الرياضية المهمة.

تعمل وظائف التفرد كتقريب جيد لتشغيل أو إيقاف تشغيل الجهد.

وظائف التفرد الثلاثة الأكثر شيوعًا هي خطوة الوحدة ونبضة الوحدة ومنحدر الوحدة.

The Unit Step

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



- **A step function is one that maintains a constant value before a certain time and then changes to another constant afterwards.**
- **The prototypical form is zero before t=0 and one afterwards.**
- **See the graph for an illustration.**

وظيفة الخطوة هي الوظيفة التي تحافظ على قيمة ثابتة قبل وقت معين ثم تتغير إلى ثابت آخر بعد ذلك.

الصيغة النموذجية هي صفر قبل **t = 0** وواحدة بعد ذلك.

انظر الرسم البياني للحصول على توضيح.

Spark POLYTECHNIC

Mathematically, the unit step is expressed as:

$$u(t)=\begin{cases}0, & t<0\\ 1, & t>0\end{cases}$$

The switching time may be shifted to t=t0 by:

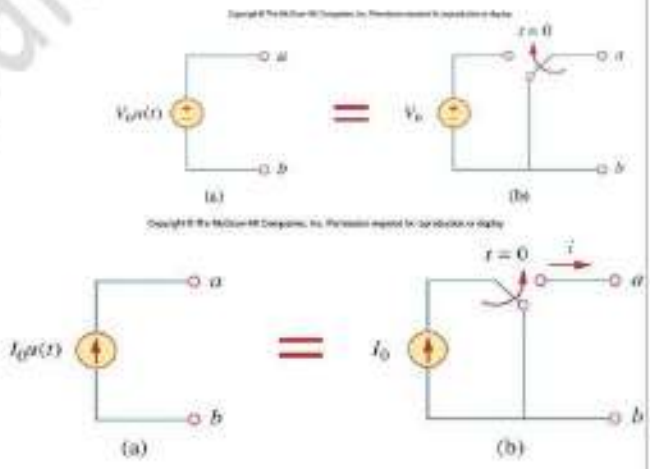
$$u(t-t_0)=\begin{cases}0, & t<t_0\\ 1, & t>t_0\end{cases}$$

Note that this results in a delay in the switch.

The unit step function is written as u(t)

Equivalent Circuit

- **The unit step function has an equivalent circuit to represent when it is used to switch on a soure.**
- **The equivalent circuits for a voltage and current source are shown.**



تحتوي وظيفة خطوة الوحدة على دائرة مكافئة لتمثيل وقت استخدامها لتشغيل مصدر.

تظهر الدوائر المكافئة للجهد والمصدر التيار

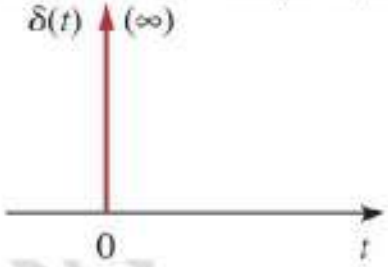
Spark POLYTECHNIC

The Unit Impulse Function

- **The derivative of the unit step function is the unit impulse function.**
- **This is expressed as:**

$$\delta(t)=\begin{cases}0 & t<0\\ \text{Undefined} & t=0\\ 0 & t>0\end{cases}$$

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



- **Voltages of this form can occur during switching operations.**

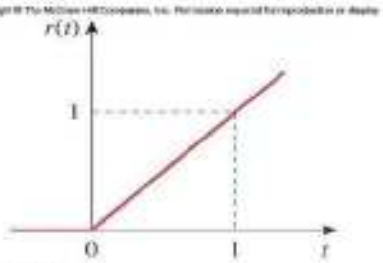
مشتق دالة خطوة الوحدة هو دالة نبضة الوحدة

يمكن أن تحدث الفولتية من هذا النموذج أثناء عمليات التحويل.

The Unit Ramp Function

- **Integration of the unit step function results in the unit ramp function:**

$$r(t)=\begin{cases}0, & t\le 0\\ t, & t\ge 0\end{cases}$$



- **Much like the other functions, the onset of the ramp may be adjusted**

ينتج عن تكامل وظيفة خطوة الوحدة وظيفة منحدر الوحدة:

مثل الكثير من الوظائف الأخرى ، يمكن تعديل بداية المنحدر

lahlaadsalam.aw

Spark POLYTECHNIC

Step Response of an RC circuit :

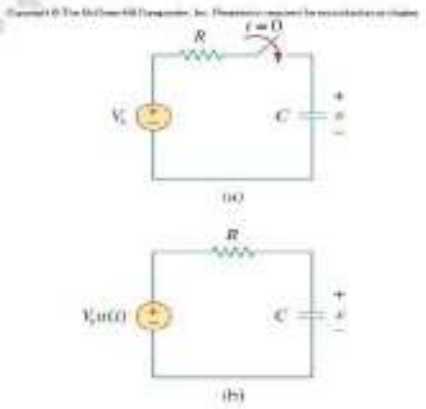
The step Response is the response of the circuit due to a sudden application a dc voltage or current source

استجابة الخطوة هي استجابة الدارة بسبب التطبيق المفاجئ لجهد التيار المستمر او مصدر التيار

Since the voltage of capacitor cannot change instantaneously .

لان جهد المكثف لا يمكن ان يتغير على الفور (او بشكل مفاجئ)

$V(t)=V(0^+)=V(0^-)=V(0)$



- **When a DC source is suddenly applied to a RC circuit, the source can be modeled as a step function.**
- **The circuit response is known as the step response.**
- **Let's consider the circuit shown here.**
- **We can find the voltage on the capacitor as a function of time.**

- عندما يتم تطبيق مصدر التيار المستمر فجأة على دائرة **RC** ، يمكن نمذجة المصدر كوظيفة متدرجة.
- تُعرف استجابة الدائرة باسم استجابة الخطوة.
- دعونا ننظر في الدائرة الموضحة هنا.
- يمكننا إيجاد الجهد على المكثف كدالة زمنية.

We assume an initial voltage of V_0 on the capacitor.

Applying KCL:

$$\frac{dv}{dt}+\frac{v}{RC}=\frac{V_s}{RC}u(t)$$

laith abuSalah awad

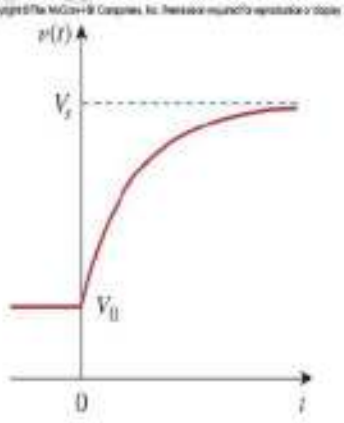
For t>0 this becomes:

$$\frac{dv}{dt}+\frac{v}{RC}=\frac{V_s}{RC}$$

Integrating both sides and introducing initial conditions finally yields:

$$v(t)=\begin{cases} V_0, & t<0 \\ V_s+(V_0-V_s)e^{-t/\tau} & t>0 \end{cases}$$

- **This is known as the complete response, or total response.**
- **We can consider the response to be broken into two separate responses:**
- **The natural response of the capacitor or inductor due to the energy stored in it.**
- **The second part is the forced response**



- يُعرف هذا بالاستجابة الكاملة .
- يمكننا اعتبار الرد مقسمًا إلى ردين منفصلين:
- الاستجابة الطبيعية للمكثف أو المحث بسبب الطاقة المخزنة فيه.
- الجزء الثاني هو الرد القسري

Forced Response

The complete response can be written as:

$$v=v_n+v_f$$

Where the nature response is:

$$v_n=V_0e^{-t/\tau}$$

faith abdulsalam awwad

And the forced response is:

$$v_f = V_s\left(1-e^{-t/\tau}\right)$$

Note that the eventual response of the circuit is to reach Vs after the natural response decays to zero.

لاحظ أن الاستجابة النهائية للدائرة هي الوصول إلى **Vs** بعد أن تتحلل الاستجابة الطبيعية إلى الصفر

Another way to look at the response is to break it up into the transient response and the steady state response:

$$v = v_t + v_{ss}$$

Where the transient is:

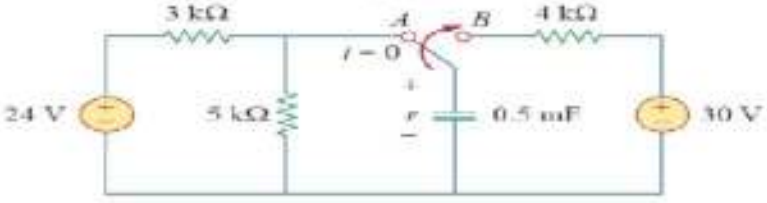
$$v_t = \left(V_0 - V_s\right)e^{-t/\tau}$$

And the steady state is:

$$v_{ss} = V_s$$

Example 7.10

Ex. 7.10: The switch in Fig. Below has been in position A for a long time. At $t=0$ the switch moves to B. Determine $v(t)$ for $t>0$ and calculate its value at $t=1$ s and 4 s.



Solution

- Voltage across the capacitor just before $t=0$. Capacitor is open circuit under dc conditions:

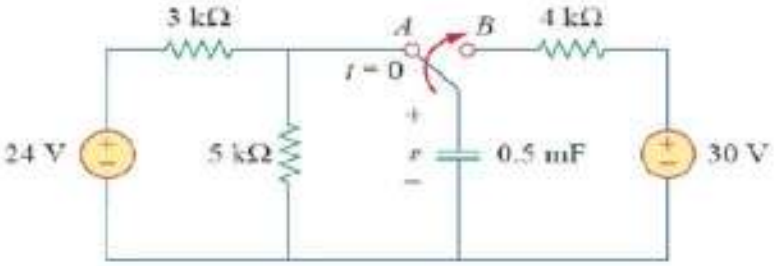
$$v(0^-) = \frac{5}{5+3}(24) = 15\ \text{V}$$

- Capacitor voltage cannot change instantaneously: $v(0) = v(0^-) = v(0^+) = 15\ \text{V}$
- For $t>0$ (switch to B). Thevenin Resistance connected to the capacitor: $R_{\text{Th}} = 4\ \text{k}\Omega$
- Time constant: $\tau = R_{\text{Th}}C = 4\times10^3 \times 0.5\times10^{-3} = 2\ \text{s}$

Spel POLYTECHNIC

dr abdulsalam alwad

Ex. 7.10: The switch in Fig. below has been in position A for a long time. At $t=0$ the switch moves to B. Determine $v(t)$ for $t>0$ and calculate its value at t 1 s and 4 s.



Solution

- Since the capacitor acts like an open circuit to dc at steady state, $v(\infty) = 30$ V. Thus,

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$
$$= 30 + (15 - 30)e^{-t/2} = (30 - 15e^{-0.5t})\ \text{V}$$

At $t = 1$,

$$v(1) = 30 - 15e^{-0.5} = 20.9\ \text{V}$$

At $t = 4$,

$$v(4) = 30 - 15e^{-2} = 27.97\ \text{V}$$

Practice problem 7.10

Find V(t) for t > 0 in the circuit of Fig. 7.44. Assume the switch has been open for a long time and is closed at t = 0. Calculate V(t) at t = 0.5.

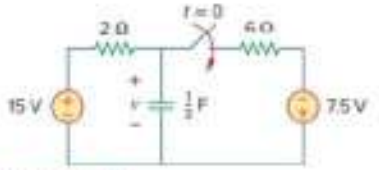


Figure 7.44
For Practice Prob. 7.10.

Answer: (9.375 + 5.625e^{-2t}) V for all t > 0, 11.444 V

Practice Problem 7.11

The switch in Fig. 7.47 is closed at $t = 0$. Find $i(t)$ and $v(t)$ for all time. Note that $u(-t) = 1$ for $t < 0$ and 0 for $t > 0$. Also, $u(-t) = 1 - u(t)$.

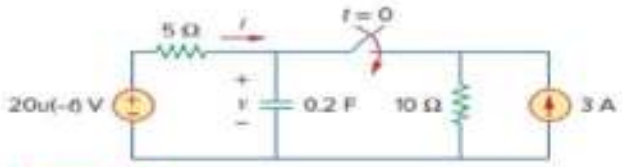


Figure 7.47
For Practice Prob. 7.11.

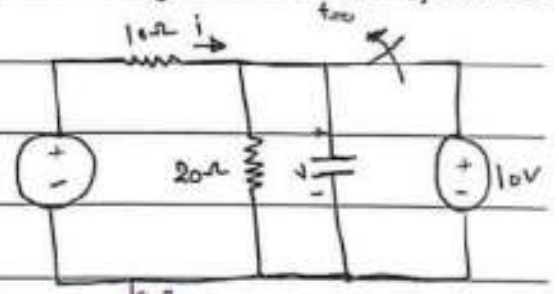
Answer: $i(t) = \begin{cases} 0, & t < 0 \\ -2(\quad + e^{-1.5t}) \text{ A}, & t > 0, \end{cases}$

$v = \begin{cases} 20 \text{ V}, & t < 0 \\ 10(1 + e^{-1.5t}) \text{ V}, & t > 0 \end{cases}$

Spear POLYTECHNIC

Example 7.11

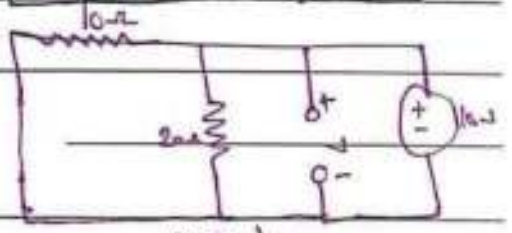
In Fig. 7.45, the switch has been closed for a long time and is opened at t=0, find i and v for all time



$$30u(-t) = \begin{cases} 0, & t<0 \\ 30, & t>0 \end{cases}$$

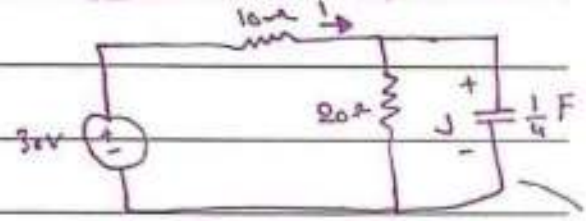
$v = 10i$, $i = \frac{v}{10} = -1A$ $t<0$

$v(0) = v(0^-) = 10V$



$v(\infty) = \frac{20}{20+10}(30) = 20V$

$R_{th} = 10 \| 20 = \frac{10 \times 20}{30} = \frac{20}{3}$



$\tau = R_{th} C = \frac{20}{3} \cdot \frac{1}{4} = \frac{5}{3}s$, $v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$

$= 20 - 10e^{-0.6t}$ V

$\because 10i = 30$

$v = \begin{cases} 10V & t<0 \\ 20 - 10e^{-0.6t} & t>0 \end{cases}$ $\quad i = \begin{cases} -1A & t<0 \\ 1 + e^{-0.6t} A & t>0 \end{cases}$

Step Response of an RL circuit

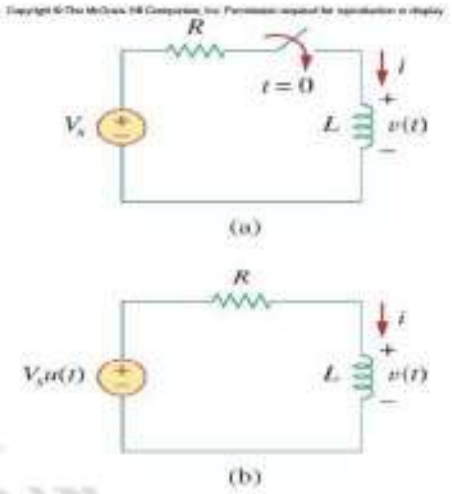
The step Response is the response of the circuit due to sudden

Application of a dc voltage or current source

Since the current of an inductor cannot change instantaneously

Spark POLYTECHNIC

$i(0)=i(0^+)=i(0^-)=i_0$



- Now we can look at the step response of a RL circuit.
- We will use the transient and steady state response approach.
- We know that the transient response will be an exponential:

$$i_t = Ae^{-t/\tau}$$

- After a sufficiently long time, the current will reach he steady state:

$$i_{ss} = \frac{V_s}{R}$$

- This yields an overall response of:

$$i = Ae^{-t/\tau} + \frac{V_s}{R}$$

- To determine the value of A we need to keep in mind that the current cannot change instantaneously.

$$i\left(0^+\right) = i\left(0^-\right) = I_0$$

- Thus we can use the t=0 time to establish A

$$A = I_0 - \frac{V_s}{R}$$

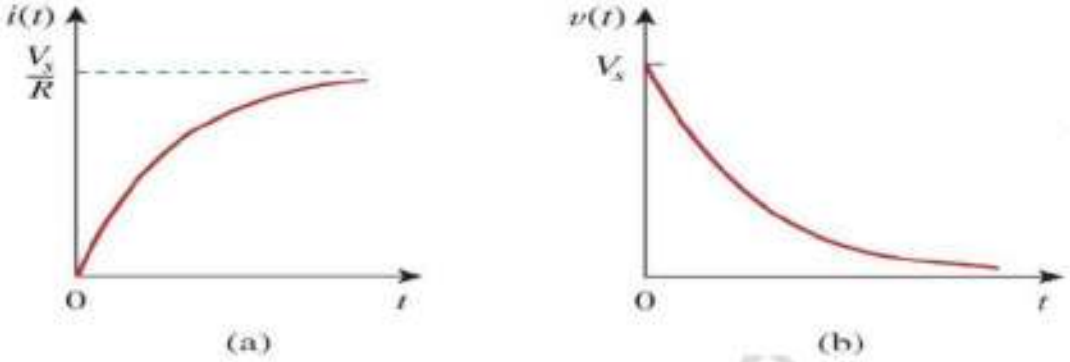
- The complete response of the circuit is thus:

fatih abdulsalam al...

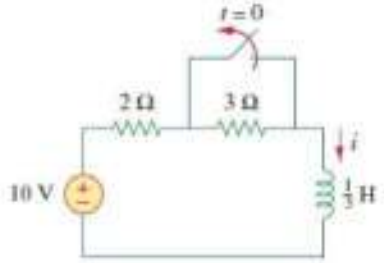
$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-t/\tau}$$

❖ **Without an initial current, the circuit response is shown here.**

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



Example 7.12 Find i(t) in the circuit of Fig for t > 0. Assume that the switch has been closed for a long time.



When $t < 0$, the 3-Ω resistor is short-circuited, and the inductor acts like a short circuit. The current through the inductor at $t = 0^-$ (i.e., just before $t = 0$) is

$$i(0^-) = \frac{10}{2} = 5\ \text{A}$$

Since the inductor current cannot change instantaneously,

$$i(0) = i(0^+) = i(0^-) = 5\ \text{A}$$

When $t > 0$, the switch is open. The 2-Ω and 3-Ω resistors are in series, so that

$$i(\infty) = \frac{10}{2+3} = 2\ \text{A}$$

The Thevenin resistance across the inductor terminals is

$$R_{\text{Th}} = 2 + 3 = 5\ \Omega$$

For the time constant,

$$\tau = \frac{L}{R_{\text{Th}}} = \frac{\frac{1}{3}}{5} = \frac{1}{15}\ \text{s}$$

Thus,

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$
$$= 2 + (5-2)e^{-15t} = 2 + 3e^{-15t}\ \text{A}, \qquad t > 0$$

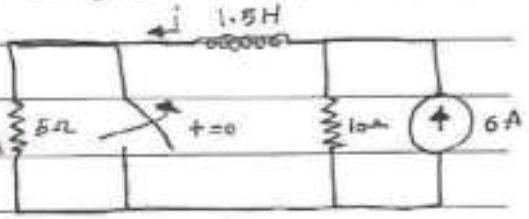
Check: In Fig., for $t > 0$, KVL must be satisfied; that is,

$$10 = 5i + L\frac{di}{dt}$$

$$5i + L\frac{di}{dt} = [10 + 15e^{-15t}] + \left[\frac{1}{3}(3)(-15)e^{-15t}\right] = 10$$

Practice Problem

The switch in Fig. 7.52 has been closed for a long time. It opens at t=0. Find i(t) for t>0



Sol:- when t<0 the 5Ω short circuit and the inductor short circuit 10Ω short circuit

$i = 6A \quad i(0) = i(0^-) = i(0^+) = 6A$

when $t>0$

Current division $\quad i(t) = i(\infty) + (i(0) - i(\infty))e^{-t/\tau} = 4 + 2e^{-10t}\ A$

$i(\infty) = \left(\frac{10}{5+10}\right)(6) = 4A \quad R_{eq} = 5+10 = 15\Omega$

$\tau = \frac{L}{R} = \frac{1.5}{15} = \frac{1}{10}s$

Example

The switch in the circuit shown in Fig. 7.19 has been in position a for a long time. At $t = 0$, the switch moves from position a to position b. The switch is a make-before-break type; that is, the connection at position b is established before the connection at position a is broken, so there is no interruption of current through the inductor.

a) Find the expression for $i(t)$ for $t \geq 0$.

b) What is the initial voltage across the inductor just after the switch has been moved to position b?

c) How many milliseconds after the switch has been moved does the inductor voltage equal 24 V?

d) Does this initial voltage make sense in terms of circuit behavior?

e) Plot both $i(t)$ and $v(t)$ versus t.

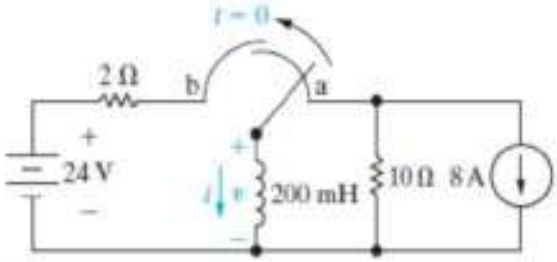


Figure 7.19 ▲ The circuit for Example 7.5.

Solution

a) The switch has been in position a for a long time, so the 200 mH inductor is a short circuit across the 8 A current source. Therefore, the inductor carries an initial current of 8 A. This current is oriented opposite to the reference direction for i; thus I_0 is −8 A. When the switch is in position b, the final value of i will be 24/2, or 12 A. The time constant of the circuit is 200/2, or 100 ms. Substituting these values into Eq. 7.35 gives

b) The voltage across the inductor is

$$v = L\frac{di}{dt}$$
$$= 0.2(200e^{-10t})$$
$$= 40e^{-10t}\ \text{V}, \quad t \geq 0^+.$$

The initial inductor voltage is

$$v(0^+) = 40\ \text{V}.$$

c) Yes; in the instant after the switch has been moved to position b, the inductor sustains a current of 8 A counterclockwise around the newly formed closed path. This current causes a 16 V drop across the 2 Ω resistor. This voltage drop adds to the drop across the source, producing a 40 V drop across the inductor.

d) We find the time at which the inductor voltage equals 24 V by solving the expression

$$24 = 40e^{-10t}$$

for t:

$$t = \frac{1}{10}\ln\frac{40}{24}$$

e) Figure 7.20 shows the graphs of $i(t)$ and $v(t)$ versus t. Note that the instant of time when the current equals zero corresponds to the instant of time when the inductor voltage equals the source voltage of 24 V, as predicted by Kirchhoff's voltage law.

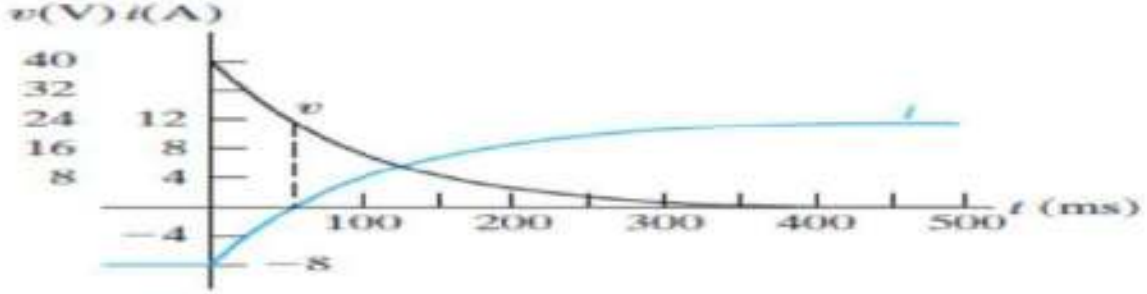
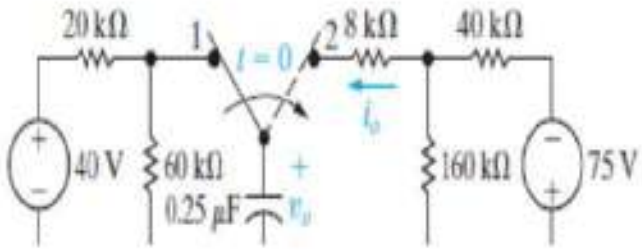


Figure 7.20 ▲ The current and voltage waveforms for Example 7.5.

Example

The switch in the circuit shown in Fig. 7.22 has been in position 1 for a long time. At $t = 0$, the switch moves to position 2. Find



Solution

a) The switch has been in position 1 for a long time, so the initial value of v_o is 40(60/80), or 30 V. To take advantage of Eqs. 7.51 and 7.53, we find the Norton equivalent with respect to the terminals of the capacitor for $t \geq 0$. To do this, we begin by computing the open-circuit voltage, which is given by the −75 V source divided across the 40 kΩ and 160 kΩ resistors:

$$V_{oc} = \frac{160 \times 10^3}{(40 + 160) \times 10^3}(-75) = -60 \text{ V}.$$

Next, we calculate the Thévenin resistance, as seen to the right of the capacitor, by shorting the −75 V source and making series and parallel combinations of the resistors:

$$R_{Th} = 8000 + 40{,}000 \,\|\, 160{,}000 = 40 \text{ k}\Omega$$

The value of the Norton current source is the ratio of the open-circuit voltage to the Thévenin resistance, or $-60/(40 \times 10^3) = -1.5$ mA. The resulting Norton equivalent circuit is shown in Fig. 7.23. From Fig. 7.23, $I_sR = -60$ V and $RC = 10$ ms. We have already noted that $v_o(0) = 30$ V, so the solution for v_o is

$$v_o = -60 + [30 - (-60)]e^{-100t}$$
$$= -60 + 90e^{-100t} \text{ V}, \quad t \geq 0.$$

Spark POLYTECHNIC

b) We write the solution for i_o directly from Eq. 7.53 by noting that $I_s = -1.5$ mA and $V_o/R = (30/40) \times 10^{-3}$, or 0.75 mA:

$$i_o = -2.25e^{-100t}\ \text{mA}, \quad t \geq 0^+.$$

We check the consistency of the solutions for v_o and i_o by noting that

$$i_o = C\frac{dv_o}{dt} = (0.25 \times 10^{-6})(-9000e^{-100t})$$
$$= -2.25e^{-100t}\ \text{mA}.$$

Because $dv_o(0^-)/dt = 0$, the expression for i_o clearly is valid only for $t \geq 0^+$.

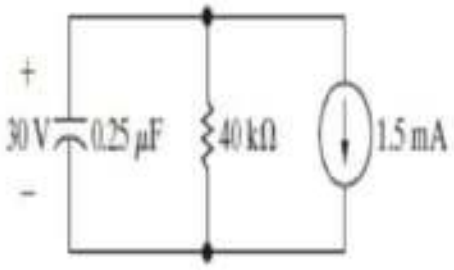
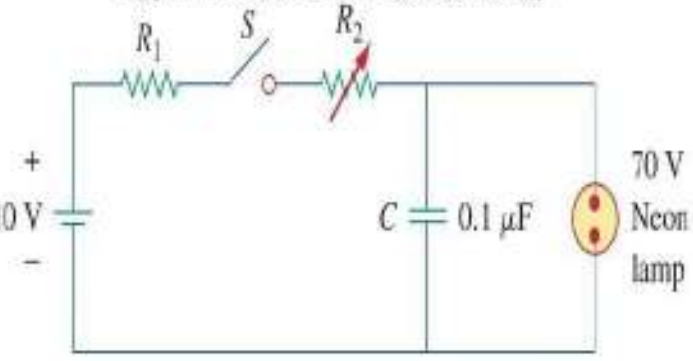


Figure 7.23 ▲ The equivalent circuit for $t > 0$ for the circuit shown in Fig. 7.22.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



Application

Delay Circuit دارة التأخير

- **The RC circuit can be used to delay the turn on of a connected device.**
- **For example, a neon lamp which only triggers when a voltage exceeds a specific value can be delayed using such a circuit.**

يمكن استخدام دائرة **RC** لتأخير تشغيل جهاز متصل.

على سبيل المثال ، يمكن تأخير مصباح النيون الذي يتم تشغيله فقط عندما يتجاوز الجهد قيمة معينة باستخدام مثل هذه الدائرة.

- **When the switch is closed, the capacitor charges.**
- **The voltage will rise at a rate determined by:**

$$\tau = (R_1 + R_2)C$$

Spar POLYTECHNIC

- **Once the voltage reaches 70 volts, the lamp triggers.**
- **Once on, the lamp has low resistance and discharges the capacitor.**
- **This shuts off the capacitor and starts the cycle over again.**

عندما يتم إغلاق المفتاح ، يتم شحن المكثف.

سيرتفع الجهد بمعدل يحدده:

بمجرد وصول الجهد إلى 70 فولت ، يتم تشغيل المصباح.

بمجرد تشغيله ، يكون للمصباح مقاومة منخفضة ويقوم بتفريغ المكثف.

هذا يغلق المكثف ويبدأ الدورة مرة أخرى.

هيك بكون خلص شرح شابتر ملاحظة بسيطة الشابتر هذا جميع الافكار فيه تعتمد على المادة السابقة يعتمد اكثر اشي على **ch6 , ch4 .ch3**

لتمكن اكثر وا اكثر من المادة عليك في اسئلة الكتاب واسئلة السنوات السابقة

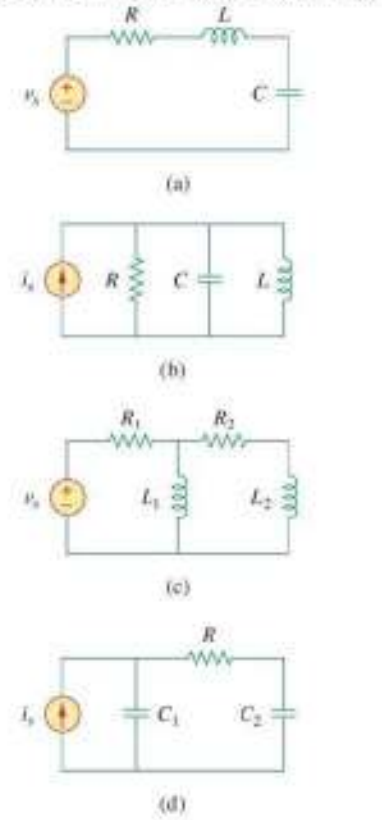
جميع الامثلة لقد طرحت في هذا الملخص اتمنى ان اكون قد طرحتها بشكل جيد

بالتوفيق للجميع يارب

laith abdulsalam awwad

Chapter 8 Second Order Circuits

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



- **The previous chapter considered circuits which only required first order differential equations to solve.**
- **However, when more than one "storage element", i.e. capacitor or inductor is present, the equations require second order differential equations**
- **The analysis is similar to what was done with first order circuits**
- **This time, though we will only consider DC independent sources**

نظرا الشابتر السابق في الدوائر التي تتطلب فقط معادلات تفاضلية من الدرجة الأولى لحلها.

ومع ذلك ، عند وجود أكثر من "عنصر تخزين" ، اي مكثف أو محث ، تتطلب المعادلات معادلات تفاضلية من الدرجة الثانية

يشبه التحليل ما تم إجراؤه باستخدام دوائر من الدرجة الأولى

هذه المرة ، على الرغم من أننا سننظر فقط في المصادر تيار المباشر المستقلة

Finding Initial and Final Values البحث عن القيم الأولية والنهائية

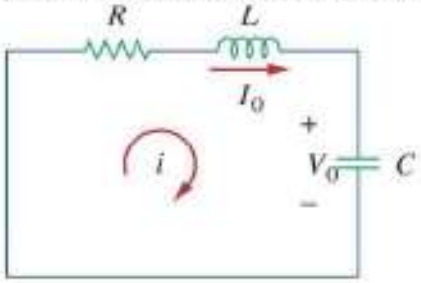
- **Working on second order system is harder than first order in terms of finding initial and final conditions.**
- **You need to know the derivatives, dv/dt and di/dt as well.**

Spark POLYTECHNIC

- ❖ **Getting the polarity across a capacitor and the direction of current through an inductor is critical.**
- ❖ **Capacitor voltage and inductor current are always continuous.**
- ❖ العمل على نظام الدرجة الثانية أصعب من الدرجة الأولى من حيث إيجاد الشروط الأولية والنهائية.
- ❖ تحتاج إلى معرفة المشتقات ، **dv/dt** و **di/dt** أيضًا.
- ❖ يعد الحصول على القطبية عبر مكثف واتجاه التيار عبر محث أمرًا بالغ الأهمية.
- ❖ جهد المكثف وتيار المحث مستمران دائمًا.

Source Free Series RLC

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



- ❖ **Consider the circuit shown.**
- ❖ **The energy at t=0 is stored in the capacitor, represented by V_0 and in the inductor, represented by I_0.**

$$v(0) = \frac{1}{C}\int_{-\infty}^{0} i\,dt = V_0$$

$$i(0) = I_0$$

- ❖ **Applying KVL around the loop:**

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} i(\tau)\,d\tau = 0$$

- ❖ **The integral can be eliminated by differentiation:**

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$$

Spark POLYTECHNIC

- **Here you can see the second order equation that results** هنا يمكنك أن ترى معادلة الدرجة الثانية التي تنتج
- **Two initial conditions are needed for solving this problem.**
- **The initial current is given.**
- **The first derivative of the current can also be had:**

$$Ri(0)+L\frac{di(0)}{dt}+V_0=0$$

- **Or**

$$\frac{di(0)}{dt}=-\frac{1}{L}(RI_0+V_0)$$

- **Based on the first order solutions, we can expect that the solution will be in exponential form.**
- **The equation will then be:**

$$Ae^{st}\left(s^2+\frac{R}{L}s+\frac{1}{LC}\right)=0$$

- **For which the solutions are:**

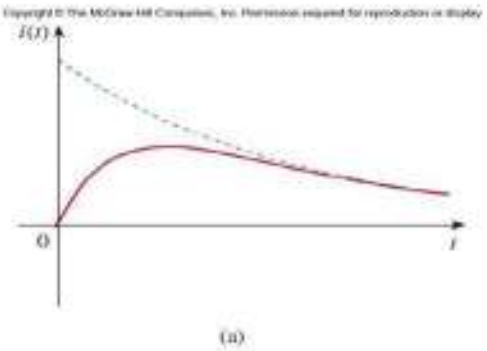
$$s_1=-\alpha+\sqrt{\alpha^2-\omega_o^2}\qquad s_2=-\alpha-\sqrt{\alpha^2-\omega_o^2}$$

$$\alpha=\frac{R}{2L}\qquad \omega_0=\frac{1}{\sqrt{LC}}$$

وهنا يوجد لدينا ثلاث حالات

Spark POLYTECHNIC

1-Over damped ($\alpha>\omega_0$)



(a)

- When $\alpha>\omega_0$, the system is over damped
- In this case, both s1 and s2 are real and negative.
- The response of the system is:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

- From this, we should not expect to see an oscillation

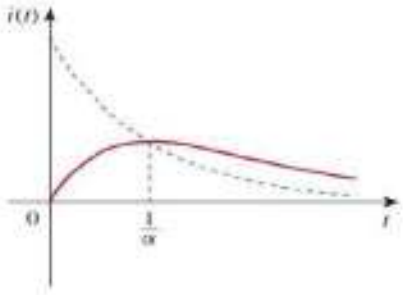
over damped عندما تكون $\alpha > \omega_0$ تسمى هذه الحالة ب

في هذه الحالة ، كلا **s1** و **s2** حقيقي وسلبي.

وتكون استجابة النظام هي:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

2-Critically Damped ($\alpha=\omega_0$)



(b)

- **When α=ω0, the system is critically damped.**
- **The differential equation becomes:**

$$\frac{d}{dt}\left(e^{\alpha t} i\right) = A_1$$

- **For which the solution is:**

$$i(t) = \left(A_2 + A_1 t\right) e^{-\alpha t}$$

Spark POLYTECHNIC

- **There are two components to the response, an exponential decay and an exponential decay multiplied by a linear term**

عندما تكون $\alpha = \omega_0$ ، يكون النظام مثبطًا بشكل خطير.

تصبح المعادلة التفاضلية:

$$\frac{d}{dt}\left(e^{\alpha t} i\right) = A_1$$

من أجلها يكون الحل:

$$i(t) = \left(A_2 + A_1 t\right) e^{-\alpha t}$$

هناك مكونان للاستجابة ، الاضمحلال الأسي والانحلال الأسي مضروبًا في مصطلح خطي

3-Underdamped ($\alpha < \omega_0$)

- **When $\alpha < \omega_0$, the system is considered to be underdamped**
- **In this case, the solution will be:**

$$i(t) = e^{-\alpha t}\left(B_1 \cos \omega_d t + B_2 \sin \omega_d t\right)$$

- **Where** $j = \sqrt{-1}$ **and** $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
- **ω_0 is often called the undamped natural frequency**
- **ωd is called the damped natural frequency**

عندما تكون $\alpha < \omega_0$ ، يعتبر النظام ناقص التخميد

في هذه الحالة يكون الحل هو:

$$i(t) = e^{-\alpha t}\left(B_1 \cos \omega_d t + B_2 \sin \omega_d t\right)$$

Spark POLYTECHNIC

غالبًا ما يسمى ω_0 بالتردد الطبيعي غير المخمد

ω_d يسمى التردد الطبيعي المخمد

Damping and RLC networks

RLC networks can be charaterized by the following:

- **The behavior of these networks is captured by the idea of damping**
- **Oscillatory response is possible due to the presence of two types of energy storage elements.**
- **It is typically difficult to tell the difference between damped and critically damped responses.**

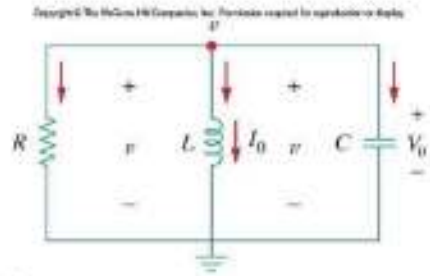
يمكن تفصيل شبكات **RLC** من خلال ما يلي:

يتم التقاط سلوك هذه الشبكات من خلال فكرة التخميد

الاستجابة التذبذبية ممكنة بسبب وجود نوعين من عناصر تخزين الطاقة.

عادة ما يكون من الصعب التمييز بين الاستجابات المخففة والمخففة بشكل حاسم.

Source Free Parallel RLC



- **Now let us look at parallel forms of RLC networks**
- **Consider the circuit shown**
- **Assume the initial current and voltage to be:**

$$i(0) = I_0 = \frac{1}{L}\int_{-\infty}^{0} v(t)\,dt$$

$$v(0) = V_0$$

Spark POLYTECHNIC

❖ **Applying KCL to the top node we get:**

$$\frac{v}{R}+\frac{1}{L}\int_{-\infty}^{t} v(\tau)d\tau + C\frac{dv}{dt}=0$$

❖ **Taking the derivative with respect to t gives:**

$$\frac{d^2v}{dt^2}+\frac{1}{RC}\frac{dv}{dt}+\frac{1}{LC}v=0$$

❖ **The characteristic equation for this is:**

$$s^2+\frac{1}{RC}s+\frac{1}{LC}=0$$

❖ **From this, we can find the roots of the characteristic equation to be:**

$$s_{1,2}=-\alpha\pm\sqrt{\alpha^2-\omega_0^2}$$

$$\alpha=\frac{1}{2RC}\quad \omega_0=\frac{1}{\sqrt{LC}}$$

❖ **As in last time, there are three scenarios to consider.**

1- <u>damping</u>

❖ **For the <u>overdamped ($\alpha>\omega_0$)</u> case, the roots are real and negetive, so the response is:**

$$v(t)=A_1e^{s_1t}+A_2e^{s_2t}$$

ibtihaj abdulsalam awwad

- For <u>critically damped ($\alpha=\omega_0$)</u>, the roots are real and equal, so the response is:

$$v(t)=(A_2+A_1t)e^{-\alpha t}$$

2-<u>Underdamped ($\alpha<\omega_0$)</u>

- In the underdamped case, the roots are complex and so the response will be:

$$v(t)=e^{-\alpha t}\left(A_1\cos\omega_d t+A_2\sin\omega_d t\right)$$

- To get the values for the constants, we need to know v(0) and dv(0)/dt.
- To find the second term, we use:

$$\frac{V_0}{R}+I_0+C\frac{dv(0)}{dt}=0$$

- The voltage waveforms will be similar to those shown for the series network.
- Note that in the series network, we first found the inductor current and then solved for the rest from that.
- Here we start with the capacitor voltage and similarly, solve for the other variables from that.

ستكون أشكال موجة الجهد مماثلة لتلك المعروضة لشبكة السلسلة.

لاحظ أنه في شبكة السلسلة ، وجدنا أولاً تيار المحرّض ثم حللنا الباقي من ذلك.

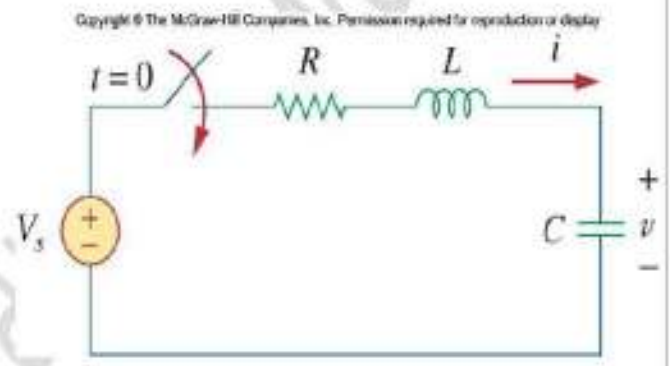
هنا نبدأ بجهد المكثف وبالمثل نحل المتغيرات الأخرى من ذلك.

Spark POLYTECHNIC

Step Response of a Series RLC Circuit

- **Now let us consider what happens when a DC voltage is suddenly applied to a second order circuit.** الآن دعونا نفكر في ما يحدث عندما يتم تطبيق جهد التيار المستمر فجأة على دائرة من الدرجة الثانية.
- **Consider the circuit shown. The switch closes at t=0.**
- **Applying KVL around the loop for t>0:**

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display



$$L\frac{di}{dt} + Ri + v = V_s$$

- **But**

$$i = C\frac{dv}{dt}$$

- **Substituting for i gives:**

$$\frac{d^2v}{dt^2} + \frac{R}{L}\frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}$$

- **This is similar to the response for the source free version of the series circuit, except the variable is different.**
- **The solution to this equation is a combination of transient response and steady state**

$$v(t) = v_t(t) + v_{ss}(t)$$

Spark POLYTECHNIC

- **The transient response is in the same form as the solutions for the source free version.**
- **The steady state response is the final value of v(t). In this case, the capacitor voltage will equal the source voltage.**

تكون الاستجابة العابرة بنفس شكل حلول **source free version**

استجابة الحالة المستقرة هي القيمة النهائية لـ **v (t)**. في هذه الحالة ، سيساوي جهد المكثف جهد المصدر .

- **The complete solutions for the three conditions of damping are:**

$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{(Overdamped)}$$

$$v(t) = V_s + (A_1 + A_2) e^{-\alpha t} \quad \text{(Critically Damped)}$$

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) \quad \text{(Underdamped)}$$

- **The variables A1 and A2 are obtained from the initial conditions, v(0) and dv(0)/dt.**

يتم الحصول على المتغيرات **A1** و **A2** من الشروط الأولية ، **v** (0) و **dv (0) / dt**.

Step Response of a Parallel RLC Circuit

- **The same treatment given to the parallel RLC circuit yields the same result.**

نفس المعاملة المعطاة لدائرة **RLC** المتوازية تؤدي إلى نفس النتيجة.

- **The response is a combination of transient and steady state responses:**

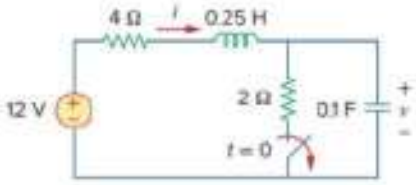
Spark POLYTECHNIC

$$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{(Overdamped)}$$
$$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t} \quad \text{(Critally Damped)}$$
$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad \text{(Underdamped)}$$

❖ **Here the variables A1 and A2 are obtained from the initial conditions, i(0) and di(0)/dt.**

هنا يتم الحصول على المتغيرين A1 و A2 من الشروط الأولية i (0) و di (0) / dt.

Example 8.1 The switch in Fig. 8.2 has been closed for a long time. It is open at t = 0. Find: (a) $i(0^+)$, $v(0^+)$, (b) $di(0^+)/dt$, $dv(0^+)/dt$, (c) $i(\infty)$, $v(\infty)$



Sol : (a) If the switch is closed a long time before t = 0, it means that the circuit has reached dc steady state at t = 0. At dc steady state, the inductor acts like a short circuit, while the capacitor acts like an open circuit, so we have the circuit in Fig. 8.3(a) at $t = 0^-$. Thus,

$$i(0^-) = \frac{12}{4+2} = 2\ \text{A}, \qquad v(0^-) = 2i(0^-) = 4\ \text{V}$$

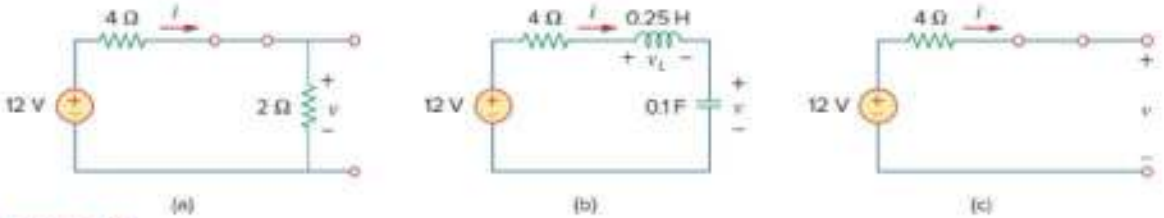


Figure 8.3
Equivalent circuit of that in Fig. 8.2 for: (a) $t = 0^-$, (b) $t = 0^+$, (c) $t \to \infty$.

$$i_C(0^+) = i(0^+) = 2\ \text{A}$$

Since $C\,dv/dt = i_C$, $dv/dt = i_C/C$, and

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = \frac{2}{0.1} = 20\ \text{V/s}$$

Similarly, since $L\,di/dt = v_L$, $di/dt = v_L/L$. We now obtain v_L by applying KVL to the loop in Fig. 8.3(b). The result is

$$-12 + 4i(0^+) + v_L(0^+) + v(0^+) = 0$$

or

$$v_L(0^+) = 12 - 8 - 4 = 0$$

Thus,

$$\frac{di(0^+)}{dt} = \frac{v_L(0^+)}{L} = \frac{0}{0.25} = 0\ \text{A/s}$$

(c) For $t > 0$, the circuit undergoes transience. But as $t \to \infty$, the circuit reaches steady state again. The inductor acts like a short circuit and the capacitor like an open circuit, so that the circuit in Fig. 8.3(b) becomes that shown in Fig. 8.3(c), from which we have

$$i(\infty) = 0\ \text{A}, \qquad v(\infty) = 12\ \text{V}$$

Practice Problem 8.1

The switch in Fig. 8.4 was open for a long time but closed at $t = 0$. Determine: (a) $i(0^+)$, $v(0^+)$, (b) $di(0^+)/dt$, $dv(0^+)/dt$, (c) $i(\infty)$, $v(\infty)$.

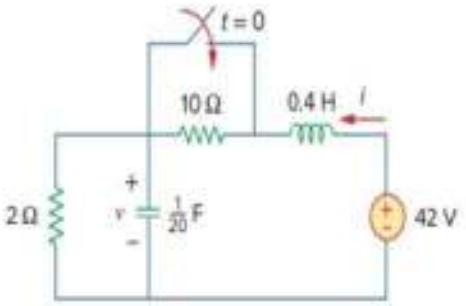


Figure 8.4
For Practice Prob. 8.1.

Answer: (a) 3.5 A, 7 V, (b) 87.5 A/s, 0 V/s, (c) 21 A, 42 V.

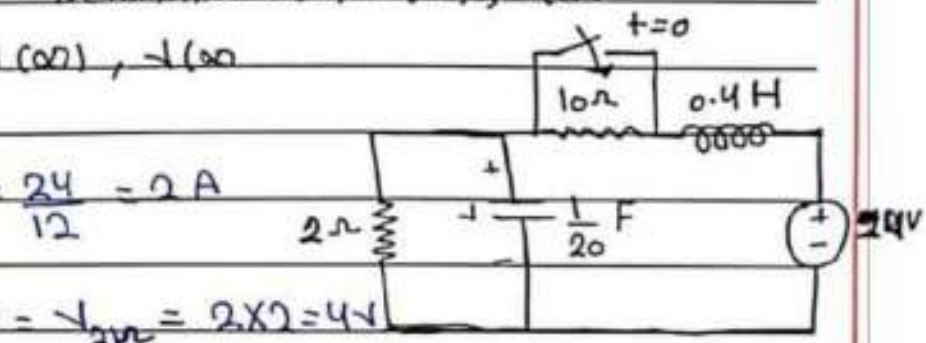
الحل في الصفحة التالية .....

Spark POLYTECHNIC

Circuit 1

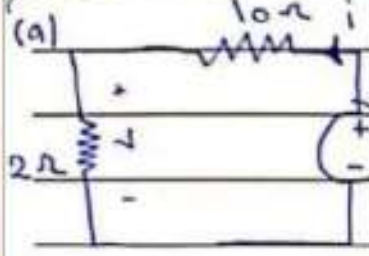
Practice Problem 8.1

The switch in Fig 8.4 was open for along time but closed at $t=0$ Determine: (a) $i(0^+), v(0^+)$
b) $\frac{di(0^+)}{dt}, \frac{dv(0^+)}{dt}$ c) $i(\infty), v(\infty)$



Sol:-

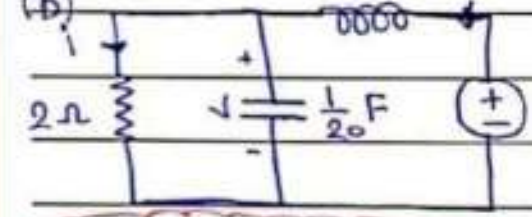
(a)



$i(0^+) = \frac{24}{12} = 2\,A$

$v(0^+) = v_{2\Omega} = 2 \times 2 = 4V$

(b)



المواسع يصبح o.c
الملف يصبح Sh.c
(نأخذ Loop لإيجاد v_L)

$L\frac{di(0^+)}{dt} = v_L$

هنا مقاومة 2Ω ما تأثر بسبب Sh.c

$-4 - V_L + 24 = 0 \rightarrow V_L = 20V$

$\frac{di(0^+)}{dt} = \frac{20}{0.4} = 50\ A/s$

(نأخذ Loop لإيجاد i_c)

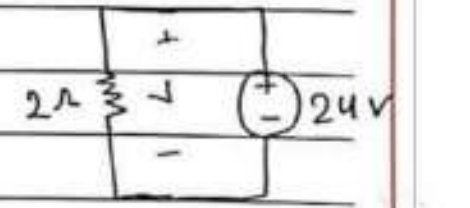
$C\frac{dv(0^+)}{dt} = i_c$

$-Ri + 4 = 0, \quad i = 2A \Rightarrow i_c = i_L - i_{2\Omega} \Rightarrow i_c = 2 - 2 = Zero$

بما أن i_L بنفس اتجاه i_c، وهو عكس اتجاه i_c

$\Rightarrow \frac{dV(0^+)}{dt} = \frac{i_c}{C} = 0\ V/s$

(c)



$\Rightarrow t > 0^+ \quad i(\infty), v(\infty)$

$i(\infty) = 24/2 = 12\ A$

$v(\infty) = 2 \times 12 = 24\ V$

Example: The Source-Free Series RLC Circuit

Example 8.3 In Fig. 8.8, R = 40 Ω, L = 4 H, and C = 1/4 F. Calculate the characteristic roots of the circuit. Is the natural response overdamped , underdamped, or critically damped?

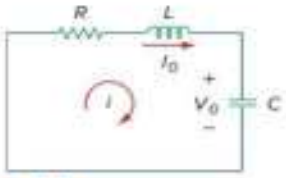


Figure 8.8
A source-free series *RLC* circuit.

Solution:
We first calculate

$$\alpha = \frac{R}{2L} = \frac{40}{2(4)} = 5, \qquad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \times \frac{1}{4}}} = 1$$

The roots are

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -5 \pm \sqrt{25 - 1}$$

or

$$s_1 = -0.101, \qquad s_2 = -9.899$$

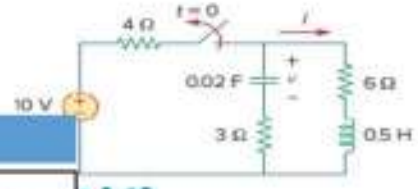
Since $\alpha > \omega_0$, we conclude that the response is overdamped. This is also evident from the fact that the roots are real and negative.

Practice Problem 8.3

If $R = 10\,\Omega$, $L = 5$ H, and $C = 2$ mF in Fig. 8.8, find α, ω_0, s_1, and s_2. What type of natural response will the circuit have?

Answer: 1, 10, $-1 \pm j9.95$, underdamped.

Example 8.4 Find i(t) in the circuit of Fig. 8.10. Assume that the circuit has reached steady state at t = 0⁻



e 8.10
ample 8.4.

شرح حل المثال السابق :

1- أولاً نجد كل من ∝, ω0 ونحدد الحالة والمعادلة الخاصة بها
2- ثم نجد A1,A2 وذالك من خلال المعادلتين
3- نعوض I(0) بقيمتها
4- ونجد $\frac{di(0)}{dt} = -\frac{RI0+V0}{L}$ حيث ان I0 للمحاثة و V0 للمواسع عند صفر
5- نكتب معادلة التيار او الجهد لتلك الدارة

Spark POLYTECHNIC

Solution:

For t < 0, the switch is closed. The capacitor acts like an open circuit while the inductor acts like a shunted circuit. The equivalent circuit is shown in Fig. 8.11(a). Thus, at t = 0,

$$i(0) = \frac{10}{4+6} = 1\text{ A}, \qquad v(0) = 6i(0) = 6\text{ V}$$

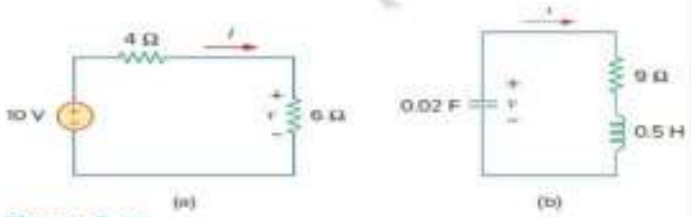


Figure 8.11
The circuit in Fig. 8.10: (a) for $t < 0$, (b) for $t > 0$.

where i(0) is the initial current through the inductor and v(0) is the initial voltage across the capacitor. For t > 0, the switch is opened and the voltage source is disconnected. The equivalent circuit is shown in Fig. 8.11(b), which is a sourcefree series RLC circuit. Notice that the 3-Ω and 6-Ω resistors, which are in series in Fig. 8.10 when the switch is opened, have been combined to give R = 9 Ω in Fig. 8.11(b). The roots are calculated as follows:

$$\alpha = \frac{R}{2L} = \frac{9}{2(\frac{1}{2})} = 9, \qquad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{1}{2} \times \frac{1}{50}}} = 10$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -9 \pm \sqrt{81 - 100}$$

or

$$s_{1,2} = -9 \pm j4.359$$

Hence, the response is underdamped ($\alpha < \omega_0$); that is,

$$i(t) = e^{-9t}(A_1 \cos 4.359t + A_2 \sin 4.359t) \qquad \textbf{(8.4.1)}$$

We now obtain A_1 and A_2 using the initial conditions. At $t = 0$,

$$i(0) = 1 = A_1 \qquad \textbf{(8.4.2)}$$

From Eq. (8.5),

$$\left.\frac{di}{dt}\right|_{t=0} = -\frac{1}{L}[Ri(0) + v(0)] = -2[9(1) - 6] = -6\text{ A/s} \qquad \textbf{(8.4.3)}$$

Note that $v(0) = V_0 = -6$ V is used, because the polarity of in Fig. 8.11(b) is opposite that in Fig. 8.8. Taking the derivative of $i(t)$ in Eq. (8.4.1),

$$\frac{di}{dt} = -9e^{-9t}(A_1 \cos 4.359t + A_2 \sin 4.359t) + e^{-9t}(4.359)(-A_1 \sin 4.359t + A_2 \cos 4.359t)$$

Imposing the condition in Eq. (8.4.3) at $t = 0$ gives

$$-6 = -9(A_1 + 0) + 4.359(-0 + A_2)$$

But $A_1 = 1$ from Eq. (8.4.2). Then

$$-6 = -9 + 4.359A_2 \qquad \rightarrow \qquad A_2 = 0.6882$$

Substituting the values of A_1 and A_2 in Eq. (8.4.1) yields the complete solution as

$$i(t) = e^{-9t}(\cos 4.359t + 0.6882 \sin 4.359t)\text{ A}$$

Spark POLYTECHNIC

Example The Source-Free Parallel RLC Circuit

Example 8.5 In the parallel circuit of Fig. 8.13, find v(t) for t >0, assuming v(0) = 5 V, i(0) = 0, L = 1 H, and C = 10 mF. Consider these cases: R = 1.923 Ω, R = 5 Ω, and R = 6.25 Ω.

Solution:

CASE 1 If $R = 1.923\ \Omega$,

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 1.923 \times 10 \times 10^{-3}} = 26$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10 \times 10^{-3}}} = 10$$

Since $\alpha > \omega_0$ in this case, the response is overdamped. The roots of the characteristic equation are

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -2, -50$$

and the corresponding response is

$$v(t) = A_1 e^{-2t} + A_2 e^{-50t} \tag{8.5.1}$$

We now apply the initial conditions to get A_1 and A_2.

$$v(0) = 5 = A_1 + A_2 \tag{8.5.2}$$

$$\frac{dv(0)}{dt} = \frac{v(0) + Ri(0)}{RC} = -\frac{5 + 0}{1.923 \times 10 \times 10^{-3}} = -260$$

But differentiating Eq. (8.5.1),

$$\frac{dv}{dt} = -2A_1 e^{-2t} - 50A_2 e^{-50t}$$

At $t = 0$,

$$-260 = -2A_1 - 50A_2 \tag{8.5.3}$$

From Eqs. (8.5.2) and (8.5.3), we obtain $A_1 = -0.2083$ and $A_2 = 5.208$. Substituting A_1 and A_2 in Eq. (8.5.1) yields

$$v(t) = -0.2083e^{-2t} + 5.208e^{-50t} \tag{8.5.4}$$

CASE 2 When $R = 5\ \Omega$,

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 5 \times 10 \times 10^{-3}} = 10$$

while $\omega_0 = 10$ remains the same. Since $\alpha = \omega_0 = 10$, the response is critically damped. Hence, $s_1 = s_2 = -10$, and

$$v(t) = (A_1 + A_2 t)e^{-10t} \tag{8.5.5}$$

To get A_1 and A_2, we apply the initial conditions

$$v(0) = 5 = A_1 \tag{8.5.6}$$

$$\frac{dv(0)}{dt} = -\frac{v(0) + Ri(0)}{RC} = -\frac{5 + 0}{5 \times 10 \times 10^{-3}} = -100$$

But differentiating Eq. (8.5.5),

$$\frac{dv}{dt} = (-10A_1 - 10A_2 t + A_2)e^{-10t}$$

Spark POLYTECHNIC

At $t = 0$,

$$-100 = -10A_1 + A_2 \tag{8.5.7}$$

From Eqs. (8.5.6) and (8.5.7), $A_1 = 5$ and $A_2 = -50$. Thus,

$$v(t) = (5 - 50t)e^{-10t}\ \text{V} \tag{8.5.8}$$

CASE 3 When $R = 6.25\ \Omega$,

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 6.25 \times 10 \times 10^{-3}} = 8$$

while $\omega_0 = 10$ remains the same. As $\alpha < \omega_0$ in this case, the response is underdamped. The roots of the characteristic equation are

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -8 \pm j6$$

Hence,

$$v(t) = (A_1 \cos 6t + A_2 \sin 6t)e^{-8t} \tag{8.5.9}$$

We now obtain A_1 and A_2, as

$$v(0) = 5 = A_1 \tag{8.5.10}$$

$$\frac{dv(0)}{dt} = -\frac{v(0) + Ri(0)}{RC} = -\frac{5 + 0}{6.25 \times 10 \times 10^{-3}} = -80$$

But differentiating Eq. (8.5.9),

$$\frac{dv}{dt} = (-8A_1 \cos 6t - 8A_2 \sin 6t - 6A_1 \sin 6t + 6A_2 \cos 6t)e^{-8t}$$

At $t = 0$,

$$-80 = -8A_1 + 6A_2 \tag{8.5.11}$$

From Eqs. (8.5.10) and (8.5.11), $A_1 = 5$ and $A_2 = -6.667$. Thus,

$$v(t) = (5 \cos 6t - 6.667 \sin 6t)e^{-8t} \tag{8.5.12}$$

:e of damping
three cases.

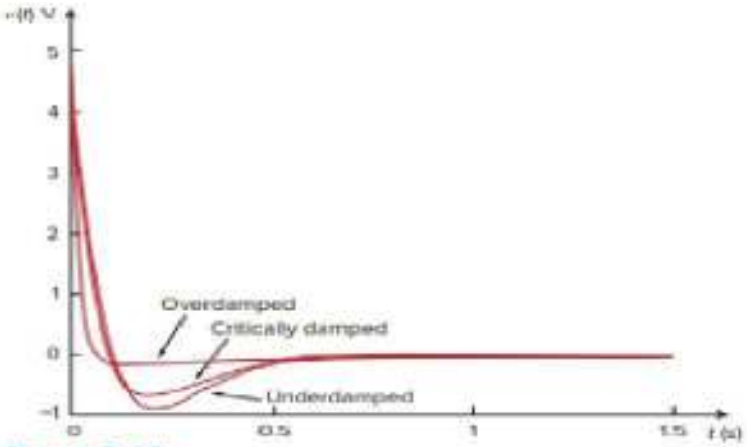


Figure 8.14
For Example 8.5: responses for three degrees of damping.

Practice Problem 8.5

In Fig. 8.13, let $R = 2\ \Omega$, $L = 0.4$H, $C = 25$mF, $v(0) = 0$, $i(0) = 50$mA. Find $v(t)$ for $t > 0$.

Answer: $-2te^{-10t}$ V.

Spark POLYTECHNIC

Example 8.6

Find $v(t)$ for $t > 0$ in the *RLC* circuit of Fig. 8.15.

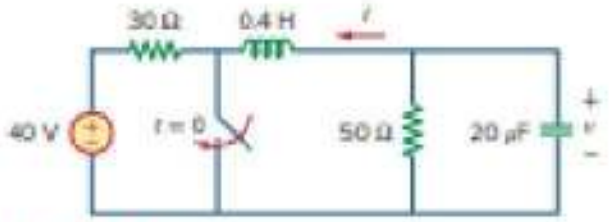


Figure 8.15
For Example 8.6.

Solution:
When $t < 0$, the switch is open; the inductor acts like a short circuit while the capacitor behaves like an open circuit. The initial voltage across the capacitor is the same as the voltage across the 50-Ω resistor; that is,

$$v(0) = \frac{50}{30 + 50}(40) = \frac{5}{8} \times 40 = 25 \text{ V} \qquad (8.6.1)$$

The initial current through the inductor is

$$i(0) = -\frac{40}{30 + 50} = -0.5 \text{ A}$$

The direction of i is as indicated in Fig. 8.15 to conform with the direction of I_0 in Fig. 8.13, which is in agreement with the convention that current flows into the positive terminal of an inductor (see Fig. 6.23). We need to express this in terms of dv/dt, since we are looking for v.

$$\frac{dv(0)}{dt} = -\frac{v(0) + Ri(0)}{RC} = -\frac{25 - 50 \times 0.5}{50 \times 20 \times 10^{-6}} = 0 \qquad (8.6.2)$$

When $t > 0$, the switch is closed. The voltage source along with the 30-Ω resistor is separated from the rest of the circuit. The parallel *RLC* circuit acts independently of the voltage source, as illustrated in Fig. 8.16. Next, we determine that the roots of the characteristic equation are

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 50 \times 20 \times 10^{-6}} = 500$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{0.4 \times 20 \times 10^{-6}} = 354$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$= -500 \pm \sqrt{250{,}000 - 124{,}997.6} = -500 \pm 354$$

or

$$s_1 = -854, \qquad s_2 = -146$$

Spark POLYTECHNIC

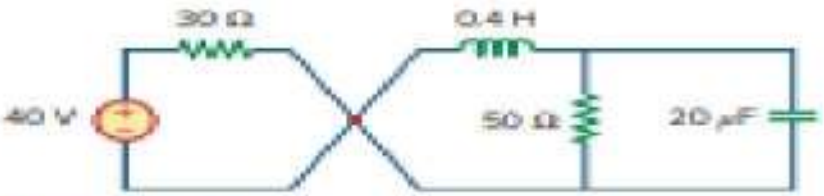


Figure 8.16
The circuit in Fig. 8.15 when $t > 0$. The parallel *RLC* circuit on the right-hand side acts independently of the circuit on the left-hand side of the junction.

Since $\alpha > \omega_0$, we have the overdamped response

$$v(t) = A_1 e^{-854t} + A_2 e^{-146t} \tag{8.6.3}$$

At $t = 0$, we impose the condition in Eq. (8.6.1),

$$v(0) = 25 = A_1 + A_2 \quad \Rightarrow \quad A_2 = 25 - A_1 \tag{8.6.4}$$

Taking the derivative of $v(t)$ in Eq. (8.6.3),

$$\frac{dv}{dt} = -854A_1 e^{-854t} - 146A_2 e^{-146t}$$

Imposing the condition in Eq. (8.6.2),

$$\frac{dv(0)}{dt} = 0 = -854A_1 - 146A_2$$

or

$$0 = 854A_1 + 146A_2 \tag{8.6.5}$$

Solving Eqs. (8.6.4) and (8.6.5) gives

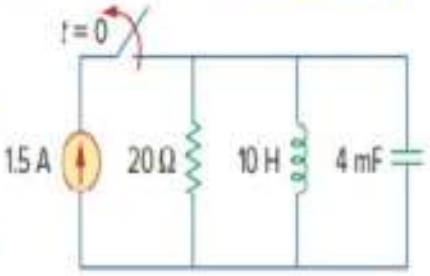
$$A_1 = -5.156, \qquad A_2 = 30.16$$

Thus, the complete solution in Eq. (8.6.3) becomes

$$v(t) = -5.156e^{-854t} + 30.16e^{-146t} \text{ V}$$

Practice Problem 8.6

Refer to the circuit in Fig. 8.17. Find $v(t)$ for $t > 0$.



Answer: $50(e^{-10t} - e^{-25t})$ V.

Spark POLYTECHNIC

Example Step Response of a Series RLC Circuit

Example 8.7 For the circuit in Fig. 8.19, find v(t) and i(t) for t>0. Consider these cases: R = 5 Ω, R = 4 Ω, and R = 1 Ω.



Figure 8.19
For Example 8.7.

CASE 1 When $R = 5\ \Omega$. For $t < 0$, the switch is closed for a long time. The capacitor behaves like an open circuit while the inductor acts like a short circuit. The initial current through the inductor is

$$i(0) = \frac{24}{5+1} = 4\ \text{A}$$

and the initial voltage across the capacitor is the same as the voltage across the 1-Ω resistor; that is,

$$v(0) = 1i(0) = 4\ \text{V}$$

For $t > 0$, the switch is opened, so that we have the 1-Ω resistor disconnected. What remains is the series RLC circuit with the voltage source. The characteristic roots are determined as follows:

$$\alpha = \frac{R}{2L} = \frac{5}{2 \times 1} = 2.5, \qquad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.25}} = 2$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -1, -4$$

Since $\alpha > \omega_0$, we have the overdamped natural response. The total response is therefore

$$v(t) = v_{ss} + (A_1 e^{-t} + A_2 e^{-4t})$$

where v_{ss} is the steady-state response. It is the final value of the capacitor voltage. In Fig. 8.19, $v_f = 24$ V. Thus,

$$v(t) = 24 + (A_1 e^{-t} + A_2 e^{-4t}) \qquad \textbf{(8.7.1)}$$

We now need to find A_1 and A_2 using the initial conditions.

$$v(0) = 4 = 24 + A_1 + A_2$$

or

$$-20 = A_1 + A_2 \qquad \textbf{(8.7.2)}$$

The current through the inductor cannot change abruptly and is the same current through the capacitor at $t = 0^+$ because the inductor and capacitor are now in series. Hence,

$$i(0) = C\frac{dv(0)}{dt} = 4 \qquad \Rightarrow \qquad \frac{dv(0)}{dt} = \frac{4}{C} = \frac{4}{0.25} = 16$$

Before we use this condition, we need to take the derivative of v in Eq. (8.7.1).

$$\frac{dv}{dt} = -A_1 e^{-t} - 4A_2 e^{-4t} \qquad \textbf{(8.7.3)}$$

At $t = 0$,

$$\frac{dv(0)}{dt} = 16 = -A_1 - 4A_2 \qquad \textbf{(8.7.4)}$$

Spark POLYTECHNIC

From Eqs. (8.7.2) and (8.7.4), $A_1 = -64/3$ and $A_2 = 4/3$. Substituting A_1 and A_2 in Eq. (8.7.1), we get

$$v(t) = 24 + \frac{4}{3}(-16e^{-t} + e^{-4t})\ \text{V} \qquad \textbf{(8.7.5)}$$

Since the inductor and capacitor are in series for $t > 0$, the inductor current is the same as the capacitor current. Hence,

$$i(t) = C\frac{dv}{dt}$$

Multiplying Eq. (8.7.3) by $C = 0.25$ and substituting the values of A_1 and A_2 gives

$$i(t) = \frac{4}{3}(4e^{-t} - e^{-4t})\ \text{A} \qquad \textbf{(8.7.6)}$$

Note that $i(0) = 4$ A, as expected.

■ **CASE 2** When $R = 4\ \Omega$. Again, the initial current through the inductor is

$$i(0) = \frac{24}{4+1} = 4.8\ \text{A}$$

and the initial capacitor voltage is

$$v(0) = 1i(0) = 4.8\ \text{V}$$

For the characteristic roots,

$$\alpha = \frac{R}{2L} = \frac{4}{2\times 1} = 2$$

while $\omega_0 = 2$ remains the same. In this case, $s_1 = s_2 = -\alpha = -2$, and we have the critically damped natural response. The total response is therefore

$$v(t) = v_{ss} + (A_1 + A_2 t)e^{-2t}$$

and, as before $v_{ss} = 24$ V,

$$v(t) = 24 + (A_1 + A_2 t)e^{-2t} \qquad \textbf{(8.7.7)}$$

To find A_1 and A_2, we use the initial conditions. We write

$$v(0) = 4.8 = 24 + A_1 \quad \Rightarrow \quad A_1 = -19.2 \qquad \textbf{(8.7.8)}$$

Since $i(0) = C\,dv(0)/dt = 4.8$ or

$$\frac{dv(0)}{dt} = \frac{4.8}{C} = 19.2$$

From Eq. (8.7.7),

$$\frac{dv}{dt} = (-2A_1 - 2tA_2 + A_2)e^{-2t} \qquad \textbf{(8.7.9)}$$

At $t = 0$,

$$\frac{dv(0)}{dt} = 19.2 = -2A_1 + A_2 \qquad \textbf{(8.7.10)}$$

Spark POLYTECHNIC

From Eqs. (8.7.8) and (8.7.10), $A_1 = -19.2$ and $A_2 = -19.2$. Thus, Eq. (8.7.7) becomes

$$v(t) = 24 - 19.2(1 + t)e^{-2t}\ \text{V} \qquad \textbf{(8.7.11)}$$

The inductor current is the same as the capacitor current; that is,

$$i(t) = C\frac{dv}{dt}$$

Multiplying Eq. (8.7.9) by $C = 0.25$ and substituting the values of A_1 and A_2 gives

$$i(t) = (4.8 + 9.6t)e^{-2t}\ \text{A} \qquad \textbf{(8.7.12)}$$

Note that $i(0) = 4.8$ A, as expected.

CASE 3 When $R = 1\ \Omega$. The initial inductor current is

$$i(0) = \frac{24}{1 + 1} = 12\ \text{A}$$

and the initial voltage across the capacitor is the same as the voltage across the 1-Ω resistor,

$$v(0) = 1i(0) = 12\ \text{V}$$

$$\alpha = \frac{R}{2L} = \frac{1}{2 \times 1} = 0.5$$

Since $\alpha = 0.5 < \omega_0 = 2$, we have the underdamped response

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -0.5 \pm j1.936$$

The total response is therefore

$$v(t) = 24 + (A_1 \cos 1.936t + A_2 \sin 1.936t)e^{-0.5t} \qquad \textbf{(8.7.13)}$$

We now determine A_1 and A_2. We write

$$v(0) = 12 = 24 + A_1 \quad \Rightarrow \quad A_1 = -12 \qquad \textbf{(8.7.14)}$$

Since $i(0) = C\,dv(0)/dt = 12$,

$$\frac{dv(0)}{dt} = \frac{12}{C} = 48 \qquad \textbf{(8.7.15)}$$

But

$$\frac{dv}{dt} = e^{-0.5t}(-1.936A_1 \sin 1.936t + 1.936A_2 \cos 1.936t) - 0.5e^{-0.5t}(A_1 \cos 1.936t + A_2 \sin 1.936t) \qquad \textbf{(8.7.16)}$$

At $t = 0$,

$$\frac{dv(0)}{dt} = 48 = (-0 + 1.936A_2) - 0.5(A_1 + 0)$$

Substituting $A_1 = -12$ gives $A_2 = 21.694$, and Eq. (8.7.13) becomes

$$v(t) = 24 + (21.694 \sin 1.936t - 12 \cos 1.936t)e^{-0.5t}\ \text{V} \qquad \textbf{(8.7.17)}$$

The inductor current is

$$i(t) = C\frac{dv}{dt}$$

Spark POLYTECHNIC

Multiplying Eq. (8.7.16) by $C = 0.25$ and substituting the values of A_1 and A_2 gives

$$i(t) = (3.1 \sin 1.936t + 12 \cos 1.936t)e^{-0.5t} \text{ A} \qquad \textbf{(8.7.18)}$$

Note that $i(0) = 12$ A, as expected.

Figure 8.20 plots the responses for the three cases. From this figure, we observe that the critically damped response approaches the step input of 24 V the fastest.

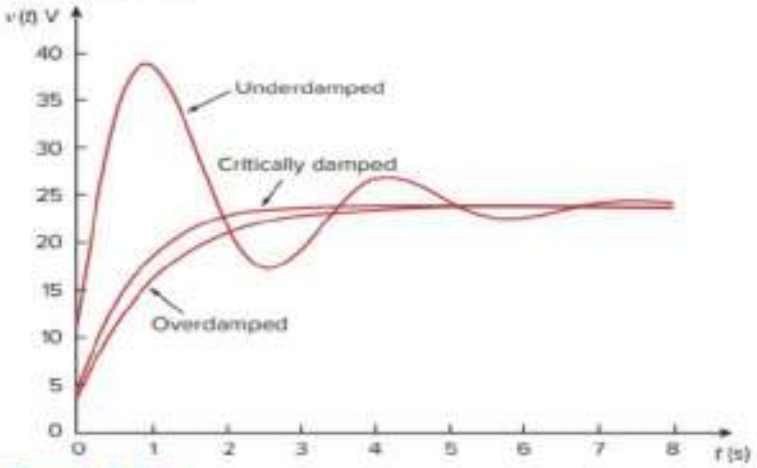


Figure 8.20
For Example 8.7: response for three degrees of damping.

Practice Problem 8.7

Having been in position a for a long time, the switch in Fig. 8.21 is moved to position b at $t = 0$. Find $v(t)$ and $v_R(t)$ for $t > 0$.

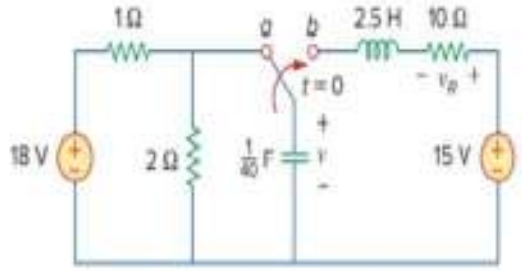


Figure 8.21
For Practice Prob. 8.7.

Answer: $15 - (1.7321 \sin 3.464t + 3 \cos 3.464t)e^{-2t}$ V, $3.464e^{-2t} \sin 3.464t$ V.

Spark POLYTECHNIC

Example Step Response of a Parallel RLC Circuit

Example 8.9 Find the complete response v and then i for t > 0 in the circuit of Fig. 8.25

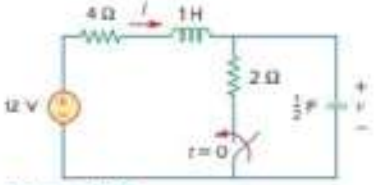


Figure 8.25
For Example 8.9.

Sol :

We first find the initial and final values. At t = 0⁻, the circuit is at steady state. The switch is open; the equivalent circuit is shown in Fig. 8.26(a). It is evident from the figure that

$v(0^-) = 12$ V, $i(0^-) = 0$

At t = 0⁺, the switch is closed; the equivalent circuit is in Fig. 8.26(b). By the continuity of capacitor voltage and inductor current, we know that

$v(0^+) = v(0^-) = 12$ V, $i(0^+) = i(0^-) = 0$

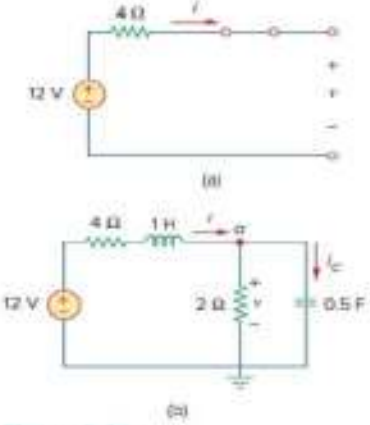


Figure 8.26
Equivalent circuit of the circuit in Fig. 8.25 for: (a) $t < 0$, (b) $t > 0$.

To get $dv(0^+)/dt$, we use $C\,dv/dt = i_C$ or $dv/dt = i_C/C$. Applying KCL at node a in Fig. 8.26(b),

$$i(0^+) = i_C(0^+) + \frac{v(0^+)}{2}$$

$$0 = i_C(0^+) + \frac{12}{2} \quad \Rightarrow \quad i_C(0^+) = -6 \text{ A}$$

Hence,

$$\frac{dv(0^+)}{dt} = \frac{-6}{0.5} = -12 \text{ V/s} \tag{8.9.2}$$

The final values are obtained when the inductor is replaced by a short circuit and the capacitor by an open circuit in Fig. 8.26(b), giving

$$i(\infty) = \frac{12}{4+2} = 2 \text{ A}, \qquad v(\infty) = 2i(\infty) = 4 \text{ V} \tag{8.9.3}$$

Next, we obtain the form of the transient response for $t > 0$. By turning off the 12-V voltage source, we have the circuit in Fig. 8.27. Applying KCL at node a in Fig. 8.27 gives

$$i = \frac{v}{2} + \frac{1}{2}\frac{dv}{dt} \tag{8.9.4}$$

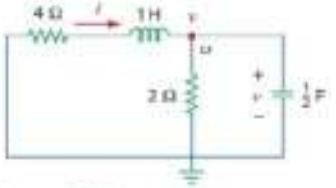


Figure 8.27
Obtaining the form of the transient response for Example 8.9.

Applying KVL to the left mesh results in

$$4i + 1\frac{di}{dt} + v = 0 \tag{8.9.5}$$

Since we are interested in v for the moment, we substitute i from Eq. (8.9.4) into Eq. (8.9.5). We obtain

$$2v + 2\frac{dv}{dt} + \frac{1}{2}\frac{dv}{dt} + \frac{1}{2}\frac{d^2v}{dt^2} + v = 0$$

$$\frac{d^2v}{dt^2} + 5\frac{dv}{dt} + 6v = 0$$

Spark POLYTECHNIC

From this, we obtain the characteristic equation as

$$s^2 + 5s + 6 = 0$$

with roots $s = -2$ and $s = -3$. Thus, the natural response is

$$v_n(t) = Ae^{-2t} + Be^{-3t} \qquad \textbf{(8.9.6)}$$

where A and B are unknown constantsto be determined later. The steady-state response is

$$v_{ss}(t) = v(\infty) = 4 \qquad \textbf{(8.9.7)}$$

The complete response is

$$v(t) = v_t + v_{ss} = 4 + Ae^{-2t} + Be^{-3t} \qquad \textbf{(8.9.8)}$$

We now determine A and B using the initial values. From Eq. (8.9.1), $v(0) = 12$. Substituting this into Eq. (8.9.8) at $t = 0$ gives

$$12 = 4 + A + B \quad \Rightarrow \quad A + B = 8 \qquad \textbf{(8.9.9)}$$

Taking the derivative of v in Eq. (8.9.8),

$$\frac{dv}{dt} = -2Ae^{-2t} - 3Be^{-3t} \qquad \textbf{(8.9.10)}$$

Substituting Eq. (8.9.2) into Eq. (8.9.10) at $t = 0$ gives

$$-12 = -2A - 3B \quad \Rightarrow \quad 2A + 3B = 12 \qquad \textbf{(8.9.11)}$$

From Eqs. (8.9.9) and (8.9.11), we obtain

$$A = 12, \qquad B = -4$$

so that Eq. (8.9.8) becomes

$$v(t) = 4 + 12e^{-2t} - 4e^{-3t}\ \text{V}, \qquad t > 0 \qquad \textbf{(8.9.12)}$$

From v, we can obtain other quantities of interest by referring to Fig. 8.26(b). To obtain i, for example,

$$\begin{aligned} i = \frac{v}{2} + \frac{1}{2}\frac{dv}{dt} &= 2 + 6e^{-2t} - 2e^{-3t} - 12e^{-2t} + 6e^{-3t} \\ &= 2 - 6e^{-2t} + 4e^{-3t}\ \text{A}, \qquad t > 0 \end{aligned} \qquad \textbf{(8.9.13)}$$

Notice that $i(0) = 0$, in agreement with Eq. (8.9.1).

انتهى هذا الشابتر ملاحظة مهمة شابتر 7 وا 8 مهمين جدا في مواد تخصص خصوصا في مادة بور الكترونيك وباقي المواد وسوف يتكرر في مختبرات

وبيقى هذا العمل جهد بشري معرض للخطأ والنقصان

Spark POLYTECHNIC